FINITENESS OF PLANAR CENTRAL CONFIGURATIONS OF FIVE BODIES FOR ALL POSITIVE MASSES

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Abstract. We prove that for every choice of five positive masses the planar five-body problem has only finitely many central configurations up to rotation, translation and scaling. This answers Smale's sixth problem for five bodies. Albouy and Kaloshin proved the statement for all masses outside a closed algebraic subset of codimension two of the mass space. We keep their classification of singular sequences and replace the construction of the exceptional set by a direct exclusion: we use finite boundary charts for seven of the eight diagrams that can carry an unbounded four-product, and an exhaustive order census for Diagram 5. We derive the necessary mass conditions where needed and show that no germ with an unbounded four-product lies on a curve of normalized central configurations through a positive configuration. Five diagrams are excluded for all complex germs, two on curves through real configurations, and one on curves through positive configurations. Parts of the argument are computer-assisted: exact algebra, interval arithmetic and exact linear programming.

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1. Introduction

A configuration of n point masses $m_1, \dots, m_n > 0$ at positions $q_1, \dots, q_n \in \mathbb{R}^2$ is a *central configuration* if there is $\lambda \in \mathbb{R}$ with

$$\lambda(q_k - c) = \sum_{l \neq k} m_l \frac{q_l - q_k}{|q_l - q_k|^3}, \qquad k = 1, \dots, n,$$

where c is the centre of mass. Central configurations give the relative equilibria of the planar n-body problem and the homographic solutions; they are invariant under rotations, translations and scalings, and one counts them modulo these symmetries. Euler and Lagrange found the five classes for $n = 3$; Moulton [12] showed that there are exactly $n!/2$ collinear classes for every n. Wintner [15] and Smale [14] asked whether the number of classes is finite for every choice of positive masses; this is Smale's sixth problem. Hampton and Moeckel [7] proved finiteness for $n = 4$. Albouy and Kaloshin [1] gave a new proof for $n = 4$ and proved the following for $n = 5$.

Theorem 1.1 ([1, Thm. 2])**.** *There is a closed algebraic subset A of codimension two such that for every $(m_1, \dots, m_5) \in \mathbb{R}^5_{>0} \setminus A$ there are finitely many positive normalized central configurations of the planar five-body problem.*

Our main result removes the exceptional set.

Theorem 1.2. *For every $(m_1, \dots, m_5) \in \mathbb{R}^5_{>0}$ there are finitely many positive normalized central configurations of the planar five-body problem. Equivalently, the planar five-body problem has finitely many central configurations up to rotation, translation and scaling.*

Related work. The generic finiteness of [1] for five bodies has been reproved by tropical geometry [8] and by an algorithmic version of the method of [1] [3, 4], which also reduces the planar six-body problem to finitely many diagrams. In the spatial five-body problem finiteness is known outside an explicit set of exceptional masses [6], and four-body central configurations are finite for all nonzero masses [16]. Computer-assisted methods classify the central configurations of five equal masses [9, 10] and, recently, give exact counts (between 294 and 450 classes) for finitely many mass vectors on the loci (23), (29) and (32) of [1, §8], the last of which lies in the exceptional set [11]; these computations treat individual mass vectors. For all positive masses, finiteness is known for special classes such as the equilateral pentagonal central configurations [5]. We are not aware of a previous proof of Theorem 1.2 on the whole exceptional set.

Strategy. Albouy and Kaloshin embed the central configuration equations in a polynomial system and study *singular sequences*: unbounded sequences of complex normalized solutions. To such a sequence they attach a two-coloured *diagram* recording which terms of the equations are of maximal order, and they prove that every singular sequence approaches one of sixteen diagrams [1, Fig. 11]. If a mass vector admits infinitely many normalized central configurations, the solutions contain an algebraic curve; along it some four-product of squared distances is unbounded, and a singular sequence along which it diverges approaches one of the first eight diagrams [1, Props. 8–9 and proof of Thm. 6]. Each diagram forces a polynomial relation among the masses. Albouy and Kaloshin then push a second quantity to a second singular limit on the same curve to obtain an independent relation, which yields the codimension-two set A.

We do not construct A. We show instead that, at every positive mass vector, none of the eight diagrams can be the first limit on a curve of normalized central configurations through a positive configuration. Our route is as follows.

(i) For Diagrams 1–4 and 6–8, build charts (weighted blow-ups of the AK system) and prove from the estimates of [1, §3] that every germ converges to a finite chart boundary; for Diagram 5, classify the orders of the outer bodies instead.
(ii) Derive the necessary mass conditions from the boundary equations, or from the isolated triangle in Diagram 5, where the exclusion needs one (Diagram 7 needs none); several conditions in [1, §8.3–8.8] are stated there with short or no proofs.
(iii) Exclude every germ with an unbounded four-product, by a finite-order obstruction, an order census, or an identity on the whole curve.

The exclusion holds at three levels. For Diagrams 1, 2, 4, 6 and 8 no germ exists at all, even in the complex domain. On curves through real configurations $U = I$ is a nonzero constant (Lemma 2.1; compare [1, Lemma 2]). The order census for Diagram 5 shows that its remaining germs on such curves have only bounded four-products, while the inertia identity for Diagram 7 forces $I \to 0$ for germs with an unbounded four-product; neither can supply the required first limit. For Diagram 3 the real level is not enough: the modified Roberts continuum [1, p. 543] is a curve of normalized central configurations at positive masses that contains an arc of real, not positive, configurations and carries a Diagram 3 germ. There we use positivity through an identity $\delta_{15} + 2\delta_{13} = 0$ that holds on the whole curve carrying the remaining germs.

Computer assistance. Several steps are computer-assisted: polynomial identities, resultants and Gröbner bases over $\mathbb{Q}$ and over number fields, Sturm sequences, interval and ball arithmetic with explicit bounds on the rounding errors, and exact rational linear programming. The statements that depend on such computations are marked as computer-assisted, and Appendix A describes the computations.

What we use from [1]**.** We use without re-proving it the classification of singular sequences (Figure 11 of [1] and its completeness), Proposition 8 of [1], the rules and estimates of [1, §3], and Lemma 1 and Proposition 1 of [1]. These are listed precisely in Section 2. We re-derive

the mass conditions of all diagrams that need one (Diagrams 1–6 and 8), several of which [1, §§8.3–8.8] state with short or no proofs, and we prove directly that U is constant along curves of solutions.

Organization. Section 2 recalls the framework of [1]. Section 3 reduces Theorem 1.2 to the exclusion of germs with an unbounded four-product. Section 4 states the exclusion theorem and proves Theorem 1.2. Sections 5–11 treat the eight diagrams. Appendix A describes the computer-assisted parts and Appendix B an independent cross-check through the exceptional set of [1].

2. The framework of Albouy and Kaloshin

2.1. **Normalized central configurations.** Fix masses $m = (m_1, \dots, m_5) \in \mathbb{R}^5_{>0}$. Following [1, §2] we use the complex coordinates $z_k = x_k + iy_k$, $w_k = x_k - iy_k$ and inverse distances $\delta_{kl} = \delta_{lk}$, and write $z_{kl} = z_l - z_k$, $w_{kl} = w_l - w_k$. The normalized system (system (4) of [1], written as system (10) there) is

$$\begin{aligned} z_k = \sum_{l\neq k} m_l\, \delta_{kl}^3\, (z_k - z_l), \qquad w_k = \sum_{l\neq k} m_l\, \delta_{kl}^3\, (w_k - w_l) \qquad (k = 1, \dots, 5), \\ \delta_{kl}^2\, z_{kl}\, w_{kl} = 1 \quad (1 \le k < l \le 5), \qquad z_{12} = w_{12}. \end{aligned} \tag{1}$$

Its zero set $A_m \subset \mathbb{C}^{20}$ is the set of *normalized central configurations*. A solution is *real* if $w_k = \bar{z}_k$ for all k and *positive* if moreover $\delta_{kl} > 0$ for all $k < l$ [1, Def. 2]. Positive normalized central configurations are in two-to-one correspondence with central configurations modulo rotation, translation and scaling, so Theorem 1.2 asserts that A_m has finitely many positive points. The map $\sigma(z, w, \delta) = (\bar{w}, \bar{z}, \bar{\delta})$ preserves (1) and fixes the real solutions. System (1) is symmetric under $z \leftrightarrow w$. A permutation π of the bodies, with the masses permuted, maps it to the same system with the gauge $z_{12} = w_{12}$ replaced by $z_{\pi(1)\pi(2)} = w_{\pi(1)\pi(2)}$; a rotation $(z, w) \mapsto (\lambda z, \lambda^{-1} w)$ restores the gauge and preserves the rest of the system, U, I, every r_{kl}^2, and the orders and diagrams of germs. We call both operations symmetries of (1).

2.2. **Potential and inertia.** Let $U = \sum_{k<l} m_k m_l \delta_{kl}$ and $I = \sum_k m_k z_k w_k$.

Lemma 2.1. *Every solution of* (1) *satisfies $U = I$, and U is constant on every irreducible curve $\Gamma \subset A_m$.*

Proof. Let $B \subset \mathbb{C}^{20}$ be the set where $\delta_{kl}^2 z_{kl} w_{kl} = 1$ for all $k < l$, and let $f = 2U + I$. On B no $z_{kl} w_{kl}$ vanishes, so B is smooth, (z, w) are local coordinates on B, and $d\delta_{kl} = -\frac{1}{2}\delta_{kl}^3 (w_{kl}\, dz_{kl} + z_{kl}\, dw_{kl})$. Hence, on B,

$$\frac{\partial f}{\partial z_k} = m_k\Big(w_k - \sum_{l\neq k} m_l \delta_{kl}^3 (w_k - w_l)\Big), \qquad \frac{\partial f}{\partial w_k} = m_k\Big(z_k - \sum_{l\neq k} m_l \delta_{kl}^3 (z_k - z_l)\Big),$$

so every point of A_m is a critical point of $f|_B$ (the gauge $z_{12} = w_{12}$ plays no role). The scaling $(z, w, \delta) \mapsto (sz, sw, s^{-1}\delta)$ preserves B and maps f to $2s^{-1}U + s^2 I$; differentiating at $s = 1$ gives $0 = -2U + 2I$ at every critical point of $f|_B$, so $U = I$ on A_m. Let $\Gamma \subset A_m$ be an irreducible curve. At a smooth point of Γ the tangent line lies in the tangent space of B, on which df vanishes; so f is locally constant on the smooth locus of Γ, which is connected, and hence constant on Γ. As $f = 3U$ on A_m, U is constant on Γ. □

The identity $U = I$ is [1, p. 541], and [1, Lemma 2] shows more: $U(A_m)$ is finite.

Fact AK1 ([1, Lemma 1])**.** *If $X \subset \mathbb{C}^N$ is a closed algebraic subset and f a polynomial, then $f(X)$ is finite or the complement of a finite set ("f is dominating").*

2.3. **Singular sequences and diagrams.** Let $\mathcal{Z} = (z_1, \dots, z_5, Z_{12}, \dots, Z_{45})$ and $\mathcal{W} = (w_1, \dots, w_5, W_{12}, \dots, W_{45})$ with $Z_{kl} = z_{kl}\,\delta_{kl}^3$ and $W_{kl} = w_{kl}\,\delta_{kl}^3$; these are, up to mass factors, the terms of the equations (1). Let $Z = \max_p |\mathcal{Z}_p|$ and $W = \max_q |\mathcal{W}_q|$. A *singular sequence* [1, Def. 3] is a sequence of normalized central configurations with Z or W unbounded, extracted so that the maximal components are fixed and $Z^{-1}\mathcal{Z}$, $W^{-1}\mathcal{W}$ converge. Its *diagram* has a z-circle at k if $\lim z_k/Z \neq 0$, a z-stroke between k and l if $\lim Z_{kl}/Z \neq 0$, and likewise in the second colour [1, §3]. After multiplying z by $a > 0$ and w by a^{-1} one may assume $Z = W = \epsilon^{-2}$ with $\epsilon \to 0$ [1, p. 547]. We write $a \prec b$, $a \preccurlyeq b$, $a \approx b$, $a \sim b$ as in [1, §3].

Fact AK2 ([1, §3])**.** *Rules 1a–1e and 2a–2e and Estimates 1–2 of* [1, §3] *hold for every singular sequence.*

Fact AK3 ([1, Prop. 1])**.** *Proposition 1 of* [1] *(localization of the origin in a cluster) holds.*

Fact AK4 ([1, §7])**.** *Every singular sequence of the five-body problem approaches, after renumbering the bodies and possibly exchanging z and w, one of the sixteen diagrams of* [1, Fig. 11].

For pairwise distinct i, j, k, l, m with $\{i, j, k, l, m\} = \{1, \dots, 5\}$ the *four-product* is $p_{ij} = r_{ij}^2 r_{kl}^2 r_{lm}^2 r_{mk}^2$, where $r_{kl}^2 = z_{kl} w_{kl}$.

Fact AK5 ([1, Prop. 8])**.** *Along a singular sequence approaching one of Diagrams 9–16 every four-product is bounded.*

Facts AK1–AK5 are the only results of [1] that the proof uses. The mass conditions of [1, §8] for Diagrams 1–6 and 8 are re-derived below (Propositions 5.10, 6.6, 7.2, 8.1, 9.5 and 11.4).

3. Reduction to germs with an unbounded four-product

Lemma 3.1. *Let $s \in (\mathbb{C}^*)^{10}$. Only finitely many points of A_m satisfy $z_{kl} w_{kl} = s_{kl}$ for all $k < l$.*

Proof. On such a point $\delta_{kl}^2 = s_{kl}^{-1}$, so δ takes finitely many values; fix one. Multiplying the z-equation of body k by m_k and summing gives $\sum_k m_k z_k = 0$ by antisymmetry, and likewise $\sum_k m_k w_k = 0$. Put $v_k = (z_k - z_1, w_k - w_1) \in \mathbb{C}^2$ and let B be the symmetric bilinear form with $B((a,b),(a,b)) = ab$. Then $B(v_k - v_l, v_k - v_l) = s_{kl}$, so the Gram matrix $B(v_k, v_l) = \frac{1}{2}(s_{1k} + s_{1l} - s_{kl})$ is fixed. The span V of the v_k is nondegenerate for B: either $V = \mathbb{C}^2$, or $V = \mathbb{C}v_2$ with $B(v_2, v_2) = s_{12} \neq 0$. If (v_k') is a second family with the same Gram matrix, its span V' is nondegenerate by the same argument, and $\sum_k c_k v_k = 0$ holds if and only if $\sum_k c_k B(v_k, v_l) = 0$ for all l, and likewise for the v_k'; so the v_k and the v_k' satisfy the same linear relations, and $v_k \mapsto v_k'$ is an isometry of nondegenerate subspaces; by Witt's theorem it extends to an element of the orthogonal group of B, which consists of the maps $(a,b) \mapsto (\lambda a, \lambda^{-1} b)$ and $(a,b) \mapsto (\lambda b, \lambda^{-1} a)$, $\lambda \in \mathbb{C}^*$. The centre of mass determines (z_1, w_1) from (v_k). For a fixed reference family (v_k^0) with first components a_2^0 and second components b_2^0 of v_2^0 (so $a_2^0 b_2^0 = s_{12} \neq 0$), the gauge $z_{12} = w_{12}$ leaves $\lambda^2 = b_2^0/a_2^0$ or $\lambda^2 = a_2^0/b_2^0$: at most four values. Hence there are finitely many solutions for each δ. □

Lemma 3.2 (Arc lemma)**.** *If A_m has infinitely many positive points, there is an irreducible algebraic curve $\Gamma \subset A_m$ such that*

(i) *Γ contains a nonconstant real-analytic arc of positive points;*
(ii) *$\sigma(\Gamma) = \Gamma$;*
(iii) *$U = I$ is constant on Γ, with value $c \in (0, \infty)$.*

Proof. In the real coordinates $(x, y, \delta) \in \mathbb{R}^{20}$ the positive points form a semialgebraic set. It has finitely many connected components, so one of them is infinite, hence of positive dimension, and the curve selection lemma [2, Thm. 2.5.5] gives a nonconstant Nash arc in it. Its Zariski closure $\Gamma \subset A_m$ (in the complexified coordinates; passing to (z, w) is linear) is an irreducible curve, which proves (i). Since (1) has real coefficients, $\sigma(\Gamma)$ is an irreducible curve containing the arc, so $\sigma(\Gamma) = \Gamma$. By Lemma 2.1, $U = I$ is constant on Γ, and on the arc $U = \sum m_k m_l \delta_{kl} > 0$. □

Lemma 3.3. *Let $\Gamma \subset A_m$ be an irreducible curve and (q_n) a singular sequence on Γ. There are a Puiseux parametrization γ of a branch of Γ at infinity, $\gamma(t) \to \infty$ as $t \to 0$, and $t_n \to 0$ with $\gamma(t_n)$ a subsequence of (q_n). Every coordinate and every component of $\mathcal{Z}$, $\mathcal{W}$ has a Puiseux expansion $c\,t^\alpha(1 + o(1))$ along γ (or vanishes identically), and the diagram of every singular sequence extracted from $(\gamma(t_n))$ is determined by these orders and leading coefficients: the circles and strokes of colour z are the components of $\mathcal{Z}$ of minimal order, and the relations $\prec$, $\preccurlyeq$, $\approx$, $\sim$ of* [1, §3] *become relations between orders and leading coefficients. In particular a four-product unbounded along (q_n) has negative order along γ.*

Proof. The closure of Γ in a projective compactification has finitely many points at infinity, each with finitely many analytic branches; a sequence tending to infinity has a subsequence on one branch, and the normalization of the branch gives the parameter t. Polynomial functions of the coordinates are meromorphic in t, which gives the expansions. If α is the least order among the components of $\mathcal{Z}$, then $Z = \max_p |\mathcal{Z}_p|$ has order α, and $\mathcal{Z}_p/Z \to 0$ exactly when $\operatorname{ord} \mathcal{Z}_p > \alpha$; this does not depend on the subsequence. The renormalization of [1, p. 547] multiplies all z-quantities by one positive factor $a(t) \approx |t|^{(\beta-\alpha)/2}$ and all w-quantities by a^{-1}, where β is the least order in $\mathcal{W}$. This preserves every relation within one colour and every product $z_{kl}w_{kl}$. Replacing a by the Puiseux series $\lambda(t) = (\mathcal{W}_q(t)/\mathcal{Z}_p(t))^{1/2}$ built from maximal components, with $|\lambda|/a$ bounded above and below, gives an analytic renormalization with the same relations; after reparametrizing $t = s^N$, ϵ is comparable to a power of s. A meromorphic function that is unbounded along $t_n \to 0$ has a pole. □

Definition 3.4. A *Diagram-r germ* is a Puiseux branch γ of a curve in A_m whose diagram, in the sense of Lemma 3.3, is Diagram r of [1, Fig. 11] after renumbering the bodies and possibly exchanging z and w. It has an *unbounded four-product* if some p_{ij} has negative order along γ.

Lemma 3.5. *On every irreducible curve $\Gamma \subset A_m$ some four-product is not constant.*

Proof. Index the squared distances and the four-products by the ten pairs. The pairs disjoint from ij are the sides of the triangle complementary to ij, so the exponent vector of p_{ij} in the variables r_{ab}^2 is the column ij of $E = I + P$, where P is the adjacency matrix of the Petersen graph (two pairs are adjacent if they are disjoint). The eigenvalues of E are 4, 2 (five times) and -1 (four times), so $\det E = 128$, and $N = 128E^{-1}$ is an integer matrix. As no r_{ab}^2 vanishes on A_m, $(r_{ab}^2)^{128} = \prod_{ij} p_{ij}^{N_{ab,ij}}$ there If every p_{ij} were constant on Γ, every r_{ab}^2 would take finitely many values on Γ and hence be constant, and Γ would be finite by Lemma 3.1. □

The lemma replaces [1, Prop. 9], whose proof inspects the diagrams of [1, Fig. 11].

Proposition 3.6 (Reduction)**.** *Let $m \in \mathbb{R}^5_{>0}$. If A_m has infinitely many positive points, there are $r \in \{1, \dots, 8\}$, a curve Γ as in Lemma 3.2, and a Diagram-r germ on Γ with an unbounded four-product.*

Proof. Let Γ be given by Lemma 3.2. By Lemma 3.5 some four-product p_{ij} is not constant on Γ, and by Fact AK1 it is dominating there. A sequence on Γ with $p_{ij} \to \infty$ is unbounded, and a singular sequence extracted from it approaches one of Diagrams 1–8 by Facts AK4 and AK5. Lemma 3.3 turns it into a germ. □

Remark 3.7. Germs at infinity never lie on the positive arc itself: on a positive arc $U = I = c$ bounds every r_{kl} above and below. So the signs of the boundary values of a germ say nothing about positivity; positivity can only enter through identities valid on all of Γ. Accordingly we exclude germs at three levels: all complex germs; germs on curves through a real point, where $I = U$ is a positive constant; and germs on curves with a positive arc.

4. The exclusion theorem

Theorem 4.1. *Let $m \in \mathbb{R}^5_{>0}$, $r \in \{1, \dots, 8\}$ and let γ be a Diagram-r germ with an unbounded four-product on an irreducible curve $\Gamma \subset A_m$.*

(a) *If* $r \in \{1, 2, 4, 6, 8\}$*, no such germ exists.*
(b) *If* $r \in \{5, 7\}$*, then* Γ *contains no real point.*
(c) *If* $r = 3$*, then* Γ *contains no arc of positive points.*

Proof of Theorem 1.2. If A_m had infinitely many positive points, Proposition 3.6 would give a germ contradicting Theorem 4.1: the curve of Lemma 3.2 contains a positive arc, hence real points. □

The proof of Theorem 4.1 occupies Sections 5–11. Table 1 summarizes it.

r	mass condition	level	method
1	(23) on the triangle	complex	triangle virial against a Morse–Bott estimate
2	K_j, L_j, P_j	complex	isolation of every boundary label at orders 1–3, with folds of corank one and two
3	(29) $m_1m_3 = m_2m_4$	positive	orders 3–6; identity $\delta_{15} + 2\delta_{13} = 0$
4	(23)	complex	w- and z-force balances at orders one and three; exact census
5	(23)	real	tropical census of the outer bodies
6	(32) $m_1 = m_2$, $m_3 = m_4$	complex	finite wing atlas
7	none needed	real	inertia identity forces $I \to 0$
8	(40)	complex	order-six and order-nine pairings; elimination on one curve of masses

TABLE 1. The eight exclusions. Numbers in parentheses refer to the equations of [1, §8].

Diagrams 1–4 and 6–8 use charts with coverage propositions and, where needed, mass conditions. Diagram 5 instead classifies the orders of the outer bodies and uses an exact linear-programming census to exclude unbounded four-products on curves through real configurations. Throughout, a statement about a diagram is meant for every renumbering of the bodies (with the masses permuted) and for both colourings; both operations are symmetries of (1).

5. Diagram 1

Diagram 1 is one of the two disconnected diagrams left by [1, §7.1, p. 565]: the z-stroke 12 with z-circles at 1 and 2, and the w-triangle with strokes $34, 35, 45$ and no circle; there are no other strokes or circles. We keep the numbering of [1, §8.1]: $P = \{1, 2\}$, $T = \{3, 4, 5\}$, the letter i denotes a body of P and k, l denote bodies of T. Put $M_P = m_1 + m_2$, $M_T = m_3 + m_4 + m_5$, $Z_P = M_P^{-1}(m_1z_1 + m_2z_2)$, $Z_T = M_T^{-1}\sum_k m_kz_k$, and define W_P, W_T likewise. On A_m we have $\sum_j m_jz_j = \sum_j m_jw_j = 0$ (proof of Lemma 3.1), so $M_PZ_P + M_TZ_T = 0 = M_PW_P + M_TW_T$. The coordinate equations of (1) read $E_j^z = E_j^w = 0$, where

$$E_j^z = z_j - \sum_{l\neq j} m_l\delta_{jl}^3(z_j - z_l), \qquad E_j^w = w_j - \sum_{l\neq j} m_l\delta_{jl}^3(w_j - w_l).$$

Along a germ, $a \prec b$, $a \preccurlyeq b$ and $a \approx b$ in the sense of [1, §3] mean $\operatorname{ord} a > \operatorname{ord} b$, $\operatorname{ord} a \geq \operatorname{ord} b$ and $\operatorname{ord} a = \operatorname{ord} b$ (Lemma 3.3).

We prove that Diagram 1 germs do not exist (Theorem 5.13). The pair torque excludes a triangle body at the w-centre of the triangle (Lemma 5.5); then every germ converges in a weighted chart (Proposition 5.6); its triangle tends to the zero-multiplier triangle, which forces relation (23) of [1, §8.1] (Proposition 5.10); and the potential of the triangle is too large for the Morse–Bott estimate at that triangle.

5.1. **The chart.** Let γ be a Diagram 1 germ at $m \in \mathbb{R}^5_{>0}$; renumbering the bodies and exchanging z and w if necessary (symmetries of (1)), its diagram is the one described above. The coordinate equations are homogeneous of degree one in z and in w separately, and the ties involve only the products $z_{jl}w_{jl}$; so $(z,w) \mapsto (cz, c^{-1}w)$ preserves them, and it does not change the diagram [1, p. 547]. By Rule 2e [1, p. 552] the three z-separations of the triangle have one order, and so do its three w-separations. Taking for c a power of the parameter t we assume that the w-separations of the triangle have order 0; the gauge $z_{12} = w_{12}$ is not used below. Define $\lambda, \mu, \rho, \sigma$ by

$$\operatorname{ord} z_{12} = -\lambda, \qquad \operatorname{ord} z_{kl} = \mu, \qquad \operatorname{ord} Z_P = \mu + \rho, \qquad \operatorname{ord} W_P = \sigma,$$

with $\rho = +\infty$ (resp. $\sigma = +\infty$) if Z_P (resp. W_P) vanishes identically, in which case the terms carrying $\hat{Z}$ (resp. $\hat{W}$) below are absent. Here σ is an exponent, not the conjugation of Section 2. After a finite ramification of t we write

$$\begin{aligned} z_i &= t^{\mu+\rho}\hat{Z} + t^{-\lambda}A_i, \qquad & w_i &= t^{\sigma}\hat{W} + t^{\lambda}B_i \qquad (i \in P), \\ z_k &= -\tfrac{M_P}{M_T}\, t^{\mu+\rho}\hat{Z} + t^{\mu}\zeta_k, \qquad & w_k &= -\tfrac{M_P}{M_T}\, t^{\sigma}\hat{W} + \eta_k \qquad (k \in T), \\ \delta_{12} &= D_{12}, \qquad & \delta_{ik} &= t^{\lambda/2}D_{ik}, \qquad \delta_{kl} = t^{-\mu/2}D_{kl}, \end{aligned} \tag{2}$$

where $(A_1, A_2) = (m_2, -m_1)\,\pi$, $(B_1, B_2) = (m_2, -m_1)\,\varpi$ and $\sum_k m_k\zeta_k = \sum_k m_k\eta_k = 0$. The chart coordinates are $\hat{Z}, \hat{W}, \pi, \varpi$, the mass-centred triangle offsets $\zeta = (\zeta_k)$, $\eta = (\eta_k)$, and the ten D_{jl}. The equations get the weights t^{λ} for E^z_i, $t^{-\lambda}$ for E^w_i, $t^{\mu/2}$ for E^z_k and $t^{3\mu/2}$ for E^w_k; the ten ties keep weight 1. The exponents of interest form the open cone

$$C = \{\lambda > 0,\ \mu > 0,\ \sigma - \lambda > 0,\ \rho + \tfrac{3}{2}\mu > 0,\ \rho + \lambda + \mu > 0\}. \tag{3}$$

For the triangle block put, with sums over T,

$$\phi_k = \sum_{l \neq k} m_l D^3_{kl}(\zeta_k - \zeta_l), \qquad \psi_k = \sum_{l \neq k} m_l D^3_{kl}(\eta_k - \eta_l), \qquad \chi_{kl} = D^2_{kl}(\zeta_l - \zeta_k)(\eta_l - \eta_k) - 1,$$

$U_T = \sum_{k<l} m_k m_l D_{kl}$ and $I_T = \sum_k m_k\zeta_k\eta_k$; note $\sum_k m_k\phi_k = \sum_k m_k\psi_k = 0$ identically. The crossed forces are $X^z_k = \sum_{i\in P} m_i\delta^3_{ik}(z_k - z_i)$ and $X^w_k = \sum_{i\in P} m_i\delta^3_{ik}(w_k - w_i)$.

Lemma 5.1 (computer-assisted)**.** *Substitute* (2) *into the ten coordinate equations and the ten ties and apply the weights. Each of the twenty results is a finite sum $\sum_e c_e\, t^e$, where the exponents e are linear forms in $(\lambda, \mu, \rho, \sigma)$, $e = 0$ occurs, and every c_e is a polynomial in the chart coordinates. Every exponent $e \neq 0$ that occurs is a combination of the five forms in* (3) *with nonnegative rational coefficients, at least one of them positive. No c_0 involves $\hat{Z}$ or $\hat{W}$. For $k \in T$,*

$$t^{\mu/2}E^z_k = t^{\mu/2}(z_k - X^z_k) - \phi_k, \qquad t^{3\mu/2}E^w_k = t^{3\mu/2}(w_k - X^w_k) - \psi_k,$$

and the exponents occurring in $t^{\mu/2}(z_k - X^z_k)$, respectively $t^{3\mu/2}(w_k - X^w_k)$, are

$$\tfrac{3}{2}\mu,\ \tfrac{1}{2}(\lambda+\mu),\ \tfrac{3}{2}(\lambda+\mu),\ \tfrac{3}{2}\mu+\rho,\ \tfrac{3}{2}(\lambda+\mu)+\rho,$$

respectively

$$\tfrac{3}{2}\mu,\ \tfrac{3}{2}(\lambda+\mu),\ \tfrac{5}{2}\lambda+\tfrac{3}{2}\mu,\ \sigma+\tfrac{3}{2}\mu,\ \tfrac{3}{2}(\lambda+\mu)+\sigma.$$

We carry out the substitution exactly, with the masses as indeterminates and the exponents as linear forms in $(\lambda, \mu, \rho, \sigma)$. The computation determines the exponents that occur, writes each $e \neq 0$ as a combination of the five forms with explicit nonnegative rational coefficients, and checks the displayed identities for the triangle equations by expansion (Appendix A).

A direct computation gives the coefficients c_0. They are $A_i(1 - M_P D^3_{12})$ and $B_i(1 - M_P D^3_{12})$ for the equations of the pair, $-\phi_k$ and $-\psi_k$ for those of the triangle, and $M^2_P D^2_{12}\pi\varpi - 1$, $-D^2_{ik}A_i\eta_k - 1$ and χ_{kl} for the ties of the pair, of the six crossed pairs and of the triangle. The *boundary fibre* $\mathcal{B}_1$ is their common zero set in the coordinates other than $\hat{Z}, \hat{W}$. On $\mathcal{B}_1$ we

have $D_{12}^3 = M_P^{-1}$ and $D_{ik}^2 A_i \eta_k = -1$, and the triangle coordinates $(\zeta, \eta, D_{34}, D_{35}, D_{45})$ solve the *zero-multiplier triangle system* $\phi = \psi = 0$, $\chi = 0$.

5.2. **Coverage.** The coverage proof uses Rules 1e and 2e and Estimates 1 and 2 of [1, §3] (Fact AK2) and [1, Prop. 1] (Fact AK3).

Lemma 5.2. *The following are identities of polynomials:*

$$m_1 E_1^w + m_2 E_2^w = m_1 w_1 + m_2 w_2 - \sum_{i\in P,\, k\in T} m_i m_k \delta_{ik}^3 (w_i - w_k),$$

$$\sum_{k\in T} m_k E_k^z = \sum_{k\in T} m_k z_k - \sum_{i\in P,\, k\in T} m_i m_k \delta_{ik}^3 (z_k - z_i),$$

$$T_P := \sum_{i\in P,\, k\in T} m_i m_k \delta_{ik}^3 (z_i w_k - w_i z_k) = -\sum_{i\in P} m_i (w_i E_i^z - z_i E_i^w).$$

In particular the torque T_P of the triangle on the pair vanishes on A_m.

Proof. In each sum the terms between two bodies of P, or between two bodies of T, cancel in pairs. □

Lemma 5.3. *Let γ be a Diagram 1 germ, normalized as above.*

(i) $\operatorname{ord}\delta_{12} = 0$ *and* $\operatorname{ord} w_{12} = \lambda > 0$*; the least order of the components of $\mathcal{Z}$ is* $\operatorname{ord} z_1 = \operatorname{ord} z_2 = \operatorname{ord} Z_{12} = -\lambda$.

(ii) $\mu > 0$, $\operatorname{ord}\delta_{kl} = \operatorname{ord} Z_{kl} = -\mu/2$, *and* $\lambda > \mu/2$.

(iii) $\operatorname{ord} z_{ik} = -\lambda$, $\operatorname{ord} w_{ik} < \lambda$ *and* $\operatorname{ord} w_i \geq \lambda$*; hence* $\operatorname{ord} W_T = \operatorname{ord} W_P \geq \lambda$.

(iv) *For $k \in T$ let o_k be the coefficient of t^0 in $w_k - W_T$. At most one o_k vanishes. If $o_k \neq 0$, then* $\operatorname{ord} w_{ik} = 0$ *and* $\operatorname{ord}\delta_{ik} = \lambda/2$. *If $o_k = 0$, then $f = \operatorname{ord}(w_k - W_T)$ satisfies $0 < f < \lambda$, and* $\operatorname{ord} w_{ik} = f$, $\operatorname{ord}\delta_{ik} = (\lambda - f)/2$.

Proof. (i) The isolated component $\{1, 2\}$ of the z-diagram has circled vertices, so it has a maximal z-stroke by Rule 1e [1, p. 546], which can only be 12. Estimate 2 [1, p. 548] gives $r_{12} \approx 1$, so $\operatorname{ord}\delta_{12} = 0$ and $\operatorname{ord} w_{12} = -\operatorname{ord} z_{12} = \lambda$. Estimate 1 [1, p. 548] with z and w exchanged gives $w_{12} \approx \epsilon^2$, and Estimate 2 with z and w exchanged gives $w_{34} \succcurlyeq \epsilon$ for the w-stroke 34. Hence $w_{12} \prec w_{34}$; a comparison within one colour is not affected by the normalizations, so $\lambda > \operatorname{ord} w_{34} = 0$. Finally $1, 2$ are z-circled and 12 is a z-stroke, and $\operatorname{ord} Z_{12} = \operatorname{ord} z_{12} + 3\operatorname{ord}\delta_{12} = -\lambda$.

(ii) Bodies $1, 2$ carry no w-stroke, so T is an isolated component of the w-diagram without w-circles. By Rule 1e it has no maximal w-stroke. Estimate 2 with z and w exchanged gives $r_{kl} \preccurlyeq 1$, with $r_{kl} \approx 1$ only for a maximal w-edge, so $r_{kl} \prec 1$; thus $\mu = \operatorname{ord}(z_{kl} w_{kl}) = 2\operatorname{ord} r_{kl} > 0$, $\operatorname{ord}\delta_{kl} = -\mu/2$ and $\operatorname{ord} Z_{kl} = \mu - \frac{3}{2}\mu = -\frac{1}{2}\mu$. Diagram 1 has no z-stroke inside T, so $\operatorname{ord} Z_{kl} > -\lambda$, that is, $\lambda > \mu/2$.

(iii) The bodies of T are not z-circled, so $\operatorname{ord} z_k > -\lambda = \operatorname{ord} z_i$ and $\operatorname{ord} z_{ik} = -\lambda$. Since $Z_{ik}^2 = (z_{ik} w_{ik}^3)^{-1}$, $\operatorname{ord} Z_{ik} = \frac{1}{2}(\lambda - 3\operatorname{ord} w_{ik})$, and the absence of a z-stroke ik gives $\operatorname{ord} Z_{ik} > -\lambda$, i.e. $\operatorname{ord} w_{ik} < \lambda$. By Estimate 1 ($z$ and w exchanged) $w_{2k} \succ \epsilon^2 \approx w_{12}$, because 12 is the only maximal z-edge, and $z_{2k} \approx z_{12}$. So [1, Prop. 1, p. 559] with z and w exchanged, applied to bodies $1, 2$ as in the proof of [1, Prop. 4, p. 564], gives $w_1, w_2 \preccurlyeq w_{12}$, i.e. $\operatorname{ord} w_i \geq \lambda$. This also follows from Lemma 5.2: every term of $M_P W_P = \sum m_i m_k \delta_{ik}^3 (w_i - w_k)$ has order $\frac{1}{2}(3\lambda - \operatorname{ord} w_{ik}) > \lambda$, and $w_i - W_P$ has order $\operatorname{ord} w_{12} = \lambda$.

(iv) Since $\sum_k m_k o_k = 0$, and the o_k do not all vanish because $\operatorname{ord} w_{kl} = 0$, at most one o_k vanishes. Now $w_k - w_i = (w_k - W_T) + (W_T - w_i)$ with $\operatorname{ord}(W_T - w_i) \geq \lambda > 0$ by (iii). If $o_k \neq 0$ this gives $\operatorname{ord} w_{ik} = 0$. If $o_k = 0$ then $f > 0$; $f \geq \lambda$ would give $\operatorname{ord} w_{ik} \geq \lambda$, against (iii), so $f < \lambda$ and $\operatorname{ord} w_{ik} = f$. In both cases $\operatorname{ord}\delta_{ik} = -\frac{1}{2}(\operatorname{ord} z_{ik} + \operatorname{ord} w_{ik})$. □

Lemma 5.4 (computer-assisted)**.** *Let $\lambda, \mu, f > 0$ with $f < \lambda$ and let ρ, σ be real. Modify* (2) *by* $(\eta_3, \eta_4, \eta_5) = (t^f \omega,\ m_5 \vartheta,\ -m_3 m_5^{-1} t^f \omega - m_4 \vartheta)$ *and* $\delta_{i3} = t^{(\lambda - f)/2} d_i$ $(i = 1, 2)$*, all other entries*

unchanged. Then

$$T_P = t^{(\lambda-f)/2}\Big(m_3 \sum_{i=1,2} m_i d_i\,(d_i^2 A_i\omega) + \sum_{g\in G} c_g\, t^g\Big),$$

where the c_g are polynomials in the chart coordinates and G consists of the thirteen exponents

$$\tfrac{1}{2}f,\ \tfrac{3}{2}f,\ \sigma-f,\ \sigma+\tfrac{1}{2}f,\ 2\lambda+\mu-f,\ 2\lambda+\mu+\tfrac{1}{2}f,\ \lambda+\mu+\rho,\ \lambda+\mu+\sigma-f,\ 2\lambda+\mu+\rho-f,$$
$$\lambda+\mu+\rho+\tfrac{1}{2}f,\ \lambda+\mu+\sigma+\tfrac{1}{2}f,\ 2\lambda+\mu+\rho+\tfrac{1}{2}f,\ \lambda+\mu+\rho+\tfrac{3}{2}f,$$

each a combination of $\lambda,\ \mu,\ \sigma-\lambda,\ \rho+\lambda+\mu,\ f,\ \lambda-f$ with nonnegative rational coefficients, at least one of them positive. Moreover, if $d_i^2A_i\omega=-1$ for $i=1,2$, then $(m_1d_1)^2-(m_2d_2)^2 = -(m_1^3+m_2^3)/(m_1A_1\omega)$.

We expand T_P exactly, with the masses as indeterminates; no relation among the masses is used, in particular not (23). Each exponent in G is written as a combination of the six forms with explicit nonnegative rational coefficients, and the last identity is checked by rational simplification (Appendix A).

Lemma 5.5. *Let $m\in\mathbb{R}^5_{>0}$. Along every Diagram 1 germ all three leading triangle w-offsets o_k are nonzero.*

Proof. Suppose not. By Lemma 5.3(iv), after renumbering T, $o_3=0\neq o_4,o_5$, $0<f<\lambda$, $\operatorname{ord}\delta_{i3}=\frac{1}{2}(\lambda-f)$ and $\operatorname{ord}\delta_{ik}=\frac{1}{2}\lambda$ for $k=4,5$. In the first two identities of Lemma 5.2 the crossed terms have orders $\frac{1}{2}(3\lambda-\operatorname{ord}w_{ik})$ and $\frac{1}{2}(\lambda-3\operatorname{ord}w_{ik})$, so $\sigma\geq\frac{1}{2}(3\lambda-f)$ and $\mu+\rho\geq\frac{1}{2}(\lambda-3f)$, that is, $\sigma-\lambda\geq\frac{1}{2}(\lambda-f)>0$ and $\rho+\lambda+\mu\geq\frac{3}{2}(\lambda-f)>0$. Hence γ is described by the chart of Lemma 5.4, with chart coordinates of order ≥ 0, of order 0 for d_i, A_i, ω (so these are nonzero at $t=0$), and every exponent in G is positive. As $\operatorname{ord}(Z_T-Z_P)=\mu+\rho>-\lambda$ and $\sigma>\lambda>f$, we have $z_3-z_i=-t^{-\lambda}A_i(1+o(1))$ and $w_3-w_i=t^f\omega(1+o(1))$, so the ties $\delta_{i3}^2(z_3-z_i)(w_3-w_i)=1$ give $d_i^2A_i\omega=-1$ at $t=0$. By Lemma 5.4, $t^{-(\lambda-f)/2}T_P$ tends along γ to $-m_3(m_1d_1+m_2d_2)$, evaluated at $t=0$, and this is nonzero because $(m_1d_1)^2-(m_2d_2)^2=-(m_1^3+m_2^3)/(m_1A_1\omega)\neq 0$. But $T_P=0$ along γ by Lemma 5.2. □

Proposition 5.6 (Coverage)**.** *Let $m\in\mathbb{R}^5_{>0}$ and let γ be a Diagram 1 germ, normalized as above.*

(i) $\lambda>\mu/2>0$.
(ii) *Every leading triangle w-offset o_k is nonzero; hence* $\operatorname{ord}w_{ik}=0$ *and* $\operatorname{ord}\delta_{ik}=\lambda/2$ *for $i\in P$, $k\in T$.*
(iii) $\sigma\geq\frac{3}{2}\lambda$ *and* $\mu+\rho\geq\frac{1}{2}\lambda$. *In particular $(\lambda,\mu,\rho,\sigma)\in C$, with $\sigma-\lambda\geq\frac{1}{2}\lambda$, $\rho+\frac{3}{2}\mu\geq\frac{1}{2}(\lambda+\mu)$ and $\rho+\lambda+\mu\geq\frac{3}{2}\lambda$.*
(iv) *Along γ every chart coordinate of* (2) *converges as $t\to 0$; the limits of π, ϖ, of the D_{jl}, of the η_k and of the differences $\zeta_k-\zeta_l$, $\eta_k-\eta_l$ $(k\neq l)$ are nonzero; and the limit point lies in the boundary fibre $\mathcal{B}_1$.*

Proof. (i) is Lemma 5.3(i),(ii), and (ii) follows from Lemmas 5.5 and 5.3(iv). (iii) On γ, Lemma 5.2 gives $M_PW_P=\sum m_im_k\delta_{ik}^3(w_i-w_k)$ and $M_TZ_T=\sum m_im_k\delta_{ik}^3(z_k-z_i)$. By (ii) and Lemma 5.3(iii) the terms of the first sum have order $\frac{3}{2}\lambda$ and those of the second $\frac{3}{2}\lambda-\lambda=\frac{1}{2}\lambda$; as $\operatorname{ord}Z_T=\operatorname{ord}Z_P=\mu+\rho$ the two bounds follow, and the three facet bounds are arithmetic. (iv) By the definitions of λ,μ,ρ,σ, by Lemma 5.3 and by (ii), every chart coordinate is a Puiseux series of order ≥ 0, of order exactly 0 for the listed ones; so all converge. By Lemma 5.1 and (iii), each weighted equation equals $c_0+\sum_{e>0}c_et^e$ along γ with bounded c_e; it vanishes identically, so c_0 vanishes at the limit point. □

Remark 5.7. (a) Lemma 5.3 and Proposition 5.6 give the clustering scheme $z:1\dots345\dots2$, $w:3\dots12\dots5\dots4$, which [1, §8.1, p. 572] derives from Rule 2e, Estimate 2 and [1, Prop. 4]. Lemma 5.5 is the torque argument of the proof of [1, Prop. 4], carried out in the chart with every competing term controlled. The proof of Theorem 5.13 does not use the clustering scheme

of [1, §8.1]. (b) If $o_3 = 0$, then $w_{34} : w_{45} : w_{53} = m_5 : -(m_4 + m_5) : m_4$ at leading order. Relation (21) of [1], which the triangle w-rows give at leading order, together with the ties and $z_{34} + z_{45} + z_{53} = 0$, then forces $m_3^2(m_4 + m_5)^4 = (m_4 m_5)^3$. On the sheet of (23) written, up to scale, as $(m_3, m_4, m_5) = (b^2, b^2(b+1)^2, (b+1)^2)$, this relation is $b^4(b-1)^2(b+1)^8(b^2+b+1)(b^4+b^3+4b^2+b+1) = 0$, and with body 4 or 5 in place of body 3 it has no positive root. So under (23) a vanishing o_k can occur only on the central-body slice $m_4 = m_5 = 4m_3$. We do not use this: Lemma 5.5 holds at every positive mass.

5.3. **The mass condition.**

Lemma 5.8 (computer-assisted)**.** *Let $m_3, m_4, m_5 > 0$ and let $(\zeta, \eta, D_{34}, D_{35}, D_{45})$, with $\sum_k m_k\zeta_k = \sum_k m_k\eta_k = 0$, solve the zero-multiplier triangle system. Put $\alpha_{kl} = \zeta_l - \zeta_k$ and $\beta_{kl} = \eta_l - \eta_k$ for $k < l$ in T. Then*

(a) *$\beta_{kl} = \Lambda^{-1}\alpha_{kl}$ for one $\Lambda \in \mathbb{C}^*$, and $(D_{kl}\alpha_{kl})^2 = \Lambda$ for all $k < l$;*
(b) *$(D_{34}\alpha_{34}, D_{35}\alpha_{35}, D_{45}\alpha_{45}) = s\,(-1, 1, -1)$ with $s^2 = \Lambda$; the other six sign patterns admit no solution;*
(c) *$m_5\alpha_{34}^2 = m_3\alpha_{45}^2 = m_4\alpha_{35}^2$ and $Q(m_3, m_4, m_5) = 0$, where*

$$Q = 2m_3m_4m_5(m_3 + m_4 + m_5) - m_3^2m_4^2 - m_4^2m_5^2 - m_5^2m_3^2.$$

The proof is an exact computation with m_3, m_4, m_5 as indeterminates. Explicit cofactor identities give (a), and in each sign pattern the relation $\alpha_{35} = \alpha_{34} + \alpha_{45}$ eliminates the separations and leaves a polynomial relation among the masses: for the six patterns excluded in (b) the polynomial is a square plus a polynomial with positive coefficients, and for the pattern $s\,(-1, 1, -1)$ it is $-Q$ (Appendix A). For positive masses $Q = 0$ is equivalent to (23): with $x_k = m_k^{-1/2}$, $Q = (m_3m_4m_5)^2(x_3 + x_4 + x_5)(-x_3 + x_4 + x_5)(x_3 - x_4 + x_5)(x_3 + x_4 - x_5)$, and the first factor is positive.

The torus $\mathbb{T} = \{(\theta_1, \theta_2, \theta_3) \in (\mathbb{C}^*)^3 : \theta_3^2 = \theta_1\theta_2\}$ acts by $(\zeta, \eta, D) \mapsto (\theta_1\zeta, \theta_2\eta, \theta_3^{-1}D)$. It multiplies ϕ by $\theta_1\theta_3^{-3}$, ψ by $\theta_2\theta_3^{-3}$ and I_T by $\theta_1\theta_2$, and it fixes χ.

Lemma 5.9 (computer-assisted)**.** *Let $b, c > 0$, $(m_3, m_4, m_5) = (b^2c^2,\, b^2(b+c)^2,\, c^2(b+c)^2)$ and $\kappa > 0$. Let y^* be the point with $\eta = \zeta$, $\sum_k m_k\zeta_k = 0$, $\zeta_4 - \zeta_3 = \kappa m_5^{-1/2}$, $\zeta_5 - \zeta_3 = -\kappa m_4^{-1/2}$, $D_{34} = m_5^{1/2}/\kappa$, $D_{35} = m_4^{1/2}/\kappa$ and $D_{45} = -m_3^{1/2}/\kappa$. In the coordinates $y = (\zeta_3, \zeta_4, \eta_3, \eta_4, D_{34}, D_{35}, D_{45})$ let $F = (\phi_3, \phi_4, \psi_3, \psi_4, \chi_{34}, \chi_{35}, \chi_{45})$. Then $F(y^*) = 0$; the Jacobian matrix $F'(y^*)$ has rank 5, and its kernel is spanned by the generators $(\zeta_3, \zeta_4, 0, 0, -\frac{1}{2}D)$ and $(0, 0, \eta_3, \eta_4, -\frac{1}{2}D)$ of the $\mathbb{T}$-action; and*

$$I_T(y^*) = \frac{\kappa^2}{m_3 + m_4 + m_5}\Big(\frac{m_3m_4}{m_5} + \frac{m_3m_5}{m_4} + \frac{m_4m_5}{m_3}\Big) > 0.$$

The computation is exact in the field $\mathbb{Q}(b, c, \kappa)$: it gives $F(y^*) = 0$, shows that $F'(y^*)$ annihilates the two generators, and factors a 5×5 minor of $F'(y^*)$ and the value $I_T(y^*)$ into factors with positive coefficients, so that both are positive for $b, c, \kappa > 0$ (Appendix A). One checks that every positive solution of $m_3^{-1/2} = m_4^{-1/2} + m_5^{-1/2}$ is of the stated form. By the equivariance above, the same rank and kernel statements hold at every point of $O = \mathbb{T} \cdot y^*$, and $I_T \neq 0$ on O.

Proposition 5.10 (First condition)**.** *Let $m \in \mathbb{R}^5_{>0}$ and let γ be a Diagram 1 germ. Then $Q(m_3, m_4, m_5) = 0$, that is, the triangle masses satisfy relation (23) of* [1, §8.1]*. After renumbering T so that $m_3^{-1/2} = m_4^{-1/2} + m_5^{-1/2}$, the limit y_0 of $y(t) = (\zeta_3, \zeta_4, \eta_3, \eta_4, D_{34}, D_{35}, D_{45})(t)$ lies on the orbit O of Lemma 5.9.*

Proof. By Proposition 5.6(iv) the limit lies in $\mathcal{B}_1$, so its triangle part solves the zero-multiplier triangle system. In the notation of Lemma 5.8, $\phi_4 = m_3\alpha_{34}D_{34}^3 - m_5\alpha_{45}D_{45}^3$ and $\phi_5 = m_4\alpha_{45}D_{45}^3 + m_3\alpha_{35}D_{35}^3$, and $m_3\phi_3 = -m_4\phi_4 - m_5\phi_5$; likewise for ψ. So $\phi = 0$ and $\psi = 0$ are relations (22) and (21) of [1, §8.1] in cross-multiplied form, and Lemma 5.8 gives $Q = 0$. After

the renumbering, Lemma 5.8(c) and $\alpha_{35} = \alpha_{34} + \alpha_{45}$ leave only $(\alpha_{34}, \alpha_{45}, \alpha_{35}) = \kappa(x_5, -x_3, -x_4)$ with $\kappa \in \mathbb{C}^*$, and Lemma 5.8(a),(b) then determine η and D from (κ, s). The torus $\mathbb{T}$ acts transitively on these pairs (κ, s), and y^* is the solution with $\kappa > 0$, $s = -1$. Hence the solutions form the single orbit O, and $y_0 \in O$. □

5.4. **Exclusion.**

Lemma 5.11 (computer-assisted)**.** *For arbitrary $\zeta, \eta \in \mathbb{C}^3$ and $D_{34}, D_{35}, D_{45} \in \mathbb{C}$,*

$$\sum_{k \in T} m_k \zeta_k \psi_k = U_T + \sum_{k<l} m_k m_l D_{kl} \chi_{kl},$$
$$2\, dU_T + \sum_{k \in T} m_k (\psi_k\, d\zeta_k + \phi_k\, d\eta_k) = \sum_{k<l} m_k m_l (D_{kl}\, d\chi_{kl} - 2\chi_{kl}\, dD_{kl}).$$

Both identities are checked by exact expansion, with the masses as indeterminates (Appendix A). On the tie manifold $\{\chi = 0\}$ they reduce to $U_T = \sum_k m_k \zeta_k \psi_k$ and $dU_T = -\frac{1}{2} \sum_k m_k (\psi_k\, d\zeta_k + \phi_k\, d\eta_k)$; both vanish on O.

Lemma 5.12. *Every $y_0 \in O$ has a neighbourhood $V \subset \mathbb{C}^7$ with a constant c_V such that $|U_T(y)| \le c_V\, |(\phi_3, \phi_4, \psi_3, \psi_4)(y)|^2$ for every $y \in V$ with $\chi(y) = 0$.*

Proof. By Lemma 5.9 and equivariance, five components F_5 of F have independent differentials at y_0. The orbit map $\theta \mapsto \theta y_0$ is an immersion of a neighbourhood of $1 \in \mathbb{T}$ onto a 2-dimensional piece of O inside the 2-dimensional manifold $F_5^{-1}(0)$ near y_0, so $F_5^{-1}(0) \cap V \subset O$ for small V, and $\operatorname{dist}(y, O) \le c\, |F_5(y)| \le c\, |(\phi_3, \phi_4, \psi_3, \psi_4)(y)|$ for $y \in V$ with $\chi(y) = 0$ (implicit function theorem). Near y_0 the manifold $\{\chi = 0\}$ is the graph of D over $(\zeta_3, \zeta_4, \eta_3, \eta_4)$, since $\partial \chi_{kl} / \partial D_{kl} = 2 D_{kl}^{-1} (\chi_{kl} + 1) \neq 0$. In these coordinates U_T and its differential vanish on O (Lemma 5.11), so Taylor's formula at a nearest point of O gives $|U_T(y)| \le c' \operatorname{dist}(y, O)^2$. □

Theorem 5.13. *Let $m \in \mathbb{R}^5_{>0}$. No curve in A_m has a Diagram 1 germ (Definition 3.4). In particular there is no Diagram 1 germ with an unbounded four-product, which is the case $r = 1$ of Theorem 4.1 (a).*

Proof. Let γ be a Diagram 1 germ, normalized as above and with T renumbered as in Proposition 5.10. Along γ the ties of the triangle give $\chi(y(t)) = 0$ exactly, and $y(t) \to y_0 \in O$.

Forcing. Along γ we have $E_k^z = E_k^w = 0$, so Lemma 5.1 gives $\phi_k = t^{\mu/2}(z_k - X_k^z)$ and $\psi_k = t^{3\mu/2}(w_k - X_k^w)$, with the exponents listed there and bounded coefficients. By Proposition 5.6(iii), $\frac{3}{2}\mu + \rho \ge \frac{1}{2}(\lambda + \mu)$. Hence $|\phi(y(t))| = O(|t|^r)$ with $r = \min(\frac{3}{2}\mu, \frac{1}{2}(\lambda + \mu)) > 0$, and $|\psi(y(t))| = O(|t|^{3\mu/2})$.

Morse–Bott. By Lemma 5.12, $|U_T(y(t))| = O(|t|^{2r})$.

Virial. By Lemma 5.11 and $\chi = 0$, $U_T(y(t)) = \sum_k m_k \zeta_k \psi_k = t^{3\mu/2} \sum_k m_k \zeta_k (w_k - X_k^w)$. Since $w_k = W_T + \eta_k$ and $\sum_k m_k \zeta_k = 0$, we get $\sum_k m_k \zeta_k w_k = I_T(y(t))$, while $\operatorname{ord} X_k^w \ge \frac{3}{2}\lambda$ by Proposition 5.6(ii). Hence $U_T(y(t)) = t^{3\mu/2}(I_T(y_0) + o(1))$ with $I_T(y_0) \neq 0$.

Contradiction. The last two steps give $\frac{3}{2}\mu \ge 2r = \min(3\mu, \lambda + \mu)$. Since $\mu > 0$, this means $\lambda + \mu \le \frac{3}{2}\mu$, i.e. $\lambda \le \frac{1}{2}\mu$, which contradicts Proposition 5.6(i). □

Remark 5.14. The window $\lambda > \mu/2$ is used only in the last step. In the chart family (2) the outer bodies are z-circled exactly when $\lambda \ge \mu/2$. The wall $\lambda = \mu/2$ is the Diagram 4 chart after a rotation; there the orders $\frac{3}{2}\mu$ and $\lambda + \mu$ coincide and the argument gives nothing, which is why Diagram 4 needs the analysis of Section 8. For $\lambda < \mu/2$ the diagram is Diagram 5. No bound on four-products is used, so Theorem 5.13 excludes Diagram 1 germs as first and as second limits.

6. Diagram 2

Diagram 2 of [1, Fig. 11] consists of the z-segment 12 and the w-triangle 345, and every vertex carries a circle of the colour of its edges [1, §8.2, p. 572]. Exchanging z and w and renumbering

the bodies are symmetries of (1), so we assume that a Diagram 2 germ has exactly this diagram. Throughout this section A, A' denote the pair bodies, $\{A, A'\} = \{1, 2\}$, and B, B' bodies of the triangle $\{3, 4, 5\}$; $M = m_1 + m_2$, $M_T = m_3 + m_4 + m_5$ and $m' = (m_3, m_4, m_5)$. The letter t is a chart coordinate; the parameter of a germ (called t in Lemma 3.3) is called s, and ord is the order in s. The letters x, y, u are those of [1, §8.2].

The proof has four steps. We blow up (1) along the diagram; every Diagram 2 germ converges to a point of the boundary fibre $t = 0$, a *label* (Proposition 6.4). A torque identity of the pair forces $\Phi = 0$ at the limit label, which puts the masses on the loci L_j, K_j, P_j of [1, §8.2] (Proposition 6.6). Where the selected twenty equations define a graph over t, the two remaining equations reduce to $tq(t) = 0$, with $q(0) = \Phi$ and $q'(0)$ given by a quadratic form of the triangle (Lemmas 6.7 and 6.10). At fold labels without such a graph we instead use one- or two-parameter local reductions. Finally every label with $\Phi = 0$ at positive masses is shown not to be a limit of germs (§6.5), in most regular cases because $q'(0) \neq 0$. No reality condition is used.

6.1. **The chart and its labels.** The chart has coordinates $(t, Z, W, D) \in \mathbb{C}^{21}$, $Z = (Z_1, \dots, Z_5)$, $W = (W_1, \dots, W_5)$, $D = (D_{kl})_{k<l}$, related to the coordinates of (1) by

$$\begin{gathered} z_A = t^{-1} Z_A, \quad w_A = t W_A, \quad z_B = t Z_B, \quad w_B = t^{-1} W_B, \\ \delta_{12} = D_{12}, \quad \delta_{BB'} = D_{BB'}, \quad \delta_{AB} = t D_{AB}. \end{gathered} \tag{4}$$

The single-index capitals Z_k, W_k are chart coordinates, unrelated to the two-index quantities Z_{kl}, W_{kl} used for diagrams. Multiplying the coordinate equations of (1) for z_A, w_A, z_B, w_B by t, t^{-1}, t^{-1}, t gives the polynomials

$$\begin{aligned} F_A^Z &= Z_A - m_{A'} D_{12}^3 (Z_A - Z_{A'}) - t^3 \textstyle\sum_B m_B D_{AB}^3 (Z_A - t^2 Z_B), \\ F_A^W &= W_A - m_{A'} D_{12}^3 (W_A - W_{A'}) + t \textstyle\sum_B m_B D_{AB}^3 (W_B - t^2 W_A), \\ F_B^Z &= Z_B - \textstyle\sum_{B' \neq B} m_{B'} D_{BB'}^3 (Z_B - Z_{B'}) + t \textstyle\sum_A m_A D_{AB}^3 (Z_A - t^2 Z_B), \\ F_B^W &= W_B - \textstyle\sum_{B' \neq B} m_{B'} D_{BB'}^3 (W_B - W_{B'}) - t^3 \textstyle\sum_A m_A D_{AB}^3 (W_B - t^2 W_A), \end{aligned} \tag{5}$$

and the distance equations become $T_{12} = D_{12}^2 (Z_2 - Z_1)(W_2 - W_1) - 1$, $T_{BB'} = D_{BB'}^2 (Z_{B'} - Z_B)(W_{B'} - W_B) - 1$ $(B < B')$ and $T_{AB} = D_{AB}^2 (t^2 Z_B - Z_A)(W_B - t^2 W_A) - 1$. The gauge of (1) is replaced by $G_1 = Z_2 - Z_1 - 1$ and $G_2 = W_5 - 1$. Let $\mathcal{X}_m \subset \mathbb{C}^{21}$ be the zero set of these 22 polynomials and $\mathcal{B}_m = \mathcal{X}_m \cap \{t = 0\}$; the points of $\mathcal{B}_m$ are the *labels*.

Definition 6.1 (Shape factors)**.** For $\varsigma \in \{\pm 1\}^3$ let $N_\varsigma(w_3, w_4, w_5)$ be the numerator of $w_3/(\varsigma_3(w_5 - w_4)^2) + w_4/(\varsigma_4(w_3 - w_5)^2) + w_5/(\varsigma_5(w_4 - w_3)^2)$. The quintics S_1, S_2, S_3 are N_ς for $\varsigma = (1, -1, -1), (1, -1, 1), (1, 1, -1)$, and $N_{(1,1,1)}$ is a constant multiple of $S_4 S_5$, where

$$\begin{gathered} S_4 = w_3^3 + w_4^3 + w_5^3 - w_3^2 w_4 - w_3 w_4^2 - w_3^2 w_5 - w_3 w_5^2 - w_4^2 w_5 - w_4 w_5^2 + 3 w_3 w_4 w_5, \\ S_5 = w_3^2 + w_4^2 + w_5^2 - w_3 w_4 - w_3 w_5 - w_4 w_5. \end{gathered}$$

The quartics S_6, S_7, S_8 are the factors of $\operatorname{Res}_p((w_4 - w_3)^3 - p^3 (w_5 - w_4)^3,\ (w_4 - w_3)^3 - (-1 - p)^3 (w_3 - w_5)^3)$ (which is $S_5^3 S_6 S_7 S_8$ up to a constant) whose coefficients of $w_3^3 w_4$ are -2, 1 and -5. These are the factors of the complex three-body central configuration locus of [1, Appendix, pp. 585–586].

Lemma 6.2 (computer-assisted)**.** *Let $m \in \mathbb{R}^5_{>0}$. A point $p = (0, Z, W, D)$ is a label if and only if*

(a) *$Z_1 = -m_2/M$, $Z_2 = m_1/M$, $D_{12}^3 = 1/M$, $W_1 = -m_2 D_{12}$ and $W_2 = m_1 D_{12}$;*
(b) *$F_B^Z = F_B^W = T_{BB'} = 0$ at $t = 0$ (system (1) for the bodies $3, 4, 5$, without its gauge) and $W_5 = 1$;*
(c) *$D_{AB}^2 Z_A W_B = -1$ for all A, B.*

At a label $\sum_B m_B Z_B = \sum_B m_B W_B = 0$, every $W_B \neq 0$, the W_B are pairwise distinct, and the triangle *$(Z_B, W_B, D_{BB'})$ is of one of two kinds.*

(i) Lagrange: $D_{BB'} = M_T^{-1/3}\omega^{a_{BB'}}$, $\omega = e^{2\pi i/3}$, $a_{BB'} \in \{0,1,2\}$ *(27 classes), and* $\kappa = (W_4 - W_3)/(W_5 - W_4)$ *is a root of* $\omega^{a_{45}}\kappa^2 + (\omega^{a_{34}} + \omega^{a_{45}} - \omega^{a_{35}})\kappa + \omega^{a_{34}}$. *The root is double exactly for the six classes with* a_{34}, a_{35}, a_{45} *distinct; these are the* fold *triangles.* (W_3, W_4, W_5) *is a zero of* S_5 *if* $a_{34} = a_{35} = a_{45}$ *or the* $a_{BB'}$ *are distinct, of* S_6 *if* $a_{34} = a_{35} \neq a_{45}$, *of* S_7 *if* $a_{34} = a_{45} \neq a_{35}$, *and of* S_8 *if* $a_{35} = a_{45} \neq a_{34}$.

(ii) Euler: $Z_B = c^{-2}W_B$ *for some* $c \in \mathbb{C}^*$, $D_{BB'} = e_{BB'}c/(W_{B'} - W_B)$ *for* $B < B'$ *with* $e_{BB'} = \pm 1$, *and* (W_3, W_4, W_5) *is a zero of* S_1, S_2, S_3 *or* S_4S_5 *according to the class of* (e_{34}, e_{35}, e_{45}) *modulo a global sign. An Euler triangle on* S_5 *is a Lagrange fold triangle.*

Proof. At $t = 0$ the crossed terms of (5) vanish and $T_{AB} = -D_{AB}^2 Z_A W_B - 1$; this gives (b), (c) and, with G_1, the system $Z_A = m_{A'}D_{12}^3(Z_A - Z_{A'})$, $W_A = m_{A'}D_{12}^3(W_A - W_{A'})$, $T_{12} = 0$, whose solutions are (a). Summing $m_B F_B^Z$ and $m_B F_B^W$ gives the centre-of-mass relations. If $W_B = 0$ then $T_{1B} = -1$, and if $W_B = W_{B'}$ then $T_{BB'} = -1$. If the vectors $(1)_B, (Z_B)_B, (W_B)_B$ are linearly independent, then $D_{BB'}^3 = 1/M_T$ for all three edges. Indeed, the linear map $X \mapsto (\sum_{B' \neq B} m_{B'} D_{BB'}^3 (X_B - X_{B'}))_B$ of $\mathbb{C}^3$ fixes (Z_B) and (W_B) by (b) and kills $(1)_B$, and so does $X \mapsto X - M_T^{-1}(\sum_B m_B X_B)(1)_B$, by the centre-of-mass relations; as the three vectors form a basis, the two maps agree, and their entries at (B, B'), $B \neq B'$, give $m_{B'}D_{BB'}^3 = m_{B'}/M_T$. Then the coordinate equations reduce to the centre-of-mass relations, the ties give $Z_{B'} - Z_B = M_T^{2/3}\omega^{a_{BB'}}/(W_{B'} - W_B)$, and $Z_5 - Z_3 = (Z_4 - Z_3) + (Z_5 - Z_4)$ is the quadratic for κ. Otherwise $Z = \alpha W + \beta$; the centre of mass gives $\beta = 0$, the ties give $\alpha \neq 0$, and with $\alpha = c^{-2}$ they give the form of $D_{BB'}$. For the double roots write $c_{BB'} = \omega^{a_{BB'}}$: if two of the $c_{BB'}$ agree, the discriminant is $b(b - 4a) \neq 0$ with $|a| = |b| = 1$ (a the repeated value, b the remaining one), while if they are distinct, $c_{34} + c_{45} = -c_{35}$ and $c_{34}c_{45} = c_{35}^2$, so the discriminant vanishes. The assignment of the shape factors is computer-assisted. □

The assignment is an exact computation over $\mathbb{Q}(\omega)$. For each of the 27 Lagrange classes we reduce the factor S_j of the class, as a polynomial in κ, modulo the quadratic for κ, and for a fold class we evaluate it at the double root; for each of the eight sign vectors (e_{34}, e_{35}, e_{45}) we identify the corresponding N_ς with one of $\pm S_1$, $\pm S_2$, $\pm S_3$, $\pm S_4S_5$; and for the Euler triangles on S_5 we check that the three numbers $D_{BB'}^3$ coincide modulo S_5 (Appendix A).

Definition 6.3 (Mass polynomials)**.** For positions W_3, W_4, W_5 put $f_B = m_B^2/W_B$, $\Lambda = f_3^2 + f_4^2 + f_5^2 - 2f_3f_4 - 2f_3f_5 - 2f_4f_5$ and $\Theta = 3(f_3^2 + f_4^2 + f_5^2) + 2f_3f_5 + 2f_4f_5 - 6f_3f_4$. Equations (25) and (28) of [1, §8.2] are

$$\Lambda = 0, \qquad \Lambda\, y^2 + 2(f_3 - f_4 + f_5)(f_3 - f_4 - f_5)\, y + \tfrac{1}{2}(\Theta - \Lambda) = 0. \tag{6}$$

For a rational expression H in W_3, W_4, W_5 whose denominator is a monomial, let $H^\flat$ be the polynomial in W_4 obtained by clearing the position denominators, substituting $W_5 = 1$ and $W_3 = -(m_4W_4 + m_5)/m_3$ (centre of mass), and clearing the resulting power of m_3. Define $L_j(m')$ and $P_j(y)$ by

$$\mathrm{Res}_{W_4}(S_j^\flat, \Lambda^\flat) = c_j\, L_j, \qquad \mathrm{Res}_{W_4}(S_j^\flat, (28)^\flat) = m_3^{2\deg S_j} P_j(y),$$

where c_j has no zero on $\mathbb{R}^3_{>0}$; for $j = 5$ the first resultant is $m_3^4 M_T^2 L_5 L_5'$. Write $P_j(y) = E_j(y^2) + yO_j(y^2)$ and $K_j = P_j(0) = E_j(0)$.

These are the factors $L_1, \dots, L_8, L_5'$ and the polynomials $P_1, \dots, P_8$ of [1, pp. 573–575], reconstructed from the appendix factors; the displayed coefficients and published symmetries of [1] are reproduced. In particular $2L_5 = (m_3 - m_4)^2 + (m_4 - m_5)^2 + (m_5 - m_3)^2$, and L_5' has degree 12 and 91 coefficients, all positive. Permutations of (m_3, m_4, m_5) permute $\{L_1, L_2, L_3\}$ and $\{L_6, L_7, L_8\}$ up to nonzero constants; the exchange of bodies 3 and 4 sends S_1 to $-S_2$, exchanges $P_1 \leftrightarrow P_2$ and $P_6 \leftrightarrow P_7$, and fixes P_3, P_4, P_5, P_8. These statements are proved by exact computation over $\mathbb{Q}$, with the masses and y as indeterminates: the resultants defining L_j and L_5' are computed and factored, those defining P_j are determined by evaluation on unisolvent

grids of points modulo three primes, using a priori bounds on their degrees and coefficients, and the decompositions $P_j = E_j + yO_j$, the leading coefficients and the symmetries are checked by expansion and substitution (Appendix A).

6.2. **Coverage.**

Proposition 6.4 (Coverage)**.** *Let $m \in \mathbb{R}^5_{>0}$ and let γ be a Diagram 2 germ on a curve in A_m. Along γ put $t = (z_{12}w_5)^{-1/2}$ (either branch) and $\lambda = (tz_{12})^{-1}$, and let (Z, W, D) be given by* (4) *applied to the rotated configuration $(\lambda z, \lambda^{-1}w, \delta)$. Then t, Z, W, D are Puiseux series in s with $\operatorname{ord} t > 0$ and nonnegative orders, $(t, Z, W, D) \in \mathcal{X}_m$ for $s \neq 0$, and the limit p as $s \to 0$ is a label at which Z_1, Z_2, W_3, W_4, W_5 and all D_{kl} are nonzero.*

Proof. The equations of (1) other than its gauge are invariant under the rotations $(z, w) \mapsto (\lambda z, \lambda^{-1}w)$, the chart rows are these equations times powers of t, and $G_1 = \lambda t z_{12} - 1 = 0$, $G_2 = \lambda^{-1}tw_5 - 1 = t^2 z_{12}w_5 - 1 = 0$; so the chart point lies in $\mathcal{X}_m$. Rotating γ beforehand changes neither t nor the chart point, so for the orders we may normalize as in Lemma 3.3 and [1, p. 547]: the maximal components of $\mathcal{Z}$ and $\mathcal{W}$ have one order, and we use units in which it is -2. By Fact AK2:

(1) The pair is an isolated component of the z-diagram with z-circles, so its z-stroke is maximal (Rule 1e): $\operatorname{ord} z_{12} = -2$; by Estimate 2, $r_{12} \approx 1$, so $\operatorname{ord} w_{12} = 2$ and $\operatorname{ord} \delta_{12} = 0$.
(2) The triangle is an isolated w-component with w-circles; it has a maximal w-stroke (Rule 1e), and by Rule 2e all three are maximal: $\operatorname{ord} w_{BB'} = -2$, $\operatorname{ord} z_{BB'} = 2$, $\operatorname{ord} \delta_{BB'} = 0$.
(3) The circles give $\operatorname{ord} z_A = \operatorname{ord} w_B = -2$. The bodies $3, 4, 5$ carry no z-stroke and $1, 2$ no w-stroke, hence no circle of that colour (Rule 1a): $\operatorname{ord} z_B > -2$, $\operatorname{ord} w_A > -2$.
(4) Hence $\operatorname{ord} z_{AB} = \operatorname{ord} w_{AB} = -2$ and $\operatorname{ord} \delta_{AB} = 2$.
(5) In $w_A = \sum_{l \neq A} m_l \delta_{Al}^3 (w_A - w_l)$ the pair term has order 2 and the crossed terms order 4, so $\operatorname{ord} w_A \geq 2$; likewise $\operatorname{ord} z_B \geq 2$.
(6) So $\operatorname{ord} t = 2$ and $\operatorname{ord} \lambda = 0$; $Z_A = \lambda t z_A$, $W_B = \lambda^{-1} t w_B$, D_{12}, $D_{BB'}$ and $D_{AB} = \delta_{AB}/t$ have order exactly 0, and $W_A = \lambda^{-1} w_A / t$, $Z_B = \lambda z_B / t$ have order ≥ 0.

Hence every chart coordinate converges, $t \to 0$, and the coordinates of order exactly 0 have nonzero limits. The limit lies in the closed set $\mathcal{X}_m$ and has $t = 0$. □

6.3. **First conditions.**

Lemma 6.5 (Syzygies)**.** *Identically in (t, Z, W, D),*

$$\sum_{k=1}^5 m_k (W_k F_k^Z - Z_k F_k^W) = 0, \qquad \sum_A m_A (W_A F_A^Z - Z_A F_A^W) = t\, Q,$$
$$Q = \sum_{A,B} m_A m_B D_{AB}^3 (t^4 W_A Z_B - Z_A W_B).$$

At $t = 0$ the only rows with a nonzero t-derivative are F_A^W and F_B^Z, with $\partial_t F_A^W = \sum_B m_B D_{AB}^3 W_B$ and $\partial_t F_B^Z = \sum_A m_A D_{AB}^3 Z_A$. At a label, $Q = \Phi := \sum_B m_B \varphi_B$ with $\varphi_B = m_1 D_{1B} + m_2 D_{2B}$, and $\partial_t F_B^Z = -\varphi_B / W_B$.

Proof. Since $W_A F_A^Z = w_A F_A^z$, $Z_A F_A^W = z_A F_A^w$, $W_B F_B^Z = w_B F_B^z$ and $Z_B F_B^W = z_B F_B^w$ for the coordinate equations F_k^z, F_k^w of (1), the first identity is the torque identity $\sum_k m_k (w_k F_k^z - z_k F_k^w) = 0$, which holds by antisymmetry. In the second sum the terms with D_{12} cancel by antisymmetry and the crossed terms give tQ. The derivatives are read off from (5), and Lemma 6.2(c) gives $D_{AB}^3 Z_A W_B = -D_{AB}$ and $D_{AB}^3 Z_A = -D_{AB}/W_B$. □

Crossed data. At a label fix v_B with $v_B^2 = 1/W_B$. By Lemma 6.2(a),(c), $D_{1B}^2 = M/(m_2 W_B)$ and $D_{2B}^2 = -M/(m_1 W_B)$, so $D_{1B} = s_B \sqrt{M/m_2}\, v_B$ and $D_{2B} = i s'_B \sqrt{M/m_1}\, v_B$ with $s_B, s'_B \in \{\pm 1\}$ (positive square roots); these are the 64 crossed-sign assignments. Put $\sigma_B = s_B s'_B$, $c_\pm = \sqrt{M}\,(m_1/\sqrt{m_2} \pm i\, m_2/\sqrt{m_1})$ and $x = i(m_2/m_1)^{3/2}$. Then $c_+ c_- = M(m_1^3 + m_2^3)/(m_1 m_2) > 0$, $\varphi_B = c_{\sigma_B} s_B v_B$, and $m_2 D_{2B}/(m_1 D_{1B}) = \sigma_B x$ with $x^2 = -m_2^3/m_1^3$. The label is *same-sign* if

$\sigma_3 = \sigma_4 = \sigma_5$ and *one-different* otherwise (two and six of the eight patterns); then the *odd body* is the B whose σ_B differs from the majority sign σ. In both cases put $\tilde{v}_B = \varphi_B/c_\sigma$, so that $\tilde{v}_B = s_B v_B$ if $\sigma_B = \sigma$ and $\tilde{v}_k = \rho\, s_k v_k$ for the odd body k, where $\rho = c_{-\sigma}/c_\sigma$ satisfies $|\rho| = 1$, $\rho \neq \pm 1$. Thus $\Phi = c_\sigma \sum_B m_B \tilde{v}_B$.

Proposition 6.6 (First conditions)**.** *Let $m \in \mathbb{R}^5_{>0}$ and let p be the limit label of a Diagram 2 germ (Proposition 6.4). Then $\Phi(p) = 0$, and:*

(a) *if p is same-sign, its triangle satisfies* (25)*, $\Lambda = 0$, and $L_j(m') = 0$ for every $j \neq 5$ with $S_j(W_3, W_4, W_5) = 0$ ($L_5 L_5' = 0$ if $j = 5$);*

(b) *if p is one-different with odd body* 5 *and majority sign σ, its triangle satisfies* (28) *with $y = \sigma(x + x^{-1})/2$, and $u := y^2 = -(T-1)^2/(4T)$, $T = (m_1/m_2)^3$. Hence $P_j(y) = 0$ for every j with $S_j(W_3, W_4, W_5) = 0$: if $m_1 = m_2$ then $K_j(m') = 0$, and if $m_1 \neq m_2$ then $u < 0$ and $E_j(u) = O_j(u) = 0$.*

Proof. Along the germ all rows vanish and $t \not\equiv 0$, so Lemma 6.5 gives $Q \equiv 0$ along it, and $\Phi(p) = Q(p) = 0$ by continuity. No Jacobian rank is used, so this holds at every label, including fold labels. In case (a), $\Phi = c_\sigma \sum_B m_B s_B v_B$ with $c_\sigma \neq 0$, and the product of $m_3 v_3 \pm m_4 v_4 \pm m_5 v_5$ over the four sign choices is Λ, since $(m_B v_B)^2 = f_B$. In case (b) put $r_B = m_1 m_B D_{1B}$, so $r_B^2 = (m_1^2 M/m_2) f_B$, and $x' = \sigma x$; then $\Phi = (r_3 + r_4)(1 + x') + r_5(1 - x')$, which is (26) of [1, §8.2], and the product over the four sign changes of r_3, r_4 is $4x'^2(m_1^2 M/m_2)^2$ times the left side of (28) with $y = (x' + 1/x')/2$. In both cases W_4 is a common root of $S_j^\flat$ and $\Lambda^\flat$, resp. $(28)^\flat$, so the resultants of Definition 6.3 vanish, and $c_j \neq 0$, $m_3 \neq 0$. Finally $y = iv$ with $v = \sigma((m_2/m_1)^{3/2} - (m_1/m_2)^{3/2})/2 \in \mathbb{R}$, which vanishes iff $m_1 = m_2$, and E_j, O_j have real coefficients; so $P_j(iv) = E_j(-v^2) + iv O_j(-v^2) = 0$ gives the last assertion. □

6.4. **The isolation mechanism.** Let F_{sel} denote the twenty rows other than F_2^Z and F_5^Z. A label p is *excluded* if no holomorphic curve germ $c\colon (\mathbb{C}, 0) \to (\mathcal{X}_m, p)$ has $t \circ c \not\equiv 0$; an isolated point of $\mathcal{X}_m$ is excluded.

Lemma 6.7 (Local analytic lemma)**.** *Let $m \in \mathbb{R}^5_{>0}$ and $p \in \mathcal{B}_m$.*

(a) *If p is excluded, it is not the limit label of a Diagram 2 germ.*

(b) *Suppose $dF_{\mathrm{sel}}(p)$ has rank* 20*, so that near p the set $\{F_{\mathrm{sel}} = 0\}$ is a smooth curve C. Then near p the set $\mathcal{X}_m$ consists of the points of C at which $tQ = 0$, and p is excluded if $Q|_C \not\equiv 0$. If moreover $\partial F_{\mathrm{sel}}/\partial(Z, W, D)(p)$ is invertible, then C is the graph of an analytic map $t \mapsto \phi(t)$, and p is an isolated point of $\mathcal{X}_m$ as soon as $q(t) = Q(t, \phi(t)) \not\equiv 0$, in particular if $q(0) = \Phi(p) \neq 0$ or $q'(0) \neq 0$.*

(c) *Suppose some twenty of the twenty-two polynomials have a differential of rank* 20 *at p, so that near p their common zeros form a smooth curve C'. If one of the other two polynomials does not vanish identically on C', then p is an isolated point of $\mathcal{X}_m$.*

Proof. (a) By Proposition 6.4, a Diagram 2 germ with limit p gives, after the substitution $s = \tau^N$ that clears the denominators of the Puiseux exponents, a holomorphic germ $\tau \mapsto (t, Z, W, D)$ in $\mathcal{X}_m$ at p with $t \not\equiv 0$. (b) On C all rows except F_2^Z, F_5^Z vanish and $W_5 = 1$, so Lemma 6.5 gives $m_2 W_2 F_2^Z = tQ = -m_5 F_5^Z$ on C; and $W_2(p) = m_1 D_{12}(p) \neq 0$. This describes $\mathcal{X}_m$ near p. A germ c in $\mathcal{X}_m$ at p lies in C, and if $t \circ c \not\equiv 0$ then $Q \circ c \equiv 0$; when $Q|_C \not\equiv 0$ its zeros on the curve C are isolated, so c is constant, which contradicts $t \circ c \not\equiv 0$. In the graph case $\mathcal{X}_m$ is, near p, the set of points $(t, \phi(t))$ with $tq(t) = 0$, which is $\{p\}$ when $q \not\equiv 0$. (c) Near p, $\mathcal{X}_m$ lies in C' and in the zero set of the other polynomial, whose restriction to the smooth curve C' is analytic and not identically zero, hence has an isolated zero at p. □

At a few labels the Jacobian of the twenty-two polynomials has rank 19. There we use the following formal obstruction, a Newton–Puiseux argument on the blow-up of a Lyapunov–Schmidt reduction.

Lemma 6.8 (Corank-two isolation)**.** *Let $m \in \mathbb{R}^5_{>0}$ and $p \in \mathcal{B}_m$. Let F_R be nineteen of the twenty-two chart polynomials, $F_{R'}$ the other three, and $y \neq t$ a chart coordinate such that the derivative of $(F_R, t, y)\colon \mathbb{C}^{21} \to \mathbb{C}^{21}$ at p is invertible. By the implicit function theorem, near p the set $\{F_R = 0\}$ is the image of an injective analytic map ϕ, defined near $0 \in \mathbb{C}^2$, with $\phi(0) = p$ and $(y,t) \circ \phi(a,t) = (y(p) + a, t)$, and near p*

$$\mathcal{X}_m = \phi(g^{-1}(0)), \qquad g = F_{R'} \circ \phi\colon (\mathbb{C}^2, 0) \to (\mathbb{C}^3, 0).$$

Let $g^{(j)}$ be the homogeneous part of degree j of g, and for $\xi \in \mathbb{C}^2$ let $\Pi_\xi = Dg^{(2)}(\xi)$, a 3×2 matrix. Suppose:

(a) *$g^{(1)} = 0$; equivalently, the Jacobian at p of all twenty-two polynomials has rank 19;*
(b) *the components of $g^{(2)}$ have only finitely many common zeros $[\xi_1], \dots, [\xi_r]$ in $\mathbb{P}^1$, and $\Pi_{\xi_j} \neq 0$ for every j;*
(c) *for every j there is $N \geq 3$ such that the vector $r_N(\xi_j)$ defined below exists and does not lie in $\Pi_{\xi_j}\mathbb{C}^2$.*

Then no nonconstant formal arc $x \in \mathbb{C}[[\tau]]^2$ with $x(0) = 0$ satisfies $g(x) = 0$. Hence p is an isolated point of $\mathcal{X}_m$; in particular p is excluded. The same holds if (a)–(c) *hold for $Kg(L\,\cdot)$, $K \in GL_3(\mathbb{C})$, $L \in GL_2(\mathbb{C})$, in place of g.*

The vectors $r_N(\xi)$ are defined for a common zero ξ of $g^{(2)}$ and a fixed $e \in \mathbb{C}^2 \setminus \mathbb{C}\xi$. As $\Pi_\xi \xi = 2g^{(2)}(\xi) = 0$ and $\Pi_\xi \neq 0$, the vector $\varpi = \Pi_\xi e$ spans the line $\Pi_\xi \mathbb{C}^2$. Put $\beta_1 = 0$. For $n \geq 1$, if β_n is defined, let $r_{n+2}(\xi)$ be the coefficient of λ^{n+2} in $g(\lambda\xi + \beta_n(\lambda)e)$; if moreover $r_{n+2}(\xi) = -b_{n+1}\varpi$ with $b_{n+1} \in \mathbb{C}$, put $\beta_{n+1}(\lambda) = \beta_n(\lambda) + b_{n+1}\lambda^{n+1}$. Thus

$$r_3(\xi) = g^{(3)}(\xi), \qquad r_4(\xi) = g^{(4)}(\xi) + b_2\, Dg^{(3)}(\xi)e + b_2^2\, g^{(2)}(e).$$

Condition (c) with $N = 3$ says that the 3×3 matrix $\big(\Pi_{\xi_j} \,|\, r_3(\xi_j)\big)$ has rank two; with $N = 4$ it says that $\big(\Pi_{\xi_j} \,|\, r_3(\xi_j)\big)$ has rank one and $\big(\Pi_{\xi_j} \,|\, r_4(\xi_j)\big)$ has rank two.

Proof. The points of $\{F_R = 0\}$ near p are the points $\phi(a,t)$, and $\phi(a,t) \in \mathcal{X}_m$ if and only if $g(a,t) = 0$. If p were not isolated in $\mathcal{X}_m$, the curve selection lemma [2, Thm. 2.5.5], applied to the real algebraic set $\mathcal{X}_m \subset \mathbb{C}^{21} = \mathbb{R}^{42}$, would give a nonconstant semialgebraic arc $\gamma\colon [0, \varepsilon) \to \mathcal{X}_m$ with $\gamma(0) = p$. Its coordinates are convergent Puiseux series, so $x = (y \circ \gamma - y(p), t \circ \gamma)$ is, after a substitution $s = \tau^{N'}$, a nonconstant arc in $\mathbb{C}[[\tau]]^2$ with $g(x) = 0$. So the first assertion implies the second. The zero set of $Kg(L\,\cdot)$ is $L^{-1}g^{-1}(0)$, which gives the last assertion.

Let x be a nonconstant formal arc with $x(0) = 0$ and $g(x) = 0$, and write $x = \tau^k\xi' + O(\tau^{k+1})$ with $\xi' \neq 0$. By (a), $g(x) = \tau^{2k}g^{(2)}(\xi') + O(\tau^{2k+1})$, so $\xi' \in \mathbb{C}^*\xi$ for one of the $\xi = \xi_j$ of (b). Write $x = u\xi + \tilde{\theta}e$; then $\operatorname{ord} u = k < \operatorname{ord}\tilde{\theta}$. Take $\lambda = u(\tau)$ as the new parameter; then τ is a power series in $\lambda^{1/k}$ with nonzero linear term, and $x = \lambda\xi + \theta e$ with $\theta \in \mathbb{C}[[\lambda^{1/k}]]$ and $\operatorname{ord}\theta > 1$ ($\theta = 0$ is allowed).

We show by induction on $n \geq 1$ that β_n is defined, that $g(\lambda\xi + \beta_n e) = O(\lambda^{n+2})$ and that $o_n = \operatorname{ord}(\theta - \beta_n) > n$. For $n = N - 2$ this contradicts (c). For $n = 1$ the claim holds by (a), $g^{(2)}(\xi) = 0$ and $\operatorname{ord}\theta > 1$. Taylor's formula in the direction e gives

$$0 = g(\lambda\xi + \theta e) = g(\lambda\xi + \beta_n e) + (\theta - \beta_n)(\lambda\varpi + O(\lambda^2)) + O(\lambda^{2o_n}),$$

because $Dg(\lambda\xi + \beta_n e)e = \lambda\Pi_\xi e + \beta_n \Pi_e e + O(\lambda^2) = \lambda\varpi + O(\lambda^2)$ by (a) and $\operatorname{ord}\beta_n \geq 2$. If $o_n < n + 1$, the term $(\theta - \beta_n)\lambda\varpi$ has order $o_n + 1 < n + 2$, every other term has larger order ($2o_n > o_n + 1$), and its leading coefficient is a nonzero multiple of $\varpi \neq 0$; this is impossible. Hence $o_n \geq n + 1$, and the coefficient of λ^{n+2} gives $r_{n+2}(\xi) + b\varpi = 0$, where b is the coefficient of λ^{n+1} in θ. So β_{n+1} is defined, $b_{n+1} = b$ since $\varpi \neq 0$, and $o_{n+1} > n + 1$. The same expansion with β_{n+1} in place of θ gives $g(\lambda\xi + \beta_{n+1}e) = (r_{n+2}(\xi) + b_{n+1}\varpi)\lambda^{n+2} + O(\lambda^{n+3}) = O(\lambda^{n+3})$. □

Remark 6.9. The ξ_j are the points where the strict transform of $g^{-1}(0)$ could meet the exceptional curve of the blow-up of $0 \in \mathbb{C}^2$, and the induction is the Newton–Puiseux algorithm in the blow-up chart at ξ_j. Arcs inside the boundary fibre $\{t = 0\}$ need no separate argument: they are tangent to the line $t = 0$, which is then one of the ξ_j, and the same induction excludes them.

The triangle block and the form Q_T. For a label p let $J_T(p)$ be the Jacobian at p of the nine rows $F_3^Z, F_3^W, F_4^Z, F_4^W, F_5^W, T_{34}, T_{35}, T_{45}, G_2$ with respect to the nine triangle coordinates $Z_B, W_B, D_{BB'}$. If $J_T(p)$ is invertible and $\xi \in \mathbb{C}^3$, let (ζ, η, ϑ) solve $J_T(p)(\zeta, \eta, \vartheta) = r(\xi)$, where $r(\xi)$ has the entries ξ_3/W_3 and ξ_4/W_4 in the rows F_3^Z, F_4^Z and 0 elsewhere, and put

$$Q_T(\xi) = -\tfrac{1}{2} \textstyle\sum_B m_B \, \xi_B \, \eta_B / W_B.$$

Q_T is a quadratic form determined by m' and the triangle of p.

Lemma 6.10 (Second-order coefficient)**.** *Let p be a label with $\Phi(p) = 0$ and $J_T(p)$ invertible. Then $\partial F_{\mathrm{sel}}/\partial(Z, W, D)(p)$ is invertible and, in the notation of Lemma 6.7 and of the crossed data,*

$$q(0) = 0, \qquad q'(0) = c_\sigma^2 \, Q_T(\tilde{v}).$$

Proof. At $t = 0$ the pair rows involve only Z_A, W_A, D_{12}, the triangle rows only the triangle coordinates, and T_{AB} involves no crossed coordinate other than D_{AB}, with $\partial T_{AB}/\partial D_{AB} = -2 D_{AB} Z_A W_B = 2/D_{AB}$; so the Jacobian is block lower triangular. The pair block (rows $F_1^Z, F_1^W, F_2^W, T_{12}, G_1$) is invertible for positive masses (computer-assisted). Differentiate $F_{\mathrm{sel}}(t, \phi(t)) = 0$ at $t = 0$, using Lemma 6.5. The pair rows are forced by $\partial_t F_A^W = -Z_A^{-1} \sum_B m_B D_{AB}$, and $\partial_t F_1^W - \partial_t F_2^W = M\Phi/(m_1 m_2) = 0$; for equal forcings the solution has $Z_A'(0) = D_{12}'(0) = 0$ (computer-assisted). The triangle rows give $J_T(p)(Z', W', D')_T(0) = r(\varphi)$, hence $W_B'(0) = c_\sigma \eta_B(\tilde{v})$. On C the ties $T_{AB} = 0$ give $Q = \sum_{A,B} m_A m_B D_{AB} + O(t^2)$ and $D_{AB}' = -\frac{1}{2} D_{AB}(Z_A'/Z_A + W_B'/W_B)$. Therefore $q'(0) = \sum_{A,B} m_A m_B D_{AB}'(0) = -\frac{1}{2} \sum_B m_B \varphi_B W_B'(0)/W_B = c_\sigma^2 Q_T(\tilde{v})$. □

The two computer-assisted steps are exact computations with the masses as indeterminates, at the pair part of Lemma 6.2(a) and modulo $M D_{12}^3 = 1$: the determinant of the pair block is a power of D_{12} times factors of fixed sign for positive masses, and for equal forcings the pair rows are solved with $Z_A'(0) = D_{12}'(0) = 0$ (Appendix A).

Definition 6.11. For a label p with $J_T(p)$ invertible put $\Psi_T(p) = Q_T(\tilde{v})$ if p is same-sign and $\Psi_1(p) = Q_T(\tilde{v})$ if p is one-different. The label is *isolated at second order* if $\Phi(p) = 0$, $J_T(p)$ is invertible and $Q_T(\tilde{v}) \neq 0$; by Lemmas 6.7 and 6.10 it is then an isolated point of $\mathcal{X}_m$, and the omitted rows F_2^Z, F_5^Z vanish to order exactly two along the curve C.

At a label with $\Phi(p) = 0$ the vector $\tilde{v}$ is determined by the triangle and the signs s_B (for an odd body k, $m_k \tilde{v}_k = -\sum_{B \neq k} m_B \tilde{v}_B$), so Ψ_T and Ψ_1 do not involve m_1, m_2. For a one-different label with odd body 5 the condition $\Phi = 0$ requires $|m_3 \tilde{v}_3 + m_4 \tilde{v}_4| = m_5 |v_5|$, i.e. the vanishing of the *sheet function* $G = |m_3 \tilde{v}_3 + m_4 \tilde{v}_4|^2 - m_5^2$ (recall $W_5 = 1$), which again depends on the triangle only; then $\tilde{v}_5 = \rho s_5 v_5$ fixes ρ up to sign, hence m_1/m_2 up to exchanging m_1 and m_2.

6.5. **The case analysis.** By Proposition 6.6 and Lemma 6.7(a) it remains to show that every label p with $\Phi(p) = 0$ at positive masses is excluded. Table 2 lists the cases; the computer-assisted statements follow. Ball arithmetic is IEEE arithmetic with proved rounding bounds (Appendix A).

Lemma 6.12 (computer-assisted)**.** (a) *The polynomials L_4, L_5', K_3, K_6, K_7 and K_8 have no zero in $\mathbb{R}^3_{>0}$. More precisely, at positive masses every S_4 triangle is real (the centre-of-mass section of S_4 is a cubic with positive discriminant) and violates* (25)*.* (b) *If $m_3 = m_4 = m_5$, no S_j with $j \neq 5$ has a root satisfying* (25)*.*

Labels	Mass locus	Method	Computation
$\Phi \neq 0$ (any)	all m	first order, $q(0) = \Phi$	exact identities
same-sign, S_4, or S_5 with $L_5' = 0$	$L_4 = 0$, $L_5' = 0$	no point in $\mathbb{R}^3_{>0}$	exact
same-sign, S_5	$L_5 = 0$: $m_3 = m_4 = m_5$	order three; corank-two blow-up	exact jets
same-sign, S_1–S_3, S_6–S_8	rays of L_1–L_3, L_6–L_8	second order, $\lvert\Psi_T\rvert > 0.3$	Bernstein subdivision, Krawczyk method; interval arithmetic
one-different, $m_1 = m_2$, S_3, S_6–S_8	$K_3, K_6, K_7, K_8 = 0$	no point in $\mathbb{R}^3_{>0}$	Bernstein coefficients
one-different, $m_1 = m_2$, S_1, S_2	$K_1 = 0$, $K_2 = 0$	second order	exact; interval cover
one-different, $m_1 = m_2$, S_4	$K_4 = 0$: four arcs	second order	exact
one-different, $m_1 = m_2$, S_5	$K_5 = 0$: $\mathbb{R}_{>0}(1, 1, \rho_*)$	orders one to four	exact; interval arithmetic
one-different, $m_1 \neq m_2$, not fold	$E_j(u) = O_j(u) = 0$	second order, $\Psi_1 \neq 0$: ball covers, exact lemmas	ball and interval arithmetic; exact
one-different, $m_1 \neq m_2$, fold	$E_5(u) = O_5(u) = 0$	corank one; $Q\vert_C$ has a simple zero	exact

TABLE 2. The Diagram 2 cases. A label with $\Phi = 0$ lies on S_j for the j given in the first column (Lemma 6.2); one-different labels are taken with odd body 5.

The proof is an exact computation over $\mathbb{Q}$. For L_4 we use a rational parametrization of S_4; on a real S_4 triangle at positive masses the positions W_B do not all have the same sign, and then Λ is a sum of nonnegative terms, one of them positive. L_5' has positive coefficients, and K_3, K_6, K_7, K_8 have nonnegative Bernstein coefficients on twelve projective charts, three for each of them, each split into the two sectors of a blow-up at a corner of the cone. For (b) we check that $S_j^\flat$ and $\Lambda^\flat$ are coprime at $m_3 = m_4 = m_5$ for $j \neq 5$ (Appendix A).

Lemma 6.13 (computer-assisted)**.** *Let $m \in \mathbb{R}^5_{>0}$ with $m_3 = m_4 = m_5$. Every label at m whose triangle lies on S_5 (an all-equal Lagrange class or a fold triangle) and which satisfies $\Phi = 0$ or is a fold label is an isolated point of $\mathcal{X}_m$. At each such m these are* 72 *labels of the all-equal classes and* 1152 *fold labels,* 1224 *in all.*

Masses. Common mass scaling is an exact isomorphism of the gauged chart, so we may take $m_3 = m_4 = m_5 = 1$; then $\delta_* := M_T^{-1/3} = 3^{-1/3}$. Every positive pair (m_1, m_2) is, for exactly one $(\nu, M) \in (0,1) \times \mathbb{R}_{>0}$,

$$m_1 = \frac{4M\nu^2}{(1+\nu^2)^2}, \qquad m_2 = \frac{M(1-\nu^2)^2}{(1+\nu^2)^2},$$

namely $M = m_1 + m_2$ and $\nu = 1/(\chi + \sqrt{1+\chi^2})$, where $\chi = \sqrt{m_2/m_1} = (1-\nu^2)/(2\nu)$.

The all-equal labels. The 72 labels of the all-equal classes have $\Phi = 0$, an invertible $J_T(p)$ and $q(0) = q'(0) = 0$, and along C we have $F_5^Z = \kappa_3 t^3 + O(t^4)$, so $q''(0) = -2\kappa_3$. For the nine labels of one sign class with $D_{BB'} = \omega^k \delta_*$, $(W_3, W_4, W_5) = (\omega, \omega^2, 1)$ and $D_{12} = \omega^j M^{-1/3}$,

$$\kappa_3 = \omega^j M^{5/3} R_1 (1 - i\chi) + 3\omega^k \delta_* M R_2 (1 - i\chi^5), \qquad R_1 = \frac{18\nu^2(1-\nu^2)}{(1+\nu^2)^3}, \quad R_2 = \frac{6\nu^2(1+\nu^2)}{(1-\nu^2)^3}.$$

These nine are the images of the case $j = k = 0$ under the automorphisms $\delta_* \mapsto \omega^k \delta_*$, $M^{1/3} \mapsto \omega^{-j} M^{1/3}$ of the coefficient field, which fix the masses and the polynomials; the other 63 labels are images of these nine under the exchange of bodies 3 and 4, conjugation and the reversal of t. Here $R_1, R_2 > 0$, and $(1 - i\chi)/(1 - i\chi^5)$ is a positive multiple of $1 + \chi^6 + i(\chi^5 - \chi)$, where

$|\chi^5-\chi|\le\chi+\chi^5\le 1+\chi^6$ because $(1+\chi^6)-(\chi+\chi^5)=(1-\chi)(1-\chi^5)\ge 0$. So the quotient of the two terms of κ_3 is ω^{j-k} times a number of argument at most $\pi/4$ in absolute value, hence is not -1. Therefore $\kappa_3\neq 0$, and each of these labels is isolated by Lemma 6.7(b).

Fold labels of rank 20. At 1080 fold labels the Jacobian of the twenty-two polynomials has rank 20. Twenty rows together with $h=W_4-W_4(p)$ have an invertible derivative: the rows of F_{sel} at 864 labels, and all rows other than F_2^Z and F_3^Z at the other 216. An omitted row has a nonzero coefficient of h^3 along the curve these rows define, so each of these labels is isolated by Lemma 6.7(c).

Fold labels of rank 19. The other 72 fold labels form one orbit under the maps listed below. Its seed $p_*=p_*(\nu,M)$ has the pair part of Lemma 6.2(a) with $D_{12}=M^{-1/3}$ and the fold triangle

$$(W_3,W_4,W_5)=(\omega,\omega^2,1),\qquad Z_B=-\delta_* W_B,\qquad (D_{34},D_{35},D_{45})=\delta_*(1,\omega,\omega^2).$$

Apply Lemma 6.8 with $F_{R'}=(F_2^W,F_4^W,F_5^W)$, F_R the other nineteen rows and $y=Z_3$. The kernel coordinates are then the chart parameter t and $a=Z_3-Z_3(p_*)$. At $t=0$ the derivative of (F_R,t,Z_3) is block lower triangular, as in the proof of Lemma 6.10. Its diagonal blocks are (1) for dt; the triangle block (rows $F_3^Z,F_3^W,F_4^Z,F_5^Z,T_{34},T_{35},T_{45},G_2,dZ_3$, columns $Z_B,W_B,D_{BB'}$), with determinant $18\delta_*$; the pair block (rows $F_1^Z,F_1^W,F_2^Z,T_{12},G_1$, columns Z_A,W_A,D_{12}), with determinant $-3M^{-1/3}$; and the diagonal crossed block $(2/D_{AB})$, with determinant $-512i\,\nu^3(1-\nu^2)^3/(1+\nu^2)^6$. So the invertibility hypothesis holds for all (ν,M). The differentials of the three omitted rows vanish on the two-dimensional kernel of $dF_R(p_*)$, so (a) holds. Write $g=(\psi_2,\psi_4,\psi_5)$ with $\psi_k=F_k^W\circ\phi$.

At $M=1$ the jet of g through degree four was computed exactly over $\mathbb{Q}(\nu)(\sqrt{3},i,\delta_*)$, and the nineteen rows of F_R vanish through degree four on the computed jet of ϕ. In the quadratic part, $\psi_2^{(2)}=0$, there is no a^2 term, and the 3×3 coefficient matrix has rank one. So $g^{(2)}=\psi_4^{(2)}\,n$ for a fixed vector $n=(0,1,n_5)$, where $\psi_4^{(2)}=t(\gamma_{11}a+\gamma_{02}t)$ and

$$\gamma_{11}=\frac{9i\delta_*(\nu-i)(\nu+i)^3\,\Upsilon(\nu)}{4\nu(\nu^2-1)(\nu^2+1)^2},\qquad \Upsilon(\nu)=\nu^4-2i\nu^3-6\nu^2+2i\nu+1=\tfrac14(3(\nu-i)^4+(\nu+i)^4).$$

For real ν, $|3(\nu-i)^4|=3(\nu^2+1)^2>|(\nu+i)^4|$, so $\Upsilon(\nu)\neq 0$ and $\gamma_{11}\neq 0$ for $0<\nu<1$. Hence the common zeros of $g^{(2)}$ are two lines: the *boundary line* $\xi_\partial=(1,0)$, tangent to $\{t=0\}$, and the *slanted line* $\xi_s=(-\gamma_{02},\gamma_{11})$. Moreover $\Pi_\xi=n\,d\psi_4^{(2)}(\xi)$, with $d\psi_4^{(2)}(\xi_\partial)=(0,\gamma_{11})$ and $d\psi_4^{(2)}(\xi_s)=\gamma_{11}(\gamma_{11},\gamma_{02})$, both nonzero. So (b) holds. On the slanted line take $N=3$: the matrix $(\Pi_{\xi_s}\,|\,r_3(\xi_s))$ has the 2×2 minor

$$\mu_3=-\frac{3^{10}(\nu-i)^4(\nu+i)^{16}\,\Upsilon(\nu)^5}{2^{12}\delta_*\,\nu^6(\nu^2-1)^6(\nu^2+1)^8}\neq 0.$$

On the boundary line take $e=(0,1)$, so that $\varpi=\gamma_{11}n$, and $N=4$. Identically in ν, $r_3(\xi_\partial)=g^{(3)}(1,0)=-b_2\varpi$, and the matrix $(\Pi_{\xi_\partial}\,|\,r_4(\xi_\partial))$ has the minor

$$\mu_4=-\frac{27i\delta_*(\nu-i)^3(\nu+i)^5\,\Upsilon(\nu)}{64\,\nu^3(\nu^2-1)^3}\neq 0.$$

So (c) holds. Every coefficient used is regular for $0<\nu<1$: the jets are polynomials in the coordinates of p_*, the masses and the entries of the inverses of the three diagonal blocks, and the only further division is by γ_{11}. Hence the identities above hold at every $\nu\in(0,1)$.

For general M, $p_*(\nu,M)$ is obtained from $p_*(\nu,1)$ by multiplying W_1,W_2 by $M^{2/3}$ and D_{12} by $M^{-1/3}$. The map $\tilde g(a,\tilde t)=(M^{-2/3}\psi_2,\psi_4,\psi_5)(a,\tilde t/M)$ was computed in exact Laurent arithmetic in $M^{1/3}$, and the selected rows vanish along both lines through the orders used. The quadratic part of $\tilde g$ does not depend on M, the compatibility $r_3(\xi_\partial)\in\mathbb{C}\varpi$ persists, and the two minors become $M^{-3}\mu_3(\nu)$ and $M^{-5/3}\mu_4(\nu)$. By Lemma 6.8, applied to $\tilde g$, $p_*(\nu,M)$ is an isolated point of $\mathcal{X}_m$ for every $0<\nu<1$ and $M>0$.

Transport. Two involutions preserve the twenty-two polynomials, since these have real coefficients and $m_3 = m_4$: the reversal $(t, D_{AB}) \mapsto (-t, -D_{AB})$, which is biholomorphic, and complex conjugation followed by the exchange of bodies 3 and 4, which is antiholomorphic. They commute, fix (ν, M), and carry p_* to the labels with the other three signs of the orbit. Multiplying t, Z_B, W_A, D_{12} and $D_{BB'}$ by ω^j multiplies each polynomial by a cube root of unity and fixes the gauges. A permutation of the bodies $3, 4, 5$, followed by the element of the chart torus that restores the gauges, is a local biholomorphism of $\mathcal{X}_m$ by the implicit function theorem, because the transversality determinant is $2(Z_2 - Z_1)W_5 = 2$. These $3 \cdot 6$ maps give eighteen labels for each of the four signs. The resulting 72 labels are exactly one orbit of the discrete label action and are disjoint from the 1080 labels above. All these maps preserve isolatedness.

The computations in the proof of Lemma 6.13, namely the coefficients κ_3, the ranks and the coefficients of h^3 at the 1080 fold labels, the determinants, jets and minors at p_*, and the orbit of p_*, are exact, over number fields containing ω, i and δ_* and over rational function fields over them in ν and $M^{1/3}$ (Appendix A).

Lemma 6.14 (computer-assisted)**.** (a) *The zero set of L_8 in $\mathbb{R}^3_{>0}$ is $\mathbb{R}_{>0}(p_*, q_*, 1) \cup \mathbb{R}_{>0}(q_*, p_*, 1)$ with $p_* = 0.67987758\ldots$ and $q_* = 0.82343785\ldots$.* (b) *The zero set of L_1 in $\mathbb{R}^3_{>0}$ is $\mathbb{R}_{>0}(a_*, b_*, 1) \cup \mathbb{R}_{>0}(a_*, 1, b_*)$ with $a_* = 1.02227400\ldots$ and $b_* = 0.90559635\ldots$; at $(a_*, b_*, 1)$ the zeros of S_1 that satisfy* (25) *form exactly one pair of complex conjugate non-real shapes.* (c) *Consequently the zero sets of $L_1L_2L_3$ and of $L_6L_7L_8$ in $\mathbb{R}^3_{>0}$ are the rays through the six permutations of $(a_*, b_*, 1)$ and of $(1, p_*, q_*)$, respectively.*

For (a) we combine exact Bernstein subdivision of the mass simplex with the Krawczyk method in rational interval arithmetic. For (b) we eliminate the shape by subresultants, locate the zeros by Bernstein subdivision and the signs of resultants, enclose them by the Krawczyk method, and show that the quadratic subresultant has negative discriminant there (Appendix A). Part (c) follows from (a), (b) and the permutation statement after Definition 6.3.

Lemma 6.15 (computer-assisted)**.** *Let $m \in \mathbb{R}^5_{>0}$ with m' on one of the rays of Lemma 6.14(c), and let p be a same-sign label at m with $\Phi(p) = 0$. Then $J_T(p)$ is invertible and $\Psi_T(p) \neq 0$, so p is isolated at second order.*

At each permutation of $(1, p_*, q_*)$ the 48 Lagrange triangles, and at each permutation of $(a_*, b_*, 1)$ the six Euler triangles on the conjugate shape pair, are enumerated; there only the ray factor itself vanishes among $L_1, \ldots, L_8, L_5'$, so by Proposition 6.6(a) no other triangle carries a same-sign label with $\Phi = 0$. For $36 + 36$ triangles a sign vector with $\sum_B m_B s_B v_B = 0$ exists, unique up to sign; J_T is invertible, and complex interval arithmetic gives $|\Psi_T| > 0.36$ (L_6–L_8) and $|\Psi_T| > 0.3$ (L_1–L_3). Common mass scaling covers the rays, and Ψ_T does not involve m_1, m_2. The shapes and the square and cube roots are enclosed by the Krawczyk method, and the solutions of the linear systems are enclosed rigorously (Appendix A).

Lemma 6.16 (computer-assisted)**.** *Let $m \in \mathbb{R}^5_{>0}$ with $m_1 = m_2$, let p be a one-different label at m with odd body 5 and $\Phi(p) = 0$, and let S_j carry its triangle, so that $K_j(m') = 0$. Then p is an isolated point of $\mathcal{X}_m$. More precisely:*

(a) *$j \notin \{3, 6, 7, 8\}$, by Lemma 6.12;*
(b) *if $j \in \{1, 2\}$: at every zero of K_1 in $\mathbb{R}^3_{>0}$ the S_1 shapes compatible with* (28) *at $y = 0$ reduce to one simple common root, which is real and lies on one of two explicit real branches; $J_T(p)$ is invertible and $q'(0) \neq 0$. The case $j = 2$ is the image of $j = 1$ under the exchange of bodies 3 and 4;*
(c) *if $j = 4$: m' lies on one of four arcs, $k \mapsto \mathrm{II}(k), \mathrm{III}(k)$ for $5^{-1/4} < k < 1$ and their images under $m_3 \leftrightarrow m_4$, where $\alpha = 4k(1+k^4)$, $\beta = (1-k^2)(1+6k^2+k^4)$, $\gamma = (1+k^2)(6k^2-1-k^4)$, $\mathrm{II} = ((\gamma^3-\beta^3)\alpha, (\alpha^3-\gamma^3)\beta, (\alpha^3-\beta^3)\gamma)$ and $\mathrm{III} = ((\gamma^3-\beta^3)\alpha, (\gamma^3+\alpha^3)\beta, (\alpha^3+\beta^3)\gamma)$; $J_T(p)$ is invertible and $q'(0) \neq 0$;*

(d) *if $j = 5$: $m' \in \mathbb{R}_{>0}(1, 1, \rho_*)$, where $\rho_* \in (1.5736409, 1.573641)$ is the unique positive root of $\rho^8 + 2\rho^7 + 3\rho^6 - 4\rho^5 - 5\rho^4 - 6\rho^3 - 3\rho^2 - 12\rho - 12$, and every label at m with a Lagrange triangle is isolated, for all 27 classes and all 64 crossed-sign assignments.*

In (b) and (c), 60 of the 64 crossed-sign assignments have $\Phi \neq 0$, and for the other four (one orbit of conjugation and $t \mapsto -t$) $q'(0) \neq 0$. In (d), the two-equal classes have $\Phi \neq 0$; for the all-equal classes 60 assignments have $\Phi \neq 0$ and 4 have $q'(0) \neq 0$; for the fold classes an augmented system in $h = W_4 - W_4(p)$ gives a nonzero coefficient of order three (60 assignments) or four (4). Exact grading makes all items uniform in $m_1 = m_2$. In (b) the common root is obtained from a linear subresultant over $\mathbb{Q}$, using the smoothness of the zero set of K_1 in $\mathbb{R}^3_{>0}$, which is proved with Bernstein coefficients; the two real branches are found from a rational parametrization of S_1 and Sturm sequences, and $q'(0) \neq 0$ follows from an exact reduction and an interval cover of the branch parameter. In (c) the computation is exact, on the rational parametrizations of the four arcs, and classifies the real S_4 triangles; by Lemma 6.12(a) every S_4 triangle at positive masses is real. In (d) the zero set of K_5 in $\mathbb{R}^3_{>0}$ is determined by exact resultants and a Sturm count, and the labels are classified in complex rational interval arithmetic, at one triangle in each orbit of conjugation and the exchange of bodies 3 and 4 (Appendix A). Renumbering the bodies as at the beginning of this section gives the same conclusions for all pair splits and odd-body placements.

Lemma 6.17 (computer-assisted)**.** *Let $m \in \mathbb{R}^5_{>0}$ with $m_1 \neq m_2$, and let p be a one-different label at m with odd body 5, $\Phi(p) = 0$, whose triangle is not a fold triangle. Then $J_T(p)$ is invertible and $\Psi_1(p) \neq 0$, so p is isolated at second order.*

We cover the simplex $m_3 + m_4 + m_5 = 1$ for S_1, S_3, S_4 (all Euler shapes, enclosed by Krawczyk or Pellet discs) and S_5, S_6, S_8 (all Lagrange shapes, up to conjugation and the cube-root action), for both relative signs $s_3 s_4$, by boxes on each of which G has a strict sign, or Ψ_1 is enclosed away from 0 in ball arithmetic, or, on boxes meeting $m_3 = 0$ or $m_4 = 0$ (where Ψ_1 vanishes), its transversal derivative is. The sets left out of the cover are treated separately: the vertex corners ($G \neq 0$); the ends $(m_3, m_4) = (0, \frac{1}{2})$ of S_1 and S_4 and $(\frac{1}{2}, 0)$ of S_4, where a majority body becomes massless (an exact edge identity and interval margins 0.787, 0.588, 0.551); the corner $(\frac{1}{2}, \frac{1}{2}, 0)$ of S_3 (an exact weighted blow-up of weight two in m_5 of the Euler shape parameter, after which the determinant and the numerator of the response are m_5^{13} and m_5^6 times factors bounded below in modulus by $6.96 \cdot 10^8$ and $1.63 \cdot 10^6$ in interval arithmetic); six real points of S_4 where the Euler label degenerates, and the corner $(\frac{1}{2}, \frac{1}{2}, 0)$ of S_4 (an exact closed form of Ψ_1 on the Euler shapes, in which the degenerating factor cancels); and S_5 by exact algebra (Appendix A). On S_5 let g be the W_4-component of the response of the triangle block (row F_5^Z omitted, $dW_5 = 0$) to a unit forcing of the row F_4^Z, and $X = \tilde{v}_4/W_4 - \tilde{v}_3/W_3$. Then $\Psi_1 = -\frac{1}{2} m_4 g X^2$; the numerator of g is a nonzero constant times $m_3(m_3^2 + m_3 m_4 + m_4^2)^2(P_0 + \omega P_1)$ with real polynomials P_0, P_1 whose only common zero in the simplex is $m_3 = m_4 = \frac{1}{3}$, and $X \neq 0$ except at the same point. There $m_3 = m_4 = m_5$, and the label with $G = 0$ has $\rho = -1$, which is impossible. S_2 and S_7 follow from S_1 and S_6 by exchanging bodies 3 and 4.

Lemma 6.18 (computer-assisted)**.** *Let $m \in \mathbb{R}^5_{>0}$ and let p be a one-different label at m with odd body 5, $\Phi(p) = 0$ and a fold triangle. Then $dF_{\mathrm{sel}}(p)$ has rank 20 and $Q|_C$ has a simple zero at p; in particular p is excluded.*

$J_T(p)$ has corank exactly one, with an explicit kernel vector and left null vector ℓ. Since the t-column of the triangle rows is $-\varphi_B/W_B$ on the Z-rows, ℓ applied to it is $c_\sigma X$ with X as in Lemma 6.17, and $X \neq 0$ at every label considered (the exceptional point $m_3 = m_4 = m_5$ carries no one-different label with $\Phi = 0$ at positive masses). So C exists, is parametrized by $h = W_4 - W_4(p)$, has $dt/dh = 0$ at p, and $dQ/dh = -\frac{1}{2} c_\sigma m_4 X \neq 0$ at p. The proof is an exact computation over $\mathbb{Q}(m_3, m_4)(\omega)$, with m_3, m_4 as indeterminates: it gives the kernel vector, the

left null vector, a minor of order eight of $J_T(p)$ that does not vanish on the simplex, and the zero set of X (Appendix A).

Theorem 6.19. *No Diagram 2 germ exists at positive masses: for $m \in \mathbb{R}^5_{>0}$ no curve in A_m has a Diagram 2 germ. In particular Theorem 4.1*(a) *holds for $r = 2$.*

Proof. Let γ be a Diagram 2 germ on a curve in A_m and p its limit label (Proposition 6.4). For each B, $\sigma_B x$ is the limit of $m_2\delta_{2B}/(m_1\delta_{1B})$ along γ, which does not depend on the chart; so the crossed pattern of p is determined by γ, and renumbering the bodies $3, 4, 5$ before applying Proposition 6.4 permutes it. We may therefore assume that p is same-sign or has odd body 5. By Lemma 6.7(a) it suffices to show that p is excluded. By Proposition 6.6, $\Phi(p) = 0$; let S_j carry the triangle of p (Lemma 6.2).

Same-sign. By Proposition 6.6(a), $L_j(m') = 0$, or $L_5L'_5 = 0$ if $j = 5$. Lemma 6.12 excludes $j = 4$ and $L'_5 = 0$. If $L_5 = 0$, then $m_3 = m_4 = m_5$, $j = 5$ by Lemma 6.12(b), and Lemma 6.13 applies. Otherwise $j \in \{1, 2, 3, 6, 7, 8\}$, m' lies on a ray of Lemma 6.14(c), and Lemma 6.15 applies.

One-different. If $m_1 = m_2$, Lemma 6.16 applies. If $m_1 \neq m_2$, Lemma 6.18 applies when the triangle is a fold triangle and Lemma 6.17 otherwise.

In every case p is excluded (by Definition 6.11 for the labels isolated at second order), which contradicts Lemma 6.7(a). □

Remark 6.20. (i) Theorem 6.19 is stronger than Theorem 4.1(a) requires: it excludes every Diagram 2 germ, with or without an unbounded four-product, and uses no reality. From [1] it uses only Fact AK2 (in Proposition 6.4), besides Lemma 3.3; the mass conditions (25)–(28) of [1, §8.2] are derived (Proposition 6.6), not assumed.

(ii) One can show that the layer $u = \infty$ of the P_j misses the positive cone, and that no non-degenerate triple of positions with positive centre-of-mass masses lies on two curves $S_i \neq S_j$ (Appendix A). We do not use this.

7. Diagram 3

Diagram 3 of [1, Fig. 11] is the diagram of Roberts' continuum at infinity [1, §7.2]. With the numbering of [1, §5.2 and §8.3], where the clustering scheme is $z : 12\ldots34$, $w : 14\ldots23$, it is the quadrilateral 1234 with z-edges 14, 23 and w-edges 12, 34; the diagonals and body 5 carry no stroke. Exchanging z and w gives the same diagram with the bodies 2 and 4 exchanged, so after renumbering a Diagram-3 germ has exactly this diagram. Unlike [1, proof of Thm. 6(iii)] we take no second singular limit.

7.1. **The crossed-cluster chart.** The equations of (1) other than the gauge are invariant under $(z, w, \delta) \mapsto (\mu z, \mu^{-1}w, \delta)$, $\mu \in \mathbb{C}^*$, and here we use the gauge $z_{13} = w_{13}$, as [1, p. 543] does for Roberts' continuum. The chart has the parameter t and the coordinates $A, B, x, y, \hat{z}_5, C, D, u, v, \hat{w}_5$ and D_{kl}:

$$
\begin{gathered}
z_1 = \tfrac{A}{t}, \quad z_2 = \tfrac{A}{t} + tx, \quad z_3 = \tfrac{B}{t}, \quad z_4 = \tfrac{B}{t} + ty, \quad z_5 = t^2\hat{z}_5, \\
w_1 = \tfrac{C}{t}, \quad w_2 = \tfrac{D}{t}, \quad w_3 = \tfrac{D}{t} + tv, \quad w_4 = \tfrac{C}{t} + tu, \quad w_5 = t^2\hat{w}_5, \\
\delta_{kl} = D_{kl} \ \ (kl = 12, 14, 23, 34), \qquad \delta_{kl} = t\, D_{kl} \ \ (kl = 13, 24, 15, 25, 35, 45).
\end{gathered} \tag{7}
$$

Multiplying the z- and w-equations of the bodies $1, \ldots, 4$ and the gauge equation by t, and the equations of body 5 by t^{-2}, gives 21 polynomial equations $F = F^0 + t^2F^2 + \cdots + t^6F^6 = 0$, with

no term linear in t. The *boundary equations* $F^0 = 0$ say that the following expressions vanish:

$$
\begin{gathered}
R_1^z = A - m_4 D_{14}^3 (A-B), \qquad R_4^z = B - m_1 D_{14}^3 (B-A), \\
R_2^z = A - m_3 D_{23}^3 (A-B), \qquad R_3^z = B - m_2 D_{23}^3 (B-A), \\
R_1^w = C - m_2 D_{12}^3 (C-D), \qquad R_2^w = D - m_1 D_{12}^3 (D-C), \\
R_3^w = D - m_4 D_{34}^3 (D-C), \qquad R_4^w = C - m_3 D_{34}^3 (C-D), \\
\hat{z}_5 + (m_1 D_{15}^3 + m_2 D_{25}^3) A + (m_3 D_{35}^3 + m_4 D_{45}^3) B, \\
\hat{w}_5 + (m_1 D_{15}^3 + m_4 D_{45}^3) C + (m_2 D_{25}^3 + m_3 D_{35}^3) D, \\
D_{12}^2 x (D-C) - 1, \quad D_{34}^2 y (C-D) - 1, \quad D_{14}^2 u (B-A) - 1, \\
D_{23}^2 v (B-A) - 1, \quad D_{13}^2 (B-A)(D-C) - 1, \quad D_{24}^2 (B-A)(C-D) - 1, \\
D_{15}^2 AC - 1, \quad D_{25}^2 AD - 1, \quad D_{35}^2 BD - 1, \quad D_{45}^2 BC - 1, \quad (B-A) - (D-C);
\end{gathered}
\tag{8}
$$

R_k^z, R_k^w come from the equations of body k. We work on the slice $B = A + 1$; for a germ we take $t = 1/z_{13}$ (Proposition 7.1).

7.2. **Coverage.** We write s for the parameter of a germ.

Proposition 7.1 (Coverage)**.** *Let γ be a Diagram-3 germ, renormalized as in Lemma 3.3.*

(i) *Bodies $1, \dots, 4$ are circled in both colours, body 5 in neither.*
(ii) *The sides are maximal edges: $r_{12}, r_{14}, r_{23}, r_{34} \approx 1$, $z_{14}, z_{23}, w_{12}, w_{34} \approx \epsilon^{-2}$ and $w_{14}, w_{23}, z_{12}, z_{34} \approx \epsilon^2$.*
(iii) *$z_{13}, w_{13}, z_{24}, w_{24}, z_{k5}, w_{k5} \approx \epsilon^{-2}$ for $k \le 4$, and $z_5, w_5 \preccurlyeq \epsilon^4$.*
(iv) *Renormalize instead by $\lambda = (w_{13}/z_{13})^{1/2}$, so that $z_{13} = w_{13}$, and put $t = 1/z_{13} \approx \epsilon^2$. Then $B - A = 1$ and every chart coordinate converges as $s \to 0$; the limit X_0 satisfies* (8)*, and A, B, C, D, x, y, u, v and all D_{kl} are nonzero at X_0.*

Proof. By Lemma 3.3 the relations of [1, §3] hold between orders along γ, and Fact AK2 applies. (i) Each side is the only stroke of its colour at its two ends, so Rule 1a [1, pp. 545–546] circles $1, \dots, 4$ in both colours; a circle needs a stroke of its colour, so 5 has none. (ii) The isolated component $\{1, 4\}$ of the z-diagram has z-circles, so its stroke 14 is maximal by Rule 1e [1, p. 546]; likewise 23, and 12, 34 in the w-diagram. The right equality case of Estimate 2 [1, p. 548] gives the orders. (iii) By (ii), $z_{13} = z_{14} - z_{34}$, $w_{13} = w_{12} + w_{23}$, $z_{24} = z_{23} + z_{34}$ and $w_{24} = w_{14} - w_{12}$ are $\approx \epsilon^{-2}$; by (i), $z_k \approx Z = \epsilon^{-2} \succ z_5$ for $k \le 4$, and likewise for w. The equation $z_5(1 - \sum_{k \le 4} m_k \delta_{k5}^3) = -\sum_{k \le 4} m_k \delta_{k5}^3 z_k$ of body 5, with $\delta_{k5} \approx \epsilon^2$, gives $z_5 \preccurlyeq \epsilon^4$; likewise for w_5. (iv) By (iii), λ differs from the factor of Lemma 3.3 by a Puiseux series of order 0, so (i)–(iii) persist. Now $B - A = t z_{13} = 1$, and $A = z_1/z_{13}$, $B = z_3/z_{13}$, $C = w_1/w_{13}$, $D = w_2/w_{13}$, $x = z_{12} z_{13}$, $y = z_{34} z_{13}$, $u = w_{14} w_{13}$, $v = w_{23} w_{13}$ and the D_{kl} ($= \delta_{kl}$ for the sides, $\delta_{kl} z_{13}$ otherwise) are ≈ 1, while $\hat{z}_5 = z_5 z_{13}^2$, $\hat{w}_5 = w_5 w_{13}^2 \preccurlyeq 1$. So all chart coordinates converge, and $F = 0$ passes to the limit. □

By Proposition 7.1(ii) and (iii), every Diagram-3 germ has an unbounded four-product: $p_{12} = r_{12}^2 r_{34}^2 r_{35}^2 r_{45}^2 \approx \epsilon^{-8}$.

7.3. **The first condition.**

Proposition 7.2 (First condition)**.** *At every point of $\{t = 0\}$ satisfying* (8)*, $m_1 m_3 = m_2 m_4$. Hence every Diagram-3 germ satisfies (29) of* [1, §8.3].

Proof. Since $m_1 R_1^z + m_4 R_4^z = m_1 A + m_4 B$ and $m_2 R_2^z + m_3 R_3^z = m_2 A + m_3 B$ identically,

$$
(B - A)(m_1 m_3 - m_2 m_4) = (m_1 + m_4)(m_2 R_2^z + m_3 R_3^z) - (m_2 + m_3)(m_1 R_1^z + m_4 R_4^z), \tag{9}
$$

and likewise $(D-C)(m_1 m_3 - m_2 m_4) = (m_1 + m_2)(m_3 R_3^w + m_4 R_4^w) - (m_3 + m_4)(m_1 R_1^w + m_2 R_2^w)$. With $T_{13} = D_{13}^2 (B-A)(D-C) - 1$ (the distance equation of 13) and $g = (B-A) - (D-C)$

(the gauge), $D_{13}^2(B-A)^2-1=T_{13}+D_{13}^2(B-A)g$. At a boundary point $T_{13}=g=0$, so $D_{13}^2(B-A)^2=1$, $B-A\neq 0$, and (9) gives $m_1m_3=m_2m_4$. The second statement follows from Proposition 7.1(iv). □

These identities are the chart form of [1, §5.2], where Rule 1c makes $m_1z_1+m_4z_4$ and $m_2z_2+m_3z_3$ non-maximal while $z_1\sim z_2$ and $z_3\sim z_4$.

7.4. **Labels and the analytic graph.** From now on $m\in\mathbb{R}^5_{>0}$ satisfies (29), $M=m_1+m_2+m_3+m_4$, and X denotes the 19 chart coordinates other than $B=A+1$. Let F_{z_k},F_{w_k} be the rescaled equations of body k, let Λ be the 19×21 matrix that forms from F the combinations $m_1F_{z_1}+m_4F_{z_4}$, $F_{z_4}-F_{z_1}$, $F_{z_3}-F_{z_2}$, $m_1F_{w_1}+m_2F_{w_2}$, $F_{w_2}-F_{w_1}$, $F_{w_3}-F_{w_4}$ and the other 13 equations, and put $\ell_zF=F_{z_1}-F_{z_2}-\frac{m_3}{m_2}F_{z_3}+\frac{m_4}{m_1}F_{z_4}$, $\ell_wF=F_{w_1}+\frac{m_2}{m_1}F_{w_2}-\frac{m_2}{m_1}F_{w_3}-F_{w_4}$.

Lemma 7.3 (computer-assisted)**.** (a) *On the slice,* (8) *has exactly* $3^4\cdot 2^6=5184$ *solutions, the* labels*: $D_{12},D_{34},D_{14},D_{23}$ are arbitrary cube roots of $\frac{1}{m_1+m_2}$, $\frac{1}{m_3+m_4}$, $\frac{1}{m_1+m_4}$, $\frac{1}{m_2+m_3}$; $D_{13},D_{24},D_{15},D_{25},D_{35},D_{45}$ are arbitrary square roots of 1, -1, $\frac{M}{m_3}$, $-\frac{M}{m_4}$, $\frac{M}{m_1}$, $-\frac{M}{m_2}$; $A=-\frac{m_4}{m_1+m_4}$, $C=-\frac{m_2}{m_1+m_2}$, $D=\frac{m_1}{m_1+m_2}$, $x=(m_1+m_2)D_{12}$, $y=-(m_3+m_4)D_{34}$, $u=(m_1+m_4)D_{14}$, $v=(m_2+m_3)D_{23}$, and $\hat z_5,\hat w_5$ follow from the equations of body 5. The mass m_5 does not enter.*

(b) *At every label*

$$\det\partial_X(\Lambda F^0)=-\frac{5184\,m_1^2m_2^3m_4^3\,(D_{12}D_{14}D_{23}D_{34})^4\,D_{13}D_{24}D_{15}D_{25}D_{35}D_{45}}{(m_1+m_2)(m_1+m_4)}\neq 0.$$

The proof is an exact computation over the field $\mathbb{Q}(m_1,m_2,m_4,m_5)$, with $m_3=m_2m_4/m_1$. On the slice, the 19 entries of ΛF^0 and the relations listed in (a) are combinations of each other with coefficients that have no poles at positive masses, and the determinant in (b) is computed by exact elimination (Appendix A). Identically in X, $\ell_zF^0=-B(m_1m_3-m_2m_4)/(m_1m_2)$ and $\ell_wF^0=-(D-C)D_{34}^3(m_1m_3-m_2m_4)/m_1$, so ℓ_z,ℓ_w annihilate $J_0=\partial_XF^0(X_0)$; as ΛJ_0 is invertible, J_0 has rank 19, ℓ_z,ℓ_w span its left kernel, and the 21 rows of Λ,ℓ_z,ℓ_w are independent.

Lemma 7.4. *Let X_0 be a label. There is a unique analytic $t\mapsto X(t)$ near 0 with $X(0)=X_0$ and $\Lambda F(t,X(t))=0$; put $G(t)=(\ell_zF(t,X(t)),\ell_wF(t,X(t)))=\sum_{k\geq 1}G_kt^k$. If the chart coordinates of a Diagram-3 germ converge to X_0, then along the germ they are $X(t)$, and $G\equiv 0$. Every analytic $Y(t)$ with $Y(0)=X_0$ and $F(t,Y(t))\equiv 0$ equals $X(t)$.*

Proof. The first statement is the analytic implicit function theorem (Lemma 7.3(b)). Along the germ, after a reparametrization $s\mapsto s^N$, the chart coordinates $Y(s)$ and $t(s)=1/z_{13}(s)$ are convergent power series (Lemma 3.3, Proposition 7.1) with $Y(0)=X_0$, $t(0)=0$, $t\not\equiv 0$. Since $\Lambda F(t(s),Y(s))=0$, uniqueness gives $Y(s)=X(t(s))$. So $F(t,X(t))$ vanishes on the image of $t(s)$, a neighbourhood of 0; hence $F(t,X(t))\equiv 0$ and $G\equiv 0$. The last statement is again uniqueness. □

So a label carries a Diagram-3 germ, ramified or not, only if $G\equiv 0$ there. As F has no term linear in t, $X(t)=X_0+X_2t^2+X_3t^3+\cdots$, and as $\ell J_0=0$ ($\ell=(\ell_z,\ell_w)$), G_k does not involve X_k: at X_0, $G_2=\ell F^2$, $G_3=\ell F^3$, $G_4=\ell(F^4+\partial F^2\cdot X_2+\frac12\partial^2F^0[X_2,X_2])$ and $G_5=\ell(F^5+\partial F^3\cdot X_2+\partial F^2\cdot X_3+\partial^2F^0[X_2,X_3])$.

7.5. **Orders three and four.**

Lemma 7.5 (computer-assisted)**.** *At every label $G_2=0$ and $G_3=(\frac{m_1+m_2}{m_1^2m_2}\Omega,\frac{m_1+m_4}{m_1^2m_4}\Omega)$, where $\Omega=\Omega_{\mathrm{re}}+\Omega_{\mathrm{im}}$,*

$$\Omega_{\mathrm{re}}=m_1m_2m_4D_{13}+m_5m_1^2D_{15}+m_5m_2m_4D_{35},$$
$$\Omega_{\mathrm{im}}=m_1m_2m_4D_{24}+m_5m_1m_2D_{25}+m_5m_1m_4D_{45}.$$

The computation is exact, over the field $\mathbb{Q}(m_1,m_2,m_4,m_5)$ with $m_3=m_2m_4/m_1$: ℓF^2 and ℓF^3 are evaluated at the labels and reduced modulo the cubic and quadratic equations satisfied by the D_{kl} (Appendix A).

Corollary 7.6. *If $G_3=0$ at some label, then $N_{\mathrm{re}}(m)=N_{\mathrm{im}}(m)=0$, where $Q=m_5^2M$ and*

$$N_{\mathrm{re}}=\left((m_1m_2m_4)^3-Q(m_1^6+m_2^3m_4^3)\right)^2-4Q^2m_1^6m_2^3m_4^3,$$
$$N_{\mathrm{im}}=\left((m_1m_2m_4)^3-Qm_1^3(m_2^3+m_4^3)\right)^2-4Q^2m_1^6m_2^3m_4^3.$$

As $N_{\mathrm{re}}-N_{\mathrm{im}}+QS=0$ with $S=(m_1^3-m_2^3)(m_1^3-m_4^3)\big(2(m_1m_2m_4)^3-Q(m_1^3+m_2^3)(m_1^3+m_4^3)\big)$, also $S(m)=0$.

Proof. At positive masses D_{13},D_{15},D_{35} are real and D_{24},D_{25},D_{45} purely imaginary (Lemma 7.3), so $\Omega_{\mathrm{re}}=\Omega_{\mathrm{im}}=0$. If $\alpha+\beta+\gamma=0$ then $(\alpha^2-\beta^2-\gamma^2)^2-4\beta^2\gamma^2=\prod_{\pm}(\alpha+\beta\pm\gamma)\prod_{\pm}(\alpha-\beta\pm\gamma)=0$. Inserting the squares $(m_1m_2m_4)^2$, $m_5^2m_1^4M/m_3$, $m_5^2m_2^2m_4^2M/m_1$ of the terms of Ω_{re}, using $m_3=m_2m_4/m_1$ and multiplying by $(m_1m_2m_4)^2$ gives $N_{\mathrm{re}}=0$; the terms of Ω_{im} are i times reals with squares $(m_1m_2m_4)^2$, $m_5^2m_1^2m_2^2M/m_4$, $m_5^2m_1^2m_4^2M/m_2$, which gives $N_{\mathrm{im}}=0$. Finally $Q>0$. □

Lemma 7.7 (computer-assisted)**.** *At every label $G_4=-\frac{\Pi}{m_1^2}\big(m_4(m_1+m_2),m_2(m_1+m_4)\big)$ with $\Pi=(m_1D_{12}-m_4D_{34})(m_1D_{14}-m_2D_{23})$.*

As for Lemma 7.5, the computation is exact and reduces modulo the equations satisfied by the D_{kl}; it solves for X_2 and evaluates the formula for G_4 above (Appendix A). At every label $(m_1D_{12})^3-(m_4D_{34})^3=m_1(m_1+m_4)(m_1-m_4)/(m_1+m_2)$ and $(m_1D_{14})^3-(m_2D_{23})^3=m_1(m_1+m_2)(m_1-m_2)/(m_1+m_4)$, so $\Pi=0$ forces $m_1=m_4$ or $m_1=m_2$; with (29), $m=(p,q,q,p,m_5)$ or (p,p,q,q,m_5).

Lemma 7.8 (computer-assisted)**.** *Let $m\in\mathbb{R}^5_{>0}$ satisfy (29), $N_{\mathrm{re}}(m)=N_{\mathrm{im}}(m)=0$ and $m_1=m_2$ (resp. $m_1=m_4$). Write $m_1=a^2$, $m_3=b^2$ (resp. $m_2=b^2$) with $a,b>0$, and $c=(2(a^2+b^2))^{1/2}$. Then*

$$m_5=\frac{a^3b^3}{c\,|a^3+b^3|}\qquad\text{or}\qquad m_5=\frac{a^3b^3}{c\,|a^3-b^3|}\quad(a\neq b).$$

Conversely N_{re}, N_{im} and S vanish on these four families, called $E^{\pm}_{12}$ (if $m_1=m_2$) and $E^{\pm}_{14}$ (if $m_1=m_4$). They meet only at $a=b$, i.e. at m proportional to $(1,1,1,1,\frac14)$, which lies on $E^+_{12}\cap E^+_{14}$.

On these mass patterns $N_{\mathrm{re}}=a^{12}\prod_{\pm}(a^6b^6-m_5^2c^2(a^3\pm b^3)^2)$, a polynomial identity in a,b,m_5. The converse is checked by exact substitution, with a,b,c given by a rational parametrization of the conic $c^2=2(a^2+b^2)$ (Appendix A).

Lemma 7.9 (computer-assisted)**.** *Let m lie on one of the four families.*

(a) *If $a\neq b$, exactly* 108 *labels have $G_3=G_4=0$: two of the eight sign choices for (D_{13},D_{15},D_{35}) give $\Omega_{\mathrm{re}}=0$, two of those for (D_{24},D_{25},D_{45}) give $\Omega_{\mathrm{im}}=0$, and $\Pi=0$ exactly when $D_{14}=D_{23}$ (on $E^{\pm}_{12}$) or $D_{12}=D_{34}$ (on $E^{\pm}_{14}$), for* 27 *of the* 81 *choices of cube roots.*

(b) *If $a=b$, exactly* 180 *labels have $G_3=G_4=0$: those with $D_{15}=D_{35}=-2D_{13}$, $D_{25}=D_{45}=-2D_{24}$, and $D_{12}=D_{34}$ or $D_{14}=D_{23}$ (*45 *choices of cube roots).*

An exact computation, with a,b,c as indeterminates, finds the sign choices for which the relevant linear forms in a^3,b^3 vanish identically (Appendix A). The count holds at every point: on $E^{\pm}_{12}$, with $D_{13}=\sigma$, $D_{15}=s_1c/b$, $D_{35}=s_2c/a$ and $\nu=\pm1$ the sign of $a^3\pm b^3$, Ω_{re} is $a^4b^2/|a^3\pm b^3|$ times $(\sigma\nu+s_1)a^3+(\pm\sigma\nu+s_2)b^3$, whose coefficients lie in $\{0,2\sigma\nu\}$ and $\{0,\pm2\sigma\nu\}$; so for $a,b>0$ (and $a\neq b$ on E^-_{12}) it vanishes only if both coefficients do. The other cases are alike.

7.6. **Orders five and six.**

Lemma 7.10 (computer-assisted)**.** *Let m lie on one of the four families with $a \neq b$. Then $G_5 \neq 0$ at each of the* 108 *labels of Lemma 7.9(a). At $a = b$, $G_5 = 0$ at all* 180 *labels of Lemma 7.9(b).*

We compute G_5 in exact arithmetic over the field $\mathbb{Q}(i, a, b, c)$, $c^2 = 2(a^2+b^2)$, modulo the cubic equations satisfied by D_{12}, D_{34}, D_{14}, and the product over the nine pairs below as an iterated resultant (Appendix A). On E_{12}^+ the labels have $D_{14} = D_{23}$, and the z-component of G_5 is a nonzero factor times $(a-b)\,L(D_{12}/D_{14}, D_{34}/D_{14})$, where $L(\xi_1, \xi_2) = a^2F_4\xi_1 - a^{-2}b^4F_4\xi_2 - b^2(a^2-b^2)F_2$, $F_4 = a^4 + a^3b + a^2b^2 + ab^3 + b^4$ and $F_2 = a^2 + ab + b^2$. Up to a nonzero monomial factor and sign, the product of L over the nine pairs with $\xi_1^3 = (a^2+b^2)/(2a^2)$, $\xi_2^3 = (a^2+b^2)/(2b^2)$ is $(a-b)^3(a+b)^3P_{66}(a, b)$, where P_{66} is homogeneous of degree 66 with 67 positive coefficients; so $G_5 \neq 0$ if $a \neq b$. On E_{12}^- the same computation gives $P_{66}(a, -b)$, and an exact Sturm count shows that $P_{66}(1, -\rho)$ has no root $\rho > 0$. Exchanging the bodies 2 and 4 together with z and w maps the chart at m onto the chart at $(m_1, m_4, m_3, m_2, m_5)$, $E_{12}^{\pm}$ onto $E_{14}^{\pm}$ and the 108 labels onto the 108 labels; the relabelling (13)(24) exchanges a and b and covers the minus families with $a < b$. At $a = b$ both components of G_5 vanish at the labels of E_{12}^+, hence, by the exchange, at all 180 labels.

Lemma 7.11 (computer-assisted)**.** *Let $m = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{8})$. Of the* 180 *labels of Lemma 7.9(b), the* 144 *labels with $D_{12} \neq D_{34}$ or $D_{14} \neq D_{23}$ have $G_6 \neq 0$.*

The proof is a computation of the jets modulo two primes (Appendix A). All data lie in $R = \mathbb{Z}[\frac{1}{2}, i, \omega]$, $\omega = e^{2\pi i/3}$; let $\mathfrak{p}$ be a prime of R above $p \in \{13, 37\}$, so $R/\mathfrak{p} = \mathbb{F}_p$ as $p \equiv 1 \pmod{12}$. If a germ passed through a label, then $F(t, X(t)) \equiv 0$ (Lemma 7.4); each X_k would be the unique solution of the order-k equations, given by any 19 rows with invertible minor, and all 21 residuals would vanish. We solve with 19 rows whose minor is a unit modulo $\mathfrak{p}$ and find a nonzero residual at order 6 for the 144 labels. Another choice of $\mathfrak{p}$ acts on the labels by $\mathrm{Gal}(\mathbb{Q}(i, \omega)/\mathbb{Q})$ ($\omega \mapsto \omega^2$, $i \mapsto -i$), which preserves the set of the 144 labels; both primes give the same set.

Lemma 7.12 (computer-assisted)**.** *Let $m = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{8})$, $\alpha, \beta \in \{1, \omega, \omega^2\}$ and $\varepsilon, \eta \in \{\pm 1\}$. Let (P, Q, Y) be the analytic solution near $\tau = 0$ of*

$$(1-P)(1+Q) = \alpha\tau^2, \qquad (1+P)(1-Q) = \beta\tau^2, \qquad 4PQY^2 + 1 = 0$$

with $(P, Q, Y)(0) = (1, 1, \eta i/2)$; the Jacobian there is $\pm 16i$. Then

$$z = \tfrac{1}{\tau}(1, P, -1, -P, 0), \qquad w = \tfrac{1}{\tau}(1, -Q, -1, Q, 0),$$

$$\delta_{12} = \delta_{34} = \alpha, \quad \delta_{14} = \delta_{23} = \beta, \quad \delta_{13} = \tfrac{\varepsilon\tau}{2}, \quad \delta_{15} = \delta_{35} = -\varepsilon\tau, \quad \delta_{24} = \tau Y, \quad \delta_{25} = \delta_{45} = -2\tau Y$$

solves (1) *with the gauge $z_{13} = w_{13}$. In the chart, with $t = -\tau/2$, it is an analytic solution of $F = 0$ through the label with $D_{12} = D_{34} = \alpha$, $D_{14} = D_{23} = \beta$, $D_{13} = -\varepsilon$ and $D_{24} = -\eta i$. These* 36 *labels are the labels of Lemma 7.9(b) with $D_{12} = D_{34}$ and $D_{14} = D_{23}$. Along these germs*

$$\delta_{15} + 2\delta_{13} = \delta_{35} + 2\delta_{13} = \delta_{25} + 2\delta_{24} = \delta_{45} + 2\delta_{24} = 0. \tag{10}$$

We substitute into (1) in exact arithmetic, with $\alpha, \beta, \varepsilon, \tau, P, Q, Y$ as indeterminates: the coordinate equations vanish identically once $\alpha^3 = \beta^3 = 1$, the distance equations reduce to the three displayed ones, and (10) holds without using them (Appendix A). The chart coordinates are analytic in t because P, Q, Y are analytic in τ^2 with $P(0) = Q(0) = 1$.

7.7. **The exclusion.**

Theorem 7.13. *Let $m \in \mathbb{R}^5_{>0}$, let $\Gamma \subset A_m$ be an irreducible curve, and let γ be a Diagram-3 germ on Γ, with the bodies numbered as in Diagram 3. Then $m_1 = m_2 = m_3 = m_4 = 4m_5$, and* (10) *holds identically on Γ. In particular Γ contains no positive point, and Theorem 4.1(c) holds.*

Proof. Exchanging z and w and renumbering if necessary, we may assume that γ has the diagram described at the beginning of this section; the conclusions are invariant under these operations. By Proposition 7.1 the chart coordinates of γ converge to a solution X_0 of (8) with $B - A = 1$. By Proposition 7.2, (29) holds, so X_0 is a label (Lemma 7.3), and $G \equiv 0$ at X_0 (Lemma 7.4). By Lemma 7.5 and Corollary 7.6, $N_{\mathrm{re}}(m) = N_{\mathrm{im}}(m) = 0$. By Lemma 7.7, $m_1 = m_4$ or $m_1 = m_2$, so m lies on one of the families of Lemma 7.8. By Lemmas 7.9(a) and 7.10, $a = b$, that is $m_1 = m_2 = m_3 = m_4 = 4m_5$.

The system (1) is covariant under $(m, z, w, \delta) \mapsto (\kappa^3 m, \kappa z, \kappa w, \kappa^{-1}\delta)$, $\kappa > 0$, which maps Diagram-3 germs to Diagram-3 germs and preserves (10). So we may take $m = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{8})$. By Lemmas 7.9(b) and 7.11, $D_{12} = D_{34}$ and $D_{14} = D_{23}$ at X_0. By Lemmas 7.12 and 7.4, the chart coordinates of γ are those of the twisted Roberts germ through X_0. The renormalizations do not change the δ_{kl}, so (10) holds along γ. Each expression in (10) is a regular function on the irreducible curve Γ that vanishes at the infinitely many points of γ, so it vanishes on Γ. At a positive point $\delta_{15} + 2\delta_{13} > 0$. □

So every Diagram-3 germ has the masses $(4, 4, 4, 4, 1)$ up to scale and is, up to the symmetries used above, one of the 36 twisted Roberts germs, which do exist. The next remark shows that the theorem cannot be weakened to curves through a real point.

Remark 7.14 (The modified Roberts continuum). Let $m = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{8})$, proportional to $(4, 4, 4, 4, 1)$ [1, Rem. 8, p. 583]. The curve

$$z = (a, b, -a, -b, 0), \quad w = (a, -b, -a, b, 0), \quad b^2 = a^2 - 1,$$

$$\delta_{12} = \delta_{23} = \delta_{34} = \delta_{14} = 1, \quad \delta_{13} = \tfrac{1}{2a}, \quad \delta_{24} = \tfrac{i}{2b}, \quad \delta_{15} = \delta_{35} = -\tfrac{1}{a}, \quad \delta_{25} = \delta_{45} = -\tfrac{i}{b}$$

is the modified Roberts continuum [1, p. 543]. It solves (1) with the gauge $z_{13} = w_{13}$, $U = I = 1$ on it, and (10) holds, so it has no positive point. Since $w_{12}/z_{12} = (a + b)^2$ on it, the rotation by $\mu = a + b$ gives an irreducible curve $\Gamma_R \subset A_m$. For $0 < a < 1$ and $b \in i\mathbb{R}$, $|\mu| = 1$ and the points are real, with $\delta_{13} > 0 > \delta_{15} = \delta_{35}$. As $a \to \infty$ with $b \sim a$, the curve is the twisted Roberts germ of Lemma 7.12 with $(\alpha, \beta, \varepsilon, \eta) = (1, 1, 1, 1)$, $\tau = 1/a$ and $P = Q = b/a$. Its strokes are z_{14}, z_{23}, w_{12}, w_{34}, so it is a Diagram-3 germ (Roberts' continuum at infinity [1, §7.2]); the sides keep length 1, $r_{13}, r_{24}, r_{k5} \to \infty$, $p_{12} = a^2(1 - a^2)$ is unbounded, and (29) holds. So Γ_R passes through real points, carries the positive constant $U = I = 1$, and has a Diagram-3 germ with an unbounded four-product. An exclusion at the level of curves through a real point is therefore false for Diagram 3. Theorem 7.13 uses the positivity of points only through the identity $\delta_{15} + 2\delta_{13} = 0$ on all of Γ, as Remark 3.7 requires.

8. Diagrams 4 and 5: the shared triangle

Diagrams 4 and 5 contain the isolated uncircled zw-triangle 345.

Proposition 8.1 (Relation (23))**.** *Let $m \in \mathbb{R}^5_{>0}$ and let γ be a Diagram 4 or Diagram 5 germ, renormalized as in Lemma 3.3 so that $\mathcal{Z}$ and $\mathcal{W}$ have the same least order $-2\nu < 0$. Let z_c and w_c be the centres of mass of $T = \{3, 4, 5\}$. Then the limits*

$$\zeta_k = \lim_{t\to 0} t^{-\nu}(z_k - z_c), \qquad \eta_k = \lim_{t\to 0} t^{-\nu}(w_k - w_c), \qquad D_{kl} = \lim_{t\to 0} t^{\nu}\delta_{kl} \qquad (k, l \in T)$$

exist and solve the zero-multiplier triangle system of Section 5, and $\sum_k m_k\zeta_k = \sum_k m_k\eta_k = 0$. Hence (ζ, η, D) is as in Lemma 5.8, and $Q(m_3, m_4, m_5) = 0$, that is, the masses satisfy (23).

Proof. The sides of T are zw-edges, so by Estimate 1 and the left equality case of Estimate 2 (Fact AK2, read through Lemma 3.3) $\operatorname{ord} z_{kl} = \operatorname{ord} w_{kl} = \nu$ and $\operatorname{ord}\delta_{kl} = -\nu$ for $k, l \in T$. As $z_k - z_c$ and $w_k - w_c$ are combinations of these differences, the limits exist, and the sums $\sum_k m_k\zeta_k$, $\sum_k m_k\eta_k$ vanish by the choice of the centres. Let $k \in T$. In the z-row $z_k = \sum_{l\neq k} m_l\delta^3_{kl}(z_k - z_l)$ the terms with $l \in T$ have order -2ν. The term z_k has larger order, because k is not circled, and so have the terms with $l \notin T$, because T is an isolated component of the diagram and no

z-stroke leaves it. Multiplying by $t^{2\nu}$ and letting $t \to 0$ gives $\phi_k = 0$; likewise $\psi_k = 0$, and the ties of the pairs in T give $\chi_{kl} = 0$. Lemma 5.8 now applies, and for positive masses $Q = 0$ is equivalent to (23). □

8.1. **Diagram 4.** Diagram 4 consists of the isolated zw-triangle 345, whose vertices are not circled, and the isolated z-stroke 12 with z-circles at 1 and 2; there are no other strokes or circles [1, §7.4, p. 567, and Fig. 11]. For this diagram [1, §8.4] record only relation (23). We use the notation of Section 5: $P = \{1,2\}$, $T = \{3,4,5\}$, $i \in P$, $k,l \in T$, M_P, M_T, Z_P, Z_T, W_P, W_T, the equations E_j^z, E_j^w, the triangle functions $\phi_k, \psi_k, \chi_{kl}$, the torque T_P of Lemma 5.2 and the orbit O of Lemma 5.9. Renumbering inside P or inside T does not change the diagram.

Theorem 8.2. *Let $m \in \mathbb{R}^5_{>0}$. No curve in A_m carries a Diagram 4 germ.*

This is part (a) of Theorem 4.1 for $r = 4$, without any assumption on the four-products. A torque argument excludes a triangle body at the w-centre of the triangle (Lemma 8.5); then every germ converges in a ramified chart to a point of a smooth boundary fibre $\mathcal{B}_4$ (Proposition 8.6 and Lemma 8.7). A Lyapunov–Schmidt lemma (Lemma 8.8) turns the existence of a germ through a boundary point into the vanishing of successive obstructions. The first says that the triangle exerts equal leading w-forces on the two outer bodies, the second vanishes identically, and the third says that it exerts equal leading z-forces (Lemma 8.9). An exact census shows that the two balances never hold together (Lemma 8.10).

8.1.1. *The chart.* The equations of (1) other than the gauge are invariant under $(z,w,\delta) \mapsto (cz, c^{-1}w, \delta)$, $c \in \mathbb{C}^*$; in this subsection we use the gauge $z_{34} = w_{34}$ instead of $z_{12} = w_{12}$. Let t be a parameter, put

$$(\zeta_3,\zeta_4,\zeta_5) = (m_5 u,\ m_5 v,\ -m_3 u - m_4 v), \qquad (\eta_3,\eta_4,\eta_5) = (m_5 p,\ m_5 q,\ -m_3 p - m_4 q),$$

so that $\sum_k m_k\zeta_k = \sum_k m_k\eta_k = 0$, and

$$\begin{aligned} z_i &= t^{-4}A_i, \qquad w_i = t^4 B_i \qquad (i \in P),\\ z_k &= t^{-1}\hat{Z} + t^2\zeta_k, \qquad w_k = t^5\hat{W} + t^2\eta_k \qquad (k \in T),\\ \delta_{12} &= D_{12}, \qquad \delta_{ik} = t\,D_{ik}, \qquad \delta_{kl} = t^{-2}D_{kl}. \end{aligned} \tag{11}$$

The twenty chart coordinates x are $A_1, A_2, B_1, B_2, \hat{Z}, \hat{W}, u, v, p, q$ and the ten D_{jl}; for $t \neq 0$ the chart is invertible. Put

$$\Sigma_i = \sum_k m_k D_{ik}^3, \qquad f_i = \sum_k m_k D_{ik}^3 \eta_k,$$
$$R_i^z = A_i - m_{i'} D_{12}^3 (A_i - A_{i'}), \qquad R_i^w = B_i - m_{i'} D_{12}^3 (B_i - B_{i'}),$$

where $\{i,i'\} = P$. Substituting (11) gives

$$\begin{aligned} t^4 E_i^z &= R_i^z - \textstyle\sum_k m_k D_{ik}^3\,(t^3 A_i - t^6\hat{Z} - t^9\zeta_k),\\ t^{-4} E_i^w &= R_i^w + t\,f_i - \textstyle\sum_k m_k D_{ik}^3\,(t^3 B_i - t^4\hat{W}),\\ t^4 E_k^z &= -\phi_k + t^3\hat{Z} + t^6\zeta_k - \textstyle\sum_i m_i D_{ik}^3\,(t^6\hat{Z} + t^9\zeta_k - t^3 A_i),\\ t^4 E_k^w &= -\psi_k + t^6\eta_k + t^9\hat{W} - \textstyle\sum_i m_i D_{ik}^3\,(t^9\eta_k + t^{12}\hat{W} - t^{11} B_i), \end{aligned} \tag{12}$$

while the ties become $D_{12}^2(A_1 - A_2)(B_1 - B_2) - 1$, the *crossed ties* $D_{ik}^2(t^3\hat{Z} + t^6\zeta_k - A_i)(\eta_k - t^2 B_i + t^3\hat{W}) - 1$ and χ_{kl}, and the gauge becomes $t^{-2}(z_{34} - w_{34}) = (\zeta_4 - \zeta_3) - (\eta_4 - \eta_3)$. These 21 polynomials form $F(t,x) = \sum_{j\geq 0} t^j F_j(x)$. The *boundary fibre* $\mathcal{B}_4 = \mathcal{B}_4(m)$ is the zero set of F_0. By (12), F_1 vanishes except in the two rows $t^{-4}E_i^w$, where it equals f_i, and F_2 vanishes except in the crossed ties, where it equals $D_{ik}^2 A_i B_i$.

8.1.2. *Coverage.* Let γ be a Diagram 4 germ at $m \in \mathbb{R}^5_{>0}$, renormalized as in Lemma 3.3 so that $Z \approx W \approx \epsilon^{-2}$. After a base change of the parameter τ of the branch, all coordinates are Laurent series in τ and $\operatorname{ord}_\tau Z = \operatorname{ord}_\tau W = -4q$ for an integer $q \geq 1$; we put $t = \tau^q$ and measure orders in units of $\operatorname{ord} t$, so that the maximal components of $\mathcal{Z}$ and $\mathcal{W}$ have order -4 and $\operatorname{ord}\epsilon = 2$.

Lemma 8.3. *Along γ, for $i \in P$ and $k \neq l$ in T:*

(i) $\operatorname{ord} z_k > -4$ *and* $\operatorname{ord} z_{ik} = -4$*;*
(ii) $\operatorname{ord} z_1 = \operatorname{ord} z_2 = \operatorname{ord} z_{12} = -4$*,* $\operatorname{ord}\delta_{12} = 0$*,* $\operatorname{ord} w_{12} = 4$*,* $\operatorname{ord} w_1, \operatorname{ord} w_2 \geq 4$*, and hence* $\operatorname{ord} W_T = \operatorname{ord} W_P \geq 4$*;*
(iii) $\operatorname{ord} z_{kl} = \operatorname{ord} w_{kl} = 2$ *and* $\operatorname{ord}\delta_{kl} = -2$.

Proof. (i) The bodies of T are not z-circled, while 1 and 2 are. (ii) Bodies $1, 2$ are z-circled, so $\operatorname{ord} z_1 = \operatorname{ord} z_2 = -4$. The isolated component $\{1, 2\}$ of the z-diagram has circled vertices, so its stroke 12 is maximal by Rule 1e [1, p. 546], and the right equality case of Estimate 2 [1, p. 548] gives $r_{12} \approx 1$, $z_{12} \approx \epsilon^{-2}$ and $w_{12} \approx \epsilon^2$. The triangle edges are not maximal, since by (iii), whose proof does not use (ii), $z_{kl} \approx \epsilon$ on them; so 12 is the only maximal z-edge and, by Estimate 1 [1, p. 548] with z and w exchanged, $w_{12} \prec w_{ik}$ for $i \in P$, $k \in T$. As in the proof of [1, Prop. 4, p. 564], [1, Prop. 1, p. 559] with z and w exchanged, applied to bodies 1 and 2, puts the origin in the w-cluster of size w_{12} formed by w_1 and w_2, so $w_1, w_2 \preccurlyeq w_{12}$. Finally $M_P W_P + M_T W_T = 0$. (iii) The triangle edges are zw-edges, so $r_{kl} \approx z_{kl} \approx w_{kl} \approx \epsilon$ by Estimate 1 and the left equality case of Estimate 2. □

Lemma 8.4 (computer-assisted)**.** *Let $0 < e < 2$. In* (11) *replace $\hat{Z}$ by $t^{-3e/2}\hat{Z}$, $\hat{W}$ by $t^{-e/2}\hat{W}$, (η_3, η_4, η_5) by $(t^e\omega,\ m_5 q,\ -m_3 m_5^{-1} t^e\omega - m_4 q)$ and D_{i3} by $t^{-e/2} d_i$ $(i \in P)$. Then*

$$T_P = t^{1-e/2}\Big(m_3 \sum_{i\in P} m_i d_i\,(d_i^2 A_i \omega) + \sum_{g\in G} c_g\, t^g\Big),$$

where the c_g are polynomials in the chart coordinates and $G = \{\frac{1}{2}e,\ \frac{3}{2}e,\ 3,\ 3-\frac{3}{2}e,\ 5-\frac{5}{2}e,\ 5-e,\ 8-e,\ 8+\frac{1}{2}e\}$; all elements of G are positive.

We obtain this expansion by an exact computation over $\mathbb{Q}$, with the masses and the chart coordinates as indeterminates and with exponents of t that are affine functions of e (Appendix A).

Lemma 8.5. *For $k \in T$ let o_k be the coefficient of t^2 in $w_k - W_T$ along γ. Then $o_3, o_4, o_5 \neq 0$.*

Proof. By Lemma 8.3(iii), $o_l - o_k$ is the leading coefficient of w_{kl}, so at most one o_k vanishes. Renumbering T, suppose that $o_3 = 0$ and let $\operatorname{ord}(w_3 - W_T) = 2+e$ with $e > 0$. By Lemma 8.3(ii), $\operatorname{ord}(W_T - w_i) \geq 4$.

If $e \geq 2$, then $\operatorname{ord} w_{i3} \geq 4$, so $\operatorname{ord}\delta_{i3} = -\frac{1}{2}(\operatorname{ord} z_{i3} + \operatorname{ord} w_{i3}) \leq 0$ and $\operatorname{ord} Z_{i3} = \operatorname{ord} z_{i3} + 3\operatorname{ord}\delta_{i3} \leq -4$. Then Z_{i3} is a maximal component of $\mathcal{Z}$, that is, a z-stroke between i and 3, which Diagram 4 does not have.

Let $0 < e < 2$. Then $\operatorname{ord} w_{i3} = 2 + e$ and $\operatorname{ord}\delta_{i3} = 1 - \frac{e}{2}$, while $\operatorname{ord} w_{ik} = 2$ and $\operatorname{ord}\delta_{ik} = 1$ for $k = 4, 5$. The first two identities of Lemma 5.2 express $M_P W_P$ and $M_T Z_T$ as sums of crossed terms $m_i m_k \delta_{ik}^3 (w_i - w_k)$ and $m_i m_k \delta_{ik}^3 (z_k - z_i)$, of orders at least $5 - \frac{e}{2}$ and $-1 - \frac{3}{2}e$; with $M_P W_P + M_T W_T = 0$ this gives $\operatorname{ord} W_T \geq 5 - \frac{e}{2}$ and $\operatorname{ord} Z_T \geq -1 - \frac{3}{2}e$. So γ is described by the chart of Lemma 8.4, with all its coordinates of nonnegative order and A_i, ω, d_i of order 0. At $t = 0$ the ties $\delta_{i3}^2 z_{i3} w_{i3} = 1$ give $d_i^2 A_i \omega = -1$, and $m_1 A_1 + m_2 A_2 = t^4(m_1 z_1 + m_2 z_2) = -t^4 M_T Z_T$ vanishes. By Lemma 8.4, $t^{e/2-1} T_P$ tends to $-m_3(m_1 d_1 + m_2 d_2)$, which is nonzero because

$$(m_1 d_1)^2 - (m_2 d_2)^2 = -\frac{m_1^2}{A_1\omega} + \frac{m_2^2}{A_2\omega} = -\frac{m_1^3 + m_2^3}{m_1 A_1 \omega} \neq 0.$$

But $T_P = 0$ on A_m by Lemma 5.2. □

Proposition 8.6 (Coverage)**.** *Let $m \in \mathbb{R}^5_{>0}$ and let γ be a Diagram 4 germ. Then m satisfies (23), and $\operatorname{ord} Z_T \geq -1$, $\operatorname{ord} W_T \geq 5$ along γ. Renormalize by $(z, w) \mapsto (cz, c^{-1}w)$ with $c =$*

$(w_{34}/z_{34})^{1/2}$, *so that* $z_{34} = w_{34}$. *After a base change of* τ, *every chart coordinate of* (11) *is a power series in* τ; *the value* y *of* x *at* $\tau = 0$ *lies in* $\mathcal{B}_4(m)$, *and* $\eta_k(y) \neq 0$ *for* $k \in T$.

Proof. Relation (23) is Proposition 8.1. By Lemmas 8.3 and 8.5, $\operatorname{ord} w_{ik} = 2$ for all $i \in P$, $k \in T$, hence $\operatorname{ord} \delta_{ik} = 1$. The first two identities of Lemma 5.2 then give $\operatorname{ord} Z_T \geq 3 - 4 = -1$ and $\operatorname{ord} W_P \geq 3 + 2 = 5$, and $\operatorname{ord} W_T = \operatorname{ord} W_P$. The factor c has order 0 by Lemma 8.3(iii), so all orders persist after the renormalization. Inverting (11), $A_i = t^4 z_i$, $B_i = t^{-4} w_i$, $\hat{Z} = t Z_T$, $\hat{W} = t^{-5} W_T$, $\zeta_k = t^{-2}(z_k - Z_T)$, $\eta_k = t^{-2}(w_k - W_T)$, $D_{12} = \delta_{12}$, $D_{ik} = t^{-1}\delta_{ik}$ and $D_{kl} = t^2 \delta_{kl}$; by Lemma 8.3 all have nonnegative order, and $\eta_k \to o_k \neq 0$. The rows of F are monomial multiples of the equations, so they vanish along γ, and $F_0(y) = 0$. □

8.1.3. *The boundary fibre.* On $\mathbb{C}^{20}$ consider the scalings

$$\alpha \cdot x : \ (A, B, D_{ik}) \mapsto (\alpha^2 A, \ \alpha^{-2} B, \ \alpha^{-1} D_{ik}),$$
$$\beta \cdot x : \ (\zeta, \eta, D_{kl}, D_{ik}) \mapsto (\beta^2 \zeta, \ \beta^2 \eta, \ \beta^{-2} D_{kl}, \ \beta^{-1} D_{ik}),$$

$\alpha, \beta \in \mathbb{C}^*$, the other coordinates being fixed, and the translations of $(\hat{Z}, \hat{W})$. They preserve F_0 up to nonzero factors on each row, hence preserve $\mathcal{B}_4(m)$; they are not symmetries of F.

Lemma 8.7. *Let* $m \in \mathbb{R}^5_{>0}$ *satisfy (23) and let* $y \in \mathcal{B}_4(m)$.

(a) $D_{12}^3 = M_P^{-1}$, $A_2 = \rho A_1$ *and* $B_2 = \rho B_1$ *with* $\rho = -m_1/m_2$, $D_{12}^2 (A_1 - A_2)(B_1 - B_2) = 1$ *and* $D_{ik}^2 A_i \eta_k = -1$; *in particular* $A_i, B_i, \eta_k \neq 0$. *The point* $(\zeta_3, \zeta_4, \eta_3, \eta_4, D_{34}, D_{35}, D_{45})$ *solves the zero-multiplier triangle system, so, after renumbering* T, *it lies on the orbit* O *of Lemma 5.9.*

(b) $dF_0(y)$ *has rank* 16. *Its kernel is spanned by* $\partial_{\hat{Z}}$, $\partial_{\hat{W}}$ *and the generators of the two scalings. Its cokernel is spanned by the linear form*

$$v_P(r) = (B_1 - B_2)\,(r_1^z - r_2^z) - (A_1 - A_2)\,(r_1^w - r_2^w),$$

where r_i^z, r_i^w *are the components of* $r \in \mathbb{C}^{21}$ *in the rows* $t^4 E_i^z$, $t^{-4} E_i^w$, *and by four linear forms that involve only the rows* $t^4 E_k^z$, $t^4 E_k^w$, χ_{kl} *and the gauge.*

(c) *Near* y, $\mathcal{B}_4(m)$ *coincides with the orbit of* y *under the scalings and the translations; it is a complex submanifold of dimension* 4 *with tangent space* $\ker dF_0(y)$ *at* y.

Proof. (a) Identically $R_1^z - R_2^z = (A_1 - A_2)(1 - M_P D_{12}^3)$ and $m_1 R_1^z + m_2 R_2^z = m_1 A_1 + m_2 A_2$, and likewise for R^w. The tie of 12 gives $A_1 \neq A_2$ and $B_1 \neq B_2$; so $F_0(y) = 0$ gives $D_{12}^3 = M_P^{-1}$, $m_1 A_1 + m_2 A_2 = 0$ and $m_1 B_1 + m_2 B_2 = 0$. The crossed ties are $D_{ik}^2 A_i \eta_k = -1$. The triangle rows of F_0 are $\phi_k, \psi_k, \chi_{kl}$ and the gauge; the last statement is the proof of Proposition 5.10. (b) Order the coordinates as $(D_{ik};\ A, B, D_{12};\ u, v, p, q, D_{kl};\ \hat{Z}, \hat{W})$ and the rows as (crossed ties; pair rows R_i^z, R_i^w and the tie of 12; triangle rows). The pair rows involve only the pair coordinates and the triangle rows only the triangle coordinates, and D_{ik} occurs only in its own crossed tie, with $\partial/\partial D_{ik} = -2 D_{ik} A_i \eta_k \neq 0$. So $dF_0(y)$ is block lower triangular, and its rank is $6 + \operatorname{rank} J_P + \operatorname{rank} J_T$, with J_P, J_T the pair and triangle blocks. At y, the differentials of $R_1^z - R_2^z$ and $R_1^w - R_2^w$ are $-3 M_P D_{12}^2$ times $(A_1 - A_2)\, dD_{12}$ and $(B_1 - B_2)\, dD_{12}$, those of $m_1 R_1^z + m_2 R_2^z$ and $m_1 R_1^w + m_2 R_2^w$ are $m_1 dA_1 + m_2 dA_2$ and $m_1 dB_1 + m_2 dB_2$, and the tie of 12 adds D_{12}^2 times $(B_1 - B_2)(dA_1 - dA_2) + (A_1 - A_2)(dB_1 - dB_2)$ modulo dD_{12}; so $\operatorname{rank} J_P = 4$, the kernel of J_P is spanned by $(A, -B, 0)$, and v_P, the derivative at y of the identity $(B_1 - B_2)(R_1^z - R_2^z) - (A_1 - A_2)(R_1^w - R_2^w) = 0$, spans its cokernel. By Lemma 5.9 the rows $\phi_3, \phi_4, \psi_3, \psi_4, \chi_{kl}$ have rank 5 at y, with kernel spanned by the generators $(\zeta, 0, -\frac{1}{2}D)$ and $(0, \eta, -\frac{1}{2}D)$ of $\mathbb{T}$; ϕ_5 and ψ_5 add nothing because $\sum_k m_k \phi_k = \sum_k m_k \psi_k = 0$; and the gauge row takes the value $\zeta_4 - \zeta_3$, which is nonzero by $\chi_{34} = 0$, on the first generator and 0 on $(\zeta, \eta, -D)$. So $\operatorname{rank} J_T = 6$, with kernel spanned by $(\zeta, \eta, -D)$. Hence the rank is 16, and the four listed vectors, completed in the D_{ik} by the crossed ties, span the kernel. A linear form vanishing on the image vanishes on the crossed ties (columns D_{ik}), so it is the sum of a form in the cokernel of J_P and one in the cokernel of J_T, which has dimension $10 - 6 = 4$. (c) The orbit map has differential onto $\ker dF_0(y)$, so the orbit

of y is a submanifold of dimension 4 near y, contained in $\mathcal{B}_4(m)$. By (b) and the rank theorem, $\mathcal{B}_4(m)$ is contained near y in a submanifold of dimension $20-16=4$. □

8.1.4. *A Lyapunov–Schmidt lemma.* For a polynomial curve $x(t)$ we write $[F]_j$ for the coefficient of t^j in $F(t,x(t))$.

Lemma 8.8 (Lyapunov–Schmidt)**.** *Let $F(t,x)=\sum_{j\ge0}t^jF_j(x)$ be a polynomial map $\mathbb{C}\times\mathbb{C}^n\to\mathbb{C}^p$, let $Z=F_0^{-1}(0)$, and let $y\in Z$ be a point near which Z is a complex submanifold with $T_{y'}Z=\ker dF_0(y')$ for all $y'\in Z$. For $y'\in Z$ let $Q_{y'}$ be the cokernel of $dF_0(y')$. Let $N\ge1$ and suppose that every $y'\in Z$ near y is the value at $t=0$ of a polynomial curve $x(t)$ with $[F]_j=0$ for $j<N$.*

(a) *The class $\Omega_N(y')\in Q_{y'}$ of $[F]_N$ does not depend on the choice of such a curve.*
(b) *If $\Omega_N(y)\neq0$, no holomorphic germ $c\colon(\mathbb{C},0)\to(F^{-1}(0),(0,y))$ has $t\circ c\not\equiv0$.*

For $N=1$ the hypothesis is empty and $\Omega_1(y')$ is the class of $F_1(y')$.

Proof. Let $K=\ker dF_0(y)$ and $R=dF_0(y)(\mathbb{C}^n)$, choose complements $\mathbb{C}^n=K\oplus K'$ and $\mathbb{C}^p=R\oplus R'$, and let π be the projection onto R along R'. Since $\pi\,dF_0(y)$ maps K' isomorphically onto R, the implicit function theorem gives a holomorphic $h(t,k)\in K'$, defined near $(0,0)\in\mathbb{C}\times K$ with $h(0,0)=0$, such that near $(0,y)$ the equation $\pi F(t,y+k+h')=0$ with small $h'\in K'$ holds exactly for $h'=h(t,k)$; uniqueness also holds for formal power series in t. Put $G(t,k)=F(t,y+k+h(t,k))\in R'$. Near $(0,y)$, $F^{-1}(0)=\{(t,y+k+h(t,k)):G(t,k)=0\}$. The map $k\mapsto y+k+h(0,k)$ parametrizes the submanifold $\{\pi F_0=0\}$, of dimension $\dim K=\dim Z$, which contains Z; so it parametrizes Z near y, and $G(0,k)=0$. Write $G=\sum_{j\ge1}t^jG_j(k)$. For $y'\in Z$ near y the space R' is also a complement of $dF_0(y')(\mathbb{C}^n)$, so $R'\to Q_{y'}$ is an isomorphism.

Let $j\le N$, let $y'=y+k'+h(0,k')$, suppose that $G_1,\dots,G_{j-1}$ vanish near k', and let $x(t)=y+\kappa(t)+\eta(t)$, with $\kappa(t)\in K$, $\eta(t)\in K'$, be a polynomial curve with $x(0)=y'$ and $[F]_i=0$ for $i<j$. Since $\pi F(t,x(t))=O(t^j)$, formal uniqueness gives $\eta(t)-h(t,\kappa(t))=e\,t^j+O(t^{j+1})$ for some e. By Taylor's formula $F(t,x(t))=F(t,y+\kappa+h(t,\kappa))+dF_0(y')(e)\,t^j+O(t^{j+1})$, and $F(t,y+\kappa+h(t,\kappa))=G(t,\kappa(t))=t^jG_j(k')+O(t^{j+1})$. So $[F]_j\equiv G_j(k')$ modulo $dF_0(y')(\mathbb{C}^n)$. For $j<N$ the curves of the hypothesis give $G_j(k')\in R'\cap dF_0(y')(\mathbb{C}^n)=0$ for all k' near 0; by induction $G_1,\dots,G_{N-1}$ vanish near 0, and the case $j=N$ gives (a), $\Omega_N(y')$ being the image of $G_N(k')$. If $\Omega_N(y)\neq0$, then $G=t^NH$ with H holomorphic and $H(0,0)=G_N(0)\neq0$, so near $(0,0)$ the equation $G=0$ forces $t=0$. A germ c as in (b) has the form $(t,y+k+h(t,k))$ with $G(t,k)=0$, so $t\circ c\equiv0$. □

8.1.5. *The first three obstructions.* For $y\in\mathcal{B}_4(m)$ put $S_i=\sum_k m_kD_{ik}$ and

$$\Theta=\sum_k m_k\,\frac{D_{1k}-D_{2k}}{\eta_k}=A_2\Sigma_2-A_1\Sigma_1;$$

the second expression and $f_i=-S_i/A_i$ follow from $1/\eta_k=-A_iD_{ik}^2$. Under the scalings, S_i/A_i is multiplied by α^{-3} and by β^{-1}, and Θ by α^{-1} and by β^{-3}; neither involves $\hat Z$, $\hat W$, D_{12} or m_1,m_2. Since the crossed forces on body i are $\sum_k m_k\delta_{ik}^3(z_i-z_k)=t^{-1}A_i\Sigma_i+O(t^2)$ and $\sum_k m_k\delta_{ik}^3(w_i-w_k)=-t^5f_i+O(t^7)$, the conditions $f_1=f_2$ and $\Theta=0$ say that the triangle pulls the two outer bodies equally, at leading order, in the w- and in the z-coordinate.

Lemma 8.9. *Let $m\in\mathbb{R}^5_{>0}$ satisfy (23), let $y\in\mathcal{B}_4(m)$, and let Ω_N be the classes of Lemma 8.8 for F and $Z=\mathcal{B}_4(m)$.*

(a) *$\Omega_1(y)=0$ if and only if $f_1=f_2$, that is, $S_1/A_1=S_2/A_2$.*
(b) *If $f_1=f_2$ at y, then the hypothesis of Lemma 8.8 holds at y for $N=3$, and*

$$v_P(\Omega_3(y))=\tfrac{3}{2}\,(A_1-A_2)\,\frac{B_1}{A_1}\,\Theta.$$

Proof. (a) By Lemma 8.7(b), $\Omega_1(y)$, the class of $F_1(y)$, vanishes if and only if $F_1(y)$ is killed by v_P and by the four triangle forms. F_1 vanishes on the triangle rows, and $v_P(F_1(y)) = -(A_1 - A_2)(f_1 - f_2)$.

(b) By homogeneity $f_1 = f_2$ on the orbit of y, which by Lemma 8.7(c) is a neighbourhood of y in $\mathcal{B}_4(m)$. Let y' be a point of it, $N \in \{2, 3\}$, and $x(t)$ a polynomial curve with $x(0) = y'$ and $F(t, x(t)) = O(t^N)$; such a curve exists for $N = 2$ by (a). We write $A_i(t), \dots$ for its components, $A_i, \dots$ for their values at y', and $\Phi(t) = 1 - M_P D_{12}(t)^3$.

Pair. By (12), $t^4 E_i^z = R_i^z + O(t^3)$. As $R_1^z - R_2^z = (A_1(t) - A_2(t))\Phi(t)$ and $m_1 R_1^z + m_2 R_2^z = m_1 A_1(t) + m_2 A_2(t)$, we get $\Phi = O(t^N)$ and $A_i(t) = A_i(1 + a(t)) + O(t^N)$ with one series $a(t) = O(t)$.

Triangle. The triangle rows of F agree with those of F_0 up to $O(t^3)$, so the triangle components of $x(t)$ solve the triangle block of F_0 modulo t^N. By the proof of Lemma 8.7(b),(c), the zero set of that block is, near y', the curve $\beta \mapsto \beta \cdot y'$, of dimension $\dim \ker J_T$; so the triangle components of $x(t)$ agree modulo t^N with a curve on it, and $\eta_k(t) = \theta(t)\eta_k + O(t^N)$ with $\theta(0) = 1$. For $N = 2$ the triangle components of $[F]_2$ therefore lie in the image of J_T.

Crossed ties. They give $D_{ik}(t)^2 = -1/(A_i(t)(\eta_k(t) - t^2 B_i)) + O(t^N)$, hence

$$D_{ik}(t)^3 \eta_k(t) = -\frac{D_{ik}(t)}{A_i(t)}\Big(1 + t^2 \frac{B_i}{\eta_k}\Big) + O(t^N), \qquad D_{ik}(t) = \frac{D_{ik}}{\sqrt{(1+a)\theta}}\Big(1 + \frac{t^2 B_i}{2\eta_k}\Big) + O(t^N),$$

and, with $f_i(t) = \sum_k m_k D_{ik}(t)^3 \eta_k(t)$,

$$f_i(t) = -\frac{S_i/A_i}{(1+a)^{3/2}\theta^{1/2}} - \frac{3}{2}\, t^2\, \frac{B_i}{A_i} \sum_k m_k \frac{D_{ik}}{\eta_k} + O(t^N).$$

Since $S_1/A_1 = S_2/A_2$ and $B_1/A_1 = B_2/A_2$, $f_1(t) - f_2(t) = -\frac{3}{2}\, t^2\, \frac{B_1}{A_1}\, \Theta + O(t^N)$.

The form v_P. By (12), v_P applied to the rows of $F(t, x(t))$, with the coefficients of v_P taken at y', equals

$$\Phi(t)\big[(B_1 - B_2)(A_1(t) - A_2(t)) - (A_1 - A_2)(B_1(t) - B_2(t))\big] - (A_1 - A_2)\, t\, (f_1(t) - f_2(t)) - t^3\, \Delta + O(t^4),$$

with $\Delta = (B_1 - B_2)(A_1\Sigma_1 - A_2\Sigma_2) - (A_1 - A_2)(B_1\Sigma_1 - B_2\Sigma_2)$, which vanishes because $A_2 = \rho A_1$ and $B_2 = \rho B_1$. The bracket vanishes at $t = 0$ and $\Phi = O(t^N)$.

For $N = 2$ this gives $v_P([F]_2) = 0$. The crossed ties are killed by every form of the cokernel, so with the triangle part $\Omega_2(y') = 0$: the curve can be corrected at order 2 to a curve with $F(t, x(t)) = O(t^3)$. This proves the hypothesis for $N = 3$. For $N = 3$ the displayed expression gives $v_P([F]_3) = -(A_1 - A_2) \cdot (-\frac{3}{2}\frac{B_1}{A_1}\Theta)$. □

8.1.6. *The census.* Let m satisfy (23). After renumbering T, $(\sqrt{m_3}, \sqrt{m_4}, \sqrt{m_5}) = \lambda\,(bc,\ b(b+c),\ c(b+c))$ with $\lambda, b, c > 0$, as after Lemma 5.9; exchanging 4 and 5 exchanges b and c, and $b = c$ is the central-body slice. By Lemma 5.8 and the proof of Proposition 5.10, the triangle offsets η_k of a point of $\mathcal{B}_4(m)$ are proportional to

$$(o_3, o_4, o_5) = \Big(-\frac{b^2 - c^2}{bc},\ \frac{c(2b+c)}{b(b+c)},\ -\frac{b(b+2c)}{c(b+c)}\Big).$$

The rational curve

$$b = 2(n^4 + 1), \qquad c = -n^4 + 6n^2 - 1, \qquad 1 < n < 1 + \sqrt{2},$$

has $b - c = 3(n^2 - 1)^2$, $2b + c = 3(n^2 + 1)^2$ and $b + 2c = 12n^2$, and b/c takes every value in $(1, \infty)$ on it. Put

$$g_3 = 1, \qquad g_4 = \frac{i(b+c)(n^2-1)}{c(n^2+1)}, \qquad g_5 = \frac{(b+c)(n^2-1)}{2nb},$$

so that $g_k^2 = o_3/o_k$, and $(m_3, m_4, m_5) = ((bc)^2, (b(b+c))^2, (c(b+c))^2)$. For a sign vector $s = (s_{13}, s_{14}, s_{15}, s_{23}, s_{24}, s_{25}) \in \{\pm 1\}^6$ and $j \in \{1, 3\}$ put

$$X_j = \sum_k s_{1k}\, m_k g_k^j, \qquad Y_j = \sum_k s_{2k}\, m_k g_k^j.$$

Lemma 8.10 (computer-assisted)**.** *Let $n \in (1, 1+\sqrt{2})$ and $\nu > 0$.*

(a) *For 48 of the 64 sign vectors, $\nu^3 X_1 + iY_1 \neq 0$ for all such (n, ν). For each of the other 16, namely those with $s_{13}s_{24} = 1$ and $s_{14}s_{23} = -1$, there is exactly one such pair (n, ν) with $\nu^3 X_1 + iY_1 = 0$.*

(b) *At each of these 16 pairs, $\nu X_3 \neq iY_3$.*

The proof of (a) is an exact computation in three steps. For each of the 64 sign vectors, multiplication by $\pm 4n(n^2+1)(n^4+1)(n^2-2n-1)(n^2+2n-1)$, which has no zero on $(1, 1+\sqrt{2})$, turns $\nu^3 X_1 + iY_1$ into a polynomial in n and ν, the census numerator. Exact real-root counting splits the census numerators into 48 classes without and 16 with a zero; each zero class has exactly one zero, by Sturm counts on the eliminant $\mathrm{Im}(-\bar{X}_1 \cdot iY_1)$ with the sign condition $\nu^3 > 0$. For (b), if $\nu X_3 = iY_3$ and $\nu^3 X_1 + iY_1 = 0$ then $Y_1 X_3^3 = X_1 Y_3^3$; the numerator of $Y_1 X_3^3 - X_1 Y_3^3$ is coprime over $\mathbb{Q}(i)$ to the minimal polynomial of the zero n in all 16 cases (Appendix A).

8.1.7. *Proof of Theorem 8.2.* Suppose that γ is a Diagram 4 germ at $m \in \mathbb{R}^5_{>0}$. By Proposition 8.6, m satisfies (23), and $\tau \mapsto (t(\tau), x(\tau))$, with $t = \tau^q$, is a holomorphic germ in $F^{-1}(0)$ with $t \not\equiv 0$ and $x(0) = y \in \mathcal{B}_4(m)$. By Lemma 8.7(c), Lemma 8.8 applies at y with $Z = \mathcal{B}_4(m)$. For $N = 1$ it gives $\Omega_1(y) = 0$, that is, $f_1 = f_2$ at y (Lemma 8.9(a)). Then Lemma 8.9(b) and Lemma 8.8 for $N = 3$ give $\Omega_3(y) = 0$; as $(A_1 - A_2)B_1/A_1 \neq 0$, $\Theta = 0$ at y.

Both conditions are symmetric in the bodies of T and homogeneous under the scalings, and they involve m_3, m_4, m_5 linearly. So we may renumber T as in §8.1.6 with $b > c$ ($b \neq c$ because $\eta_k(y) \neq 0$ by Proposition 8.6 and $o_3 = 0$ for $b = c$), choose $n \in (1, 1+\sqrt{2})$ with b/c equal to the ratio of the curve, replace (m_3, m_4, m_5) by the values on the curve, and apply the scalings so that $A_1 = 1$ and $\eta_3 = -1$. Then $\eta_k = -o_k/o_3$ and $D_{1k}^2 = -1/\eta_k = g_k^2$; with $\nu = (m_1/m_2)^{1/2}$ we have $A_2 = -\nu^2$ and $D_{2k}^2 = -D_{1k}^2/\nu^2$. Hence $D_{1k} = s_{1k} g_k$ and $D_{2k} = s_{2k}\, i g_k/\nu$ for a sign vector s, and $S_1 = X_1$, $S_2 = iY_1/\nu$, $\Sigma_1 = X_3$, $\Sigma_2 = -iY_3/\nu^3$. The condition $S_1/A_1 = S_2/A_2$ becomes $\nu^3 X_1 + iY_1 = 0$, and $\Theta = A_2\Sigma_2 - A_1\Sigma_1 = 0$ becomes $\nu X_3 = iY_3$. This contradicts Lemma 8.10. □

Remark 8.11. (a) Positivity of the masses enters through (23) and the parametrization of the triangle, through $m_1^3 + m_2^3 \neq 0$, and through $\nu > 0$; no reality of the germ is used, so Theorem 8.2 excludes all complex germs. (b) The two obstructions involve the outer masses only through $\rho = -m_1/m_2$, and not through the ratio M_P/M_T.

8.2. **Diagram 5.** Diagram 5 is the isolated zw-triangle 345 alone: its only strokes are the six strokes of the three triangle edges, and no vertex is circled [1, §7.4 and Fig. 11]. We write $T = \{3, 4, 5\}$ and call 1 and 2 the *outer bodies*. For this diagram [1, §§8.4–8.5] record only relation (23), and [1, §9] avoid estimating the distances that carry no edge. We build no chart. We classify the orders of the outer bodies relative to the triangle and decide by exact linear programming which are compatible with the equations on a curve through a real point, where $U = I$ has a nonzero constant value.

Theorem 8.12. *Let $m \in \mathbb{R}^5_{>0}$. No Diagram 5 germ with an unbounded four-product lies on a curve through a real point: if an irreducible curve $\Gamma \subset A_m$ carries such a germ, then Γ contains no real point.*

This is part (b) of Theorem 4.1 for $r = 5$. Germs with all four-products bounded are not excluded, and need not be (Proposition 3.6); they do occur on curves through real points, e.g. at the collision end of the modified Roberts continuum at the masses $(4, 4, 4, 4, 1)$ [1, p. 543], which is real but not positive.

8.2.1. *The frame.* Let γ be a Diagram 5 germ, numbered as in [1, Fig. 11] and renormalized as in Lemma 3.3, with orders measured in units of $\operatorname{ord}\epsilon$, so that $\mathcal{Z}$ and $\mathcal{W}$ have least order -2. After a constant reparametrization the Puiseux parameter t has $\epsilon \approx |t|$, and we write orders in t. By Fact AK2: $-2 \le \operatorname{ord} r_{kl} \le 1$, with $\operatorname{ord} r_{kl} = 1$ exactly for zw-edges, and $-2 \le \operatorname{ord} z_{kl}, \operatorname{ord} w_{kl} \le 2$ (Estimate 1); $\operatorname{ord} z_{kl} = \operatorname{ord} w_{kl} = 1$ and $\operatorname{ord}\delta_{kl} = -1$ on T (Estimate 2); $\operatorname{ord} z_k, \operatorname{ord} w_k > -2$ for all k, as nothing is circled (Rule 1a at the outer vertices; on T the strokes of the zw-edges are not maximal by Estimates 1 and 2, so Rule 1e applies); and $\operatorname{ord} Z_{kl}, \operatorname{ord} W_{kl} > -2$ unless $k, l \in T$.

Put $M_T = m_3 + m_4 + m_5$, $M = \sum_k m_k$, $z_c = \sum_{k\in T} m_k z_k / M_T$, w_c likewise, $\xi_k = z_k - z_c$ and $\eta_k = w_k - w_c$ ($k \in T$), and let $F_j = \sum_{k\in T} m_k \delta_{jk}^3 (z_j - z_k)$ and $G_j = \sum_{k\in T} m_k \delta_{jk}^3 (w_j - w_k)$ be the forces of the triangle on the outer body j. By Proposition 8.1, (23) holds and $(\xi/t, \eta/t)$ tends to a zero-multiplier triangle. Renumbering T, $(\sqrt{m_3}, \sqrt{m_4}, \sqrt{m_5}) = \lambda(b, b(b+1), b+1)$ with $b, \lambda > 0$. This triangle has real, mass-centred, pairwise distinct offsets s_3, s_4, s_5 with distinct cubes; $s_4, s_5 \neq 0$, and $s_3 = 0$ exactly for $b = 1$, the *central-body slice* $m_4 = m_5 = 4m_3$. Its z- and w-shapes agree, and it is unique up to translations, the scalings $(\xi, \eta) \mapsto (a\xi, a'\eta)$ and the sign of its inverse distances. A constant reparametrization and a constant renormalization $(z, w) \mapsto (az, a^{-1}w)$, which preserves rows, ties, orders, the r_{kl}^2, U and I, give

$$\xi_k = t\,(s_k + \sigma_k), \qquad \eta_k = t\,(s_k + \tau_k), \qquad \sigma_k, \tau_k \to 0. \tag{13}$$

For $j \in \{1, 2\}$ let $\alpha_{jc}, \beta_{jc} > -2$ be the orders and $\hat z_j, \hat w_j$ the leading coefficients of $z_j - z_c$, $w_j - w_c$; for $P \in \{12, 1c, 2c\}$ let α_P, β_P be the orders of the z- and w-differences, r_P^2 their product and $\rho_P = \frac12(\alpha_P + \beta_P) = \operatorname{ord} r_P$. The *pattern* of γ is the word $X_{12}X_{1c}X_{2c}$, where X_P is C, R or E as $\rho_P > 0$, $= 0$ or < 0.

8.2.2. *The reality input.*

Lemma 8.13. *Let $\Gamma \subset A_m$ be an irreducible curve containing a real point q. Then $U = I = I_0$ on Γ, where $I_0 = \sum_k m_k |z_k(q)|^2 > 0$, and along every Diagram 5 germ of Γ, in the frame above,*

$$I_T + I_3 = I_0, \qquad I_T = \sum_{k\in T} m_k \xi_k \eta_k, \qquad I_3 = \frac{m_1 m_2 r_{12}^2 + m_1 M_T r_{1c}^2 + m_2 M_T r_{2c}^2}{M}, \tag{14}$$

with $\operatorname{ord} I_T = 2$ and hence $\operatorname{ord} I_3 = 0$.

Proof. By Lemma 2.1, U is constant on Γ, and $U = I$ on A_m. At q, $w_k = \bar z_k$ and $z_1(q) \neq z_2(q)$ (tie of 12), so $I(q) > 0$. The renormalization preserves U and I. As $\sum_k m_k z_k = \sum_k m_k w_k = 0$ on A_m, $I - I_T - I_3$ vanishes identically, and $I_T = t^2(\sum_k m_k s_k^2 + o(1))$ with $\sum_k m_k s_k^2 > 0$ by (13). □

This is the only input from reality. It enters through the conditions (A) and (B) of §8.2.4, that two quantities have order exactly 0; so only $I_0 \neq 0$ is used, not its sign, nor $\sigma(\Gamma) = \Gamma$.

8.2.3. *Regimes of an outer body.*

Definition 8.14. Let $j \in \{1, 2\}$, $\alpha = \alpha_{jc}$ and $\beta = \beta_{jc}$. The outer body j is

(a) *far* if $\alpha, \beta < 1$. Then $\delta_{jk}/\delta_{jc} \to \varepsilon_{jk} = \pm 1$ for a fixed branch δ_{jc} of $((z_j - z_c)(w_j - w_c))^{-1/2}$. The body is a *monopole* if $\mu_j = \sum_k m_k \varepsilon_{jk} \neq 0$. Otherwise it is *dipole-z* if $\alpha > \beta$, *dipole-w* if $\beta > \alpha$, and, if $\alpha = \beta$, *dipole-equal-F*, *dipole-equal-G* or *dipole-equal* as $\hat w_j + 3\hat z_j$, $3\hat w_j + \hat z_j$ or neither vanishes, the leading coefficients of F_j, G_j at the monopole orders raised by $1 - \alpha$ being nonzero multiples of these.

(b) *near in z* if $\alpha \ge 1 > \beta$. For $\alpha = 1$ let $\theta_j = \lim t^{-1}(z_j - z_c)$. The body is *$z$-dominant-coincidence* if $\alpha = 1$ and $\theta_j = s_k$, its *special body k*, so that $\operatorname{ord}(z_j - z_k) = 1 + e_j$ with $e_j > 0$; *z-coincident* if $\alpha = 1$ and $\theta_j \notin \{s_3, s_4, s_5\}$, with the variants *$z$-coincident-$\Phi$*, *-$\Psi$*, *-both* if $\Phi_j = 0 \neq \Psi_j$, $\Psi_j = 0 \neq \Phi_j$, $\Phi_j = \Psi_j = 0$, where $\Phi_j = \sum_k m_k y_{jk}$, $\Psi_j = \sum_k m_k y_{jk}^3$, $\delta_{jk} = y_{jk}\nu_j(1 + o(1))$, $y_{jk}^{-2} = \theta_j - s_k$ and ν_j is a branch of $(t(w_j - w_c))^{-1/2}$; *z-inner* if

$\alpha > 1$ off the slice; and *z-dominant-central* if $\alpha > 1$ on the slice, with special body 3, $1 + e_j = \operatorname{ord}(z_j - z_3)$ and $g = \operatorname{ord}\xi_3$.

(c) *near in w* if $\beta \geq 1 > \alpha$, with the seven mirror regimes.

Lemma 8.15. *Let $m \in \mathbb{R}^5_{>0}$. Each outer body lies in exactly one of these 20 regimes, and z-coincident-both and w-coincident-both are empty. The dipoles need $m_a = m_b + m_c$ for the minority body a, which fails on the slice; and for a z-dominant-central body the least of α_{jc}, $1 + e_j$, g is attained at least twice.*

Proof. For $k \in T$, $z_j - z_k = (z_j - z_c) - \xi_k$ and $w_j - w_k = (w_j - w_c) - \eta_k$, with $\operatorname{ord}\xi_k, \operatorname{ord}\eta_k \geq 1$. If $\alpha, \beta \geq 1$, then $\operatorname{ord} r_{jk} \geq 1$ and jk would be a zw-edge. If $\alpha, \beta < 1$, then $\delta^2_{jk}/\delta^2_{jc} \to 1$; $\mu_j = 0$ needs a single minority body a with $m_a = m_b + m_c$, false for the slice masses $(1, 4, 4)$; and $\hat{w}_j + 3\hat{z}_j = 3\hat{w}_j + \hat{z}_j = 0$ would give $\hat{z}_j = \hat{w}_j = 0$. Let $\alpha = 1 > \beta$: at most one s_k equals $\theta_j \neq 0$, and otherwise every $z_j - z_k$ has order 1; at positive masses on (23), $\Phi_j = \Psi_j = 0$ forces $y = 0$, i.e. $\theta_j = \infty$. Let $\alpha > 1 > \beta$: then $z_j - z_k = -ts_k(1 + o(1))$ when $s_k \neq 0$; on the slice $\operatorname{ord}(z_j - z_3) \geq \min(\alpha, g) > 1$, and $z_j - z_c = (z_j - z_3) + \xi_3$ gives the tropical condition, including the deep case $\alpha = g < 1 + e_j$. The rest is symmetric. □

If $z_j \equiv z_c$ or $\xi_3 \equiv 0$, the order α_{jc} or g is infinite. Such an order enters the constraints below only through conditions that remain true when it is replaced by any sufficiently large finite value, and the census, whose variables are free, includes such values. The order α_{12} is always finite, as the tie of 12 gives $z_1 \neq z_2$. Up to the exchanges $1 \leftrightarrow 2$ and $z \leftrightarrow w$, which permute patterns and four-products, there are 111 pairs of regimes.

8.2.4. *Identities and orders.* Let $\Pi = \delta^3_{12}(z_1 - z_2)$, $U_T = \sum_{k<l\in T} m_k m_l \delta_{kl}$ and, summing over $j \in \{1, 2\}$, $k \in T$, $V_1 = \sum m_j m_k \delta^3_{jk} \xi_k (w_k - w_j)$, $V_2 = \sum m_j m_k \delta^3_{jk} \eta_k (z_k - z_j)$. The census uses the conditions below; the identities in (R), (T), the virial identity $I_T = U_T + \frac{1}{2}(V_1 + V_2)$ and the pair-centre identity of Proposition 8.22 are exact combinations of rows and ties, as a symbolic computation shows (Appendix A).

(R) $z_1 - z_2 = (m_1 + m_2)\Pi + F_1 - F_2$, $z_1 - z_c = m_2\Pi + (1 + \frac{m_1}{M_T})F_1 + \frac{m_2}{M_T}F_2$ and $z_2 - z_c = -m_1\Pi + \frac{m_1}{M_T}F_1 + (1 + \frac{m_2}{M_T})F_2$, with their w-analogues; the three forms in (Π, F_1, F_2) are pairwise independent on each coordinate plane, with common kernel $(1, -m_2, m_1)$;

(C) $z_1 - z_2 = (z_1 - z_c) - (z_2 - z_c)$, and likewise in w;

(T) $V_1 = V_2$: the outer bodies exert no torque on the triangle about its centre;

(A) $\operatorname{ord} I_3 = 0$ (Lemma 8.13);

(B) $\sum_{j,k} m_j m_k \delta_{jk}\big(1 + \frac{1}{2}\big(\frac{\xi_k}{z_j - z_k} + \frac{\eta_k}{w_j - w_k}\big)\big) + m_1 m_2 \delta_{12} = I_0 - I_T$, of order exactly 0, from $U = I_0$ and the virial identity;

(V) $V_1 = V_2 = I_T - U_T$, where $\operatorname{ord} I_T = 2$ and $\operatorname{ord} U_T \geq \min(5, -1 + 2\kappa)$ by (16);

(M) for two outer bodies near in the same colour, the relations of Lemma 8.19.

Each regime fixes the orders of F_j, G_j, of the shares of body j in the torque $V_1 - V_2$, in (B), V_1 and V_2, of its forcings of the triangle bodies and of the r^2_{jk}: exactly with nonzero leading coefficient, by lower bounds, by the vanishing of the generic coefficient (of F_j in dipole-equal-F and z-coincident-Φ, of G_j in dipole-equal-G and z-coincident-Ψ), or not at all. Nonvanishing comes from $\mu_j \neq 0$, from the dipole moment $\sum_k m_k \varepsilon_{jk} \xi_k = -2m_a\xi_a$ (a the body of minority sign; of order 1 unless a is central), from $\sum_k m_k s_k^2 > 0$, from the definitions, from $s_k \neq 0$ at a dominant special body, and, for z-inner, from the following observation: off the slice a single s_k has the minority sign, and as $y^{-2}_{jk} = -s_k$, its term alone forms the real or the imaginary part of Φ_j and of Ψ_j. A second computation re-derives these orders in the 12 regimes of the z-side (w by symmetry) from exact first-order expansions with remainder bounds, and checks them, every census constraint and the formulas for distances and four-products by exact linear programming over each regime; it finds one exception (§8.2.7). Valid constraints that the census omits, such as $\alpha_{jc}, \beta_{jc} > -2$, only weaken it.

A vanishing sum of Puiseux series attains its least order twice; a sum of order exactly 0 has least order ≤ 0, attained twice if negative; and in (R), at the least order of Π, F_1, F_2, at most one equation cancels if two of them attain it, and none, one or all three if all do. So each identity becomes a finite disjunction of rational linear systems, with strict and non-strict inequalities, in the orders. The census explores the disjuncts depth first and decides each system exactly over $\mathbb{Q}$ by the simplex method, with free variables and a common slack $0 < s \le 1$ for the strict inequalities.

8.2.5. *The Morse–Bott lemma.* In reduced coordinates $\hat{x}_k = \xi_k/t$, $\hat{y}_k = \eta_k/t$, $D_{kl} = t\delta_{kl}$ ($k, l \in T$) we have $U_T = t^{-1}\hat{U}$, $\hat{U} = \sum_{k<l} m_k m_l D_{kl}$, with $D_{kl} = ((\hat{x}_k - \hat{x}_l)(\hat{y}_k - \hat{y}_l))^{-1/2}$ on the branch of the limit. For $\psi_k = \sum_l m_l D_{kl}^3(\hat{y}_k - \hat{y}_l)$ and $\phi_k = \sum_l m_l D_{kl}^3(\hat{x}_k - \hat{x}_l)$, $\partial\hat{U}/\partial\hat{x}_k = -\frac{m_k}{2}\psi_k$, $\partial\hat{U}/\partial\hat{y}_k = -\frac{m_k}{2}\phi_k$ and, by homogeneity, $\hat{U} = \sum_k m_k\hat{x}_k\psi_k = \sum_k m_k\hat{y}_k\phi_k$. The triangle rows read exactly

$$\psi_k = t^3\hat{y}_k - \hat{g}_k, \qquad \phi_k = t^3\hat{x}_k - \hat{g}_k^z, \tag{15}$$

where $g_k = t^2\sum_j m_j\delta_{jk}^3(w_k - w_j)$ is the crossed forcing of the w-row of k, g_k^z its z-analogue, and the hat is the deviation from the m-weighted mean over T. Let $O = \{(as + u\mathbf{1}, a's + v\mathbf{1})\}$ be the orbit of the zero-multiplier triangle under translations and scalings; the reduced triangle tends to $x_0 = (s, s) \in O$.

Lemma 8.16 (computer-assisted)**.** *At each point $(as, a's)$ of O, $\hat{U} = 0$, $d\hat{U} = 0$, the translations and scalings lie in the Hessian kernel, and on $(\hat{x}_3, \hat{y}_3)$ the Hessian H has entries $3X/(\varrho a^2)$, $X/(\varrho aa')$, $3X/(\varrho a'^2)$, where $\varrho^2 = aa'$ and $X = b^4(b+1)^6/4$ for masses normalized to $b^2, b^2(b+1)^2, (b+1)^2$; its determinant is $8X^2/\varrho^6$, and $(H^{-1})_{\hat{x}\hat{x}} = 3a^2\varrho/(8X) \ne 0$.*

The proof is an exact computation in the field $\mathbb{Q}(b, a, \varrho)$ of rational functions (Appendix A).

Lemma 8.17 (Morse–Bott)**.** *If $x(t) \to x_0$ is a germ of reduced triangles, $\kappa_\psi = \operatorname{ord}\psi(x(t))$ and $\kappa_\phi = \operatorname{ord}\phi(x(t))$, then $\operatorname{ord}\hat{U}(x(t)) \ge 2\min(\kappa_\psi, \kappa_\phi)$, and $\kappa_\psi < \kappa_\phi$ implies $\kappa_\phi \le 2\kappa_\psi$ (and symmetrically).*

Proof. As $s_4 \ne s_5$, the $\hat{x}_3$- and $\hat{y}_3$-directions span a complement of the tangent space of O, which by Lemma 8.16 is thus the Hessian kernel; write points near x_0 as $o(u) + v_1 e_3^x + v_2 e_3^y$, $o(u) \in O$. Taylor's formula gives $\hat{U} = \frac{1}{2}v^T H(u)v + O(|v|^3)$ and $\partial_v\hat{U} = H(u)v + O(|v|^2) = -\frac{m_3}{2}(\psi_3, \phi_3)$ with $H(u)$ invertible, so $\operatorname{ord}\hat{U} \ge 2\operatorname{ord} v = 2\operatorname{ord}\partial_v\hat{U} \ge 2\min(\kappa_\psi, \kappa_\phi)$. If $\kappa_\psi < \kappa_\phi$, the leading coefficient of ψ satisfies $\sum_k m_k\psi_k = 0$ and, as $\hat{U} = \sum_k m_k\hat{x}_k\psi_k$ has order $> \kappa_\psi$, $\sum_k m_k s_k\psi_k = 0$; as $s_4 \ne s_5$, its third component is nonzero, so $\partial_v\hat{U} = t^{\kappa_\psi}(\chi, 0) + o(t^{\kappa_\psi})$ with $\chi \ne 0$ and $\hat{U} = \frac{1}{2}(\partial_v\hat{U})^T H^{-1}\partial_v\hat{U} + O(|v|^3) = \frac{1}{2}(H^{-1})_{\hat{x}\hat{x}}\chi^2 t^{2\kappa_\psi} + o(t^{2\kappa_\psi})$. But $\hat{U} = \sum_k m_k\hat{y}_k\phi_k$ has order $\ge \kappa_\phi$. □

By Lemma 8.17, as the self-terms in (15) have order 3, along every Diagram 5 germ

$$\operatorname{ord} U_T \ge \min(5, -1 + 2\kappa), \qquad \kappa = \min(\operatorname{ord}\hat{g}, \operatorname{ord}\hat{g}^z). \tag{16}$$

Corollary 8.18 (uniform forcing)**.** *Let both outer bodies be near in z, and let κ_* be the least order of the terms $t^2 m_j\delta_{jk}^3(w_k - w_j)$ ($j \in \{1, 2\}$, $k \in T$). Then the coefficient of t^{κ_*} in g_k is the same for $k = 3, 4, 5$; likewise with z and w exchanged.*

Proof. Let $\operatorname{ord}(z_j - z_k) = 1 + e_{jk}$, $e_{jk} \ge 0$. The term of body j in g_k has order $\kappa_{jk} = \frac{1}{2}(1 - \beta_{jc} - 3e_{jk})$, in g_k^z the order $\frac{1}{2}(3 - 3\beta_{jc} - e_{jk}) = 2\kappa_{jk} + \frac{1}{2}(1 - \beta_{jc} + 5e_{jk})$. As $\beta_{jc} > -2$ and $e_{jk} = 0$ for some k, $\kappa_* < \frac{3}{2}$. Unequal coefficients would give $\kappa_\psi = \kappa_*$ (as $3 > \kappa_*$), while $\hat{g}^z$ has order $> 2\kappa_*$ (as $\beta_{jc} < 1$) and $3 > 2\kappa_*$; so $\kappa_\phi > 2\kappa_\psi$, with $\kappa_\psi > 0$ as $\psi \to 0$, against Lemma 8.17. □

Lemma 8.19 (computer-assisted)**.** *Let both outer bodies be near in z (or, mirrored, in w). (a) A forcing from one source, $k \mapsto c(\theta - s_k)^{-3/2}$ with $c \ne 0$, or supported on fewer than three*

triangle bodies, is not uniform. (b) Two z-dominant-coincidence bodies share their special body; so $\alpha_{12} > 1$. *(c) A z-coincident-*Φ *or z-coincident-*Ψ *body does not occur with a z-dominant-coincidence body. (d) A z-inner body does not occur with a z-dominant-central body. (e) Two z-dominant-central bodies have the same g; their forcings of body* 3, *of orders* $\varphi_j = \operatorname{ord}(\delta^3_{j3}(w_3 - w_j)) = -\frac{3}{2}(1+e_j) - \frac{1}{2}\beta_{jc}$, *tie and cancel;* $g > 3 + \varphi_1$; *and* $1 + e_j = g < \alpha_{jc}$ *with* $\beta_{jc} < -g$ *for both j is impossible.*

The lemma follows from Corollary 8.18 by exact computations: (a) through the resultant $9(s_4 - s_5)^2 \cdot \frac{1}{2}\sum_{k<l}(s_k - s_l)^2 \neq 0$ of $(\theta - s_3)^3 - (\theta - s_4)^3$ and $(\theta - s_3)^3 - (\theta - s_5)^3$, (c) through twelve gcd computations over $\mathbb{Q}(\omega)$ at the points $(s_k - \omega s_{k'})/(1-\omega)$, $\omega^3 = 1 \neq \omega$, followed by exact real-root counting in b (Appendix A); (d) is the definition, and the last case of (e) falls as in Proposition 8.22.

Proof of the tie and of $g > 3 + \varphi_1$ *in (e).* The term of body j in g_k has order $2 + \varphi_j$ for $k = 3$, and order $\frac{1}{2} - \frac{1}{2}\beta_{jc} > 2 + \varphi_j$ for $k = 4, 5$, where $\operatorname{ord}(z_j - z_k) = 1$ and $e_j > 0$. So the leading crossed forcing is supported on body 3, and by Corollary 8.18 and (a) the two forcings of body 3 have one order $\varphi := \varphi_1 = \varphi_2$ and cancel; hence $\operatorname{ord}\hat{g}_3 > 2 + \varphi$. The z-forcing of body 3 by body j has order $2 + \varphi + (1 + e_j) - \beta_{jc} > 2 + \varphi$, as $\beta_{jc} < 1$, and the self-terms in (15) have order $\geq 3 > 2 + \varphi$, as $\varphi < -\frac{1}{2}$ by $\beta_{jc} > -2$. So $\operatorname{ord}\psi_3, \operatorname{ord}\phi_3 > 2 + \varphi$. In the proof of Lemma 8.17, $\partial_v \hat{U} = H(u)v + O(|v|^2) = -\frac{m_3}{2}(\psi_3, \phi_3)$ with $H(u)$ invertible and $v \to 0$, so $\operatorname{ord} v \geq \min(\operatorname{ord}\psi_3, \operatorname{ord}\phi_3) > 2 + \varphi$. On the slice $s_3 = 0$, so the mass-centred point $o(u)$ of the orbit has third $\hat{x}$-component 0, and mass-centring gives $\hat{x}_3 = (1 - m_3/M_T)v_1$. Therefore $g = \operatorname{ord}\xi_3 = 1 + \operatorname{ord}\hat{x}_3 \geq 1 + \operatorname{ord} v > 3 + \varphi$. □

8.2.6. *The census and over-rotation.*

Lemma 8.20 (computer-assisted)**.** *For each of the* 111 *pairs of regimes and each of the* 27 *patterns, the conditions of the frame and of the two regimes together with (R), (C), (T), (A), (B), (V), (M) are infeasible, except for the pairs and patterns of Table 3 and for pairs containing an empty regime. For the five monopole pairs of §8.2.7 this holds with the rows corrected there.*

The proof is a finite computation: for each of the 111 pairs of regimes and each of the 27 patterns, the finite disjunction of rational linear systems obtained above is decided by exact linear programming over $\mathbb{Q}$, with the orders checked by the second computation and, for the five monopole pairs, with the corrected rows (Appendix A).

pair of regimes (up to the exchanges)	surviving patterns
(monopole, monopole); (monopole, X) for X = dipole-z, dipole-equal, dipole-equal-F; (dipole-z, dipole-z); (dipole-equal, X) for X = dipole-equal, dipole-equal-F; (dipole-equal-F, X) for X = dipole-equal-F, dipole-equal-G	RRR
(X, Y) for $X, Y \in$ {z-coincident, z-coincident-Φ, z-coincident-Ψ}	RRR
(z-inner, z-inner)	RRR
(z-inner, X) for $X \in$ {z-coincident, z-coincident-Φ, z-coincident-Ψ}	RCR
(z-dominant-coincidence, z-dominant-coincidence)	CEE, CRR
(z-dominant-central, z-dominant-central)	RRR, RRC, RCR, CEE, CRR
every other pair of nonempty regimes	none

TABLE 3. The census of Lemma 8.20. A pattern lists the signs of ρ_{12}, ρ_{1c}, ρ_{2c}: C positive, R zero, E negative.

Each four-product contains a triangle edge, of order 2, and its order is the sum of its edge orders: $\alpha_{12}+\beta_{12}$ for r_{12}^2; for r_{jk}^2, $\alpha_{jc}+\beta_{jc}$ if j is far and $1+\beta_{jc}$ (or $1+e_j+\beta_{jc}$ at the special body) if j is near in z.

Lemma 8.21 (computer-assisted)**.** *For every pair and pattern of Table 3 other than (z-inner, z-inner) with RRR, adding* $\operatorname{ord} p_{ab}<0$ *for any one of the ten four-products makes the constraints infeasible. For (z-inner, z-inner) with RRR exactly the six four-products* p_{jk}, $j\in\{1,2\}$, $k\in T$, *can have negative order.*

The proof is a computation of the same kind: for each surviving pair and pattern and each of the ten four-products, 260 cases in all, we add $\operatorname{ord} p_{ab}<0$, with $\operatorname{ord} p_{ab}$ expressed through the edge orders as above, and decide the resulting systems by exact linear programming over $\mathbb{Q}$ (Appendix A).

By Lemmas 8.20 and 8.21 (and §8.2.7 for pairs with a monopole), a germ with an unbounded four-product would be *over-rotated*: both outer bodies inner in one colour, say z, and all $\rho_P=0$, so that $\beta_{jc}=-\alpha_{jc}<-1$ and the crossed distances, of order $\frac12(1+\beta_{jc})<0$, are unbounded.

Proposition 8.22. *Let* $m\in\mathbb{R}^5_{>0}$. *No Diagram 5 germ has both outer bodies z-inner and pattern RRR, nor both w-inner and pattern RRR.*

Proof. Here $\beta_{jc}=-\alpha_{jc}<-1$ and every z_j-z_k ($k\in T$) has order 1; as $\delta^3_{jk}(w_j-w_k)=\delta_{jk}/(z_j-z_k)$, $\operatorname{ord} G_j\ge-\frac32-\frac12\beta_{jc}=\beta_{jc}+\frac32(\alpha_{jc}-1)>\beta_{jc}$. By (R) in w, the pair-centre identity $m_1(w_1-w_c)+m_2(w_2-w_c)=\frac{M}{M_T}(m_1G_1+m_2G_2)$ holds. If $\beta_{1c}<\beta_{2c}$ (the other case is symmetric), its left side has order $-\alpha_{1c}$ and its right side order at least $-\frac32+\frac12\alpha_{2c}$, larger by $(\alpha_{1c}-1)+\frac12(\alpha_{2c}-1)>0$. So $\beta_{1c}=\beta_{2c}=:\beta$ and $m_1\hat w_1+m_2\hat w_2=0$. Next, $z_k-z_j=ts_k(1+o(1))$ and $w_k-w_j=-\hat w_j t^\beta(1+o(1))$, so $(t^2m_j\delta^3_{jk}(w_k-w_j))^2=-(m_j^2/\hat w_j)s_k^{-3}t^{1-\beta}(1+o(1))$, and the coefficient of $t^{(1-\beta)/2}$ in g_k is $(c_1\varepsilon'_{1k}+c_2\varepsilon'_{2k})s_k^{-3/2}$, with fixed branches, $c_j^2=-m_j^2/\hat w_j$ and signs ε'_{jk}. By Corollary 8.18 it is one value f for $k=3,4,5$. If $f\neq0$, then $f^2s_k^3=(c_1\varepsilon'_{1k}+c_2\varepsilon'_{2k})^2$ depends only on $\varepsilon'_{1k}\varepsilon'_{2k}$, so two of the cubes s_k^3 coincide, which is false. Hence $f=0$ and $c_1^2=c_2^2$, that is, $m_1^2\hat w_2=m_2^2\hat w_1$. The two linear relations for $(\hat w_1,\hat w_2)$ have determinant $m_1^3+m_2^3>0$, so $\hat w_1=\hat w_2=0$, which is absurd. □

No reality input is used here.

8.2.7. *The central-body slice.* On the slice body 3 sits at the centre of the zero-multiplier triangle; this gives the dominant-central regimes, voids z-inner, and causes the one exception found by the second computation: for a monopole whose minority crossed sign is on body 3 the dipole moment $-2m_3\xi_3$ has order $g>1$, so for its shares in V_1, V_2 the orders the census uses as exact are only lower bounds (the coefficients there vanish). Its other rows are those of every monopole: the leading coefficients of F_j, G_j and of its term in (B) are multiples of $\mu_j\neq0$, and its torque row is a lower bound. The correction can matter only for a monopole facing a monopole, a z-coincident (or -Φ, -Ψ) or a z-dominant-coincidence body (elsewhere the masses avoid the slice, the regime is void on it, or there is no virial row); recomputed with it, these five pairs keep their surviving patterns.

Remark 8.23. Suppose that we drop (T), (B), (A), (V), and keep of (M) only its exact part (z-inner is void on the slice; two z-dominant-central bodies share g and satisfy the tropical condition of $z_1-z_2=(z_1-z_3)-(z_2-z_3)$). Then, among all pairs of nonempty regimes, patterns and four-products, a four-product of negative order is feasible only for (z-inner, z-inner) with RRR, (z-dominant-coincidence, z-dominant-coincidence) with EEE or REE, and (z-dominant-central, z-dominant-central) with RRR, RRC, RCR, CEE, CRR or CCC. This shows where (A), (B) and the Morse–Bott relations are needed. None of these pairs contains a monopole, so the slice exception cannot create an unbounded four-product even before the correction. We do not use this.

8.2.8. *Proof of Theorem 8.12.* Let γ be a Diagram 5 germ with an unbounded four-product on an irreducible curve $\Gamma \subset A_m$ containing a real point. By Lemma 8.13, (A) and (B) hold along γ; the other conditions of §8.2.4 hold along every germ, and all of them are necessary conditions on the orders. By Lemma 8.15 and the exchanges, the outer bodies form one of the 111 pairs of regimes, both nonempty, and some p_{ab} has negative order (Definition 3.4). If the pair contains a monopole, Lemma 8.20, with the rows of §8.2.7 for the five slice-sensitive pairs, leaves only pairs of two far bodies with pattern RRR (Table 3). Then $\rho_{12} = \rho_{1c} = \rho_{2c} = 0$, so r_{12}^2 and every r_{jk}^2 with $j \in \{1,2\}$, $k \in T$ have order 0, and every four-product, which contains a triangle edge of order 2, has order at least 2: a contradiction. Otherwise every order used by the census is confirmed by the second computation, and Lemmas 8.20 and 8.21 leave only (z-inner, z-inner) with RRR or its mirror, which Proposition 8.22 excludes. □

Remark 8.24. We do not prove the *outer-triple regularity* of Diagram 5 germs, that is, that their pattern is RRR: the census leaves RCR, CEE, CRR and, on the slice, RRC, RCR, CEE, CRR. Theorem 8.12 does not use it.

9. Diagram 6

9.1. **The diagram and the butterfly chart.** Diagram 6 is the *butterfly* of [1, §7.4], a z-triangle and a w-triangle (the *wings*) with one common vertex, with both wings circled. In the numbering of [1, §8.6] the z-wing $\{1,2,5\}$ has the z-strokes $12, 15, 25$ and z-circles at $1, 2$; the w-wing $\{3,4,5\}$ has the w-strokes $34, 35, 45$ and w-circles at $3, 4$; vertex 5 is not circled, and no stroke joins $\{1,2\}$ to $\{3,4\}$. The butterfly has three circlings, Diagrams 6–8, and 5 is never circled [1, §7.4] (Fact AK4); in Diagram 6 both wings are circled, and a circled z-wing has z-circles exactly at $1, 2$ (Rules 1a, 1d), and likewise for w.

Permutations of the bodies (with the masses) and $z \leftrightarrow w$ are symmetries of (1). The thirty coloured labels of Diagram 6 (a common vertex and an ordered pair of wings) are images of this numbering, and $z \leftrightarrow w$ composed with (13)(24) preserves it. The chart below uses rotation invariants only, so the gauge $z_{12} = w_{12}$ of A_m plays no role, and it suffices to treat this numbering at arbitrary positive masses.

Definition 9.1. The *butterfly chart* has the 21 coordinates t, $Z_1, \dots, Z_5$, $W_1, \dots, W_5$, D_{kl}, related to (1) by $(z_k, w_k) = (t^{-1}Z_k, tW_k)$ for $k = 1, 2$, $(z_k, w_k) = (tZ_k, t^{-1}W_k)$ for $k = 3, 4$, $(z_5, w_5) = (tZ_5, tW_5)$, $\delta_{kl} = tD_{kl}$ on the *crossed pairs* $13, 14, 23, 24$ and $\delta_{kl} = D_{kl}$ on the *wing pairs* $12, 15, 25, 34, 35, 45$. Its *rows* are the z- and w-equations $z_k - \sum_{l \neq k} m_l \delta_{kl}^3 (z_k - z_l) = 0$ (likewise for w) of body k, multiplied by (t, t^{-1}) for $k = 1, 2$, by (t^{-1}, t) for $k = 3, 4$ and by (t, t) for $k = 5$ and denoted R_k^z, R_k^w; the ten distance equations $\delta_{kl}^2 z_{kl} w_{kl} - 1 = 0$; and the gauges $Z_2 - Z_1 - 1 = 0$, $W_4 - W_3 - 1 = 0$. We write $F = 0$ for these 22 rows and $\mathcal{B} = F^{-1}(0) \cap \{t = 0\}$ for the *boundary fibre*. (Z_k, W_k are not the terms $Z_{kl} = z_{kl}\delta_{kl}^3$ of [1, §3].)

Lemma 9.2 (computer-assisted)**.** *Let $m \in \mathbb{R}^5_{>0}$.*

(i) *The rows are polynomials in the 21 coordinates, of degree at most 5 in t.*

(ii) *Let $(z, w, \delta) \in A_m$ (so $z_{12}w_{34} \neq 0$), choose t with $t^2 = (z_{12}w_{34})^{-1}$ and put $\nu = (tz_{12})^{-1}$. The chart point of the rotated solution $(\nu z, \nu^{-1} w, \delta)$ has $Z_k = z_k/z_{12}$, $W_k = w_k z_{12}$ $(k = 1, 2)$, $Z_k = z_k w_{34}$, $W_k = w_k/w_{34}$ $(k = 3, 4)$, $Z_5 = z_5 w_{34}$, $W_5 = w_5 z_{12}$, $D_{kl} = \delta_{kl}/t$ on crossed pairs and $D_{kl} = \delta_{kl}$ on wing pairs, and satisfies $F = 0$: each coordinate row is $t^{\pm 1}\nu^{\pm 1}$ times an equation of (1), the distance rows are those of (1), and both gauges hold.*

(iii) *At $t = 0$ the crossed distance rows read $D_{kl}^2 Z_k W_l = -1$, and the wing distance rows read $D_{12}^2(Z_2 - Z_1)(W_2 - W_1) = 1$, $-D_{15}^2 Z_1 (W_5 - W_1) = 1$, $-D_{25}^2 Z_2 (W_5 - W_2) = 1$, $D_{34}^2 (Z_4 - Z_3)(W_4 - W_3) = 1$, $-D_{35}^2 (Z_5 - Z_3) W_3 = 1$, $-D_{45}^2 (Z_5 - Z_4) W_4 = 1$. So every D_{kl} is nonzero on $\mathcal{B}$.*

(iv) *Identically,* $m_1R_1^z+m_2R_2^z+m_5R_5^z+t^2(m_3R_3^z+m_4R_4^z) = m_1Z_1+m_2Z_2+t^2(m_3Z_3+m_4Z_4+m_5Z_5)$*, and likewise after* $z \leftrightarrow w$, $1 \leftrightarrow 3$, $2 \leftrightarrow 4$*. Hence on* $\mathcal{B}$*:* $Z_1 = -m_2/(m_1+m_2)$, $Z_2 = m_1/(m_1+m_2)$, $W_3 = -m_4/(m_3+m_4)$, $W_4 = m_3/(m_3+m_4)$.

(v) *At* $t = 0$ *the rows* $R_1^z, R_2^z, R_5^z, R_1^w, R_2^w$*, the distance rows of* $12, 15, 25$ *and the first gauge involve only the* z-wing coordinates $Z_1, Z_2, W_1, W_2, W_5, D_{12}, D_{15}, D_{25}$*; the remaining rows except the crossed distance rows involve only the* w-wing coordinates $Z_3, Z_4, Z_5, W_3, W_4, D_{34}, D_{35}, D_{45}$*; and* $D_{13}, D_{14}, D_{23}, D_{24}$ *occur only in the crossed distance rows.*

We check (i)–(iv) by exact expansion over $\mathbb{Q}$, with the masses and the coordinates as indeterminates; for (ii) we substitute the chart point of the rotated solution into the 22 rows (Appendix A). Item (v) holds because every term coupling the two wings carries a positive power of t.

9.2. **Coverage.**

Proposition 9.3 (Coverage)**.** *Let* $m \in \mathbb{R}^5_{>0}$ *and let* $\gamma(s)$, $s \to 0$*, be a Diagram-6 germ, in the numbering above, on a curve in* A_m*. Along* γ *define* t *and* ν *as in Lemma 9.2(ii), and let* $\tilde\gamma(s)$ *be the chart point of the rotated solution. Then* $t \to 0$*, every chart coordinate is bounded along* $\tilde\gamma$*, and* $\tilde\gamma(s)$ *converges to a point* $p \in \mathcal{B}$ *at which* Z_1, Z_2, W_3, W_4 *and all ten* D_{kl} *are nonzero.*

Proof. Normalize as in [1, p. 547] so that the largest components of $\mathcal{Z}$ and $\mathcal{W}$ both have modulus ϵ^{-2}; by Lemma 3.3 this is realized along γ, with ϵ comparable to a positive power of s. Let ord be the order in ϵ ($\operatorname{ord} f > \alpha$ means $f\epsilon^{-\alpha} \to 0$). The chart coordinates are invariant under $z \mapsto az$, $w \mapsto a^{-1}w$, so we compute in this frame.

By the circles, $\operatorname{ord} z_1 = \operatorname{ord} z_2 = \operatorname{ord} w_3 = \operatorname{ord} w_4 = -2$, and the uncircled $z_3, z_4, z_5, w_1, w_2, w_5$ have order > -2; so $\operatorname{ord} z_{15} = \operatorname{ord} z_{25} = -2$. A z-stroke kl means $\operatorname{ord}(z_{kl}\delta_{kl}^3) = \operatorname{ord}(z_{kl}^{-1/2}w_{kl}^{-3/2}) = -2$, that is, $\operatorname{ord} z_{kl} + 3\operatorname{ord} w_{kl} = 4$. So $\operatorname{ord} w_{15} = \operatorname{ord} w_{25} = 2$, hence $\operatorname{ord} w_{12} \ge 2$, and the stroke 12 gives $\operatorname{ord} z_{12} \le -2$; with $\operatorname{ord} z_{12} \ge -2$ (Estimate 1, Fact AK2) we get $\operatorname{ord} z_{12} = -2$, $\operatorname{ord} w_{12} = 2$. Thus $r_{kl}^2 = z_{kl}w_{kl}$ and δ_{kl} have order 0 on the three z-wing pairs (maximal edges, $r \approx 1$ as in Estimate 2), and symmetrically on the w-wing. On a crossed pair z_{kl} and w_{kl} are differences of a circled and an uncircled coordinate, so both have order -2 and $\operatorname{ord}\delta_{kl} = 2$. In the w-equation $w_1 = \sum_{l\neq1} m_l\delta_{1l}^3(w_1 - w_l)$ the terms have orders 2 ($l = 2, 5$) and 4 ($l = 3, 4$); so $\operatorname{ord} w_1 \ge 2$, likewise $\operatorname{ord} w_2, \operatorname{ord} z_3, \operatorname{ord} z_4 \ge 2$, and $w_5 = w_1 + w_{15}$, $z_5 = z_3 + z_{35}$ have order ≥ 2. (These are the estimates $A_w, B_z, m_1w_1 + m_2w_2, w_5, B_w, A_z \preccurlyeq \epsilon^2$ of [1, §8.6].)

Finally $\operatorname{ord} t = -\frac12(\operatorname{ord} z_{12} + \operatorname{ord} w_{34}) = 2$ and $\operatorname{ord}\nu = -\operatorname{ord} t - \operatorname{ord} z_{12} = 0$. By Lemma 9.2(ii), Z_1, Z_2, W_3, W_4 and all ten D_{kl} have order exactly 0 (crossed pairs: $2 + \frac12(-2-2) = 0$), and $W_1, W_2, W_5, Z_3, Z_4, Z_5$ have order ≥ 0. Bounded Puiseux series in s converge, so $\tilde\gamma(s) \to p \in \mathcal{B}$, and the coordinates of order exactly 0 have nonzero limits. □

In particular $D_{12} = \delta_{12}$ and $D_{34} = \delta_{34}$ have finite nonzero limits along every Diagram-6 germ, at every mass.

9.3. **The first condition.** Put $\Sigma = \sum_{k\in\{1,2\},\, l\in\{3,4\}} m_km_lD_{kl}^3Z_kW_l$.

Lemma 9.4 (computer-assisted)**.** *There is a polynomial* Q *in the chart coordinates, with coefficients polynomial in* m*, such that identically*

$$\sum_{k=1,2} m_k(W_kR_k^z - Z_kR_k^w) + m_5W_5R_5^z = -t\,\Sigma + t^2Q. \tag{17}$$

Expanding the left-hand side of (17) in exact arithmetic over $\mathbb{Q}$, with the masses and the chart coordinates as indeterminates, we find that its coefficients of t^0 and t^1 are 0 and $-\Sigma$ (Appendix A). With ρ_k^z, ρ_k^w the left-hand sides of (1), the pair torque $\sum_{k=1,2} m_k(w_k\rho_k^z - z_k\rho_k^w)$ equals $\sum_{k=1,2}\sum_{l=3,4,5} m_km_l\delta_{kl}^3(w_kz_l - z_kw_l)$ (the terms with $l \in \{1,2\}$ cancel), which is $2i$ times (31) of [1, §8.6]. In the chart it is the first sum in (17); its terms with $l = 3, 4$ are

$-t\Sigma + t^5 \sum m_k m_l D_{kl}^3 W_k Z_l$, those with $l = 5$ are $\sum_{k=1,2} m_k m_5 D_{k5}^3 (t^2 W_k Z_5 - Z_k W_5)$, and $R_5^z - (m_1 D_{15}^3 Z_1 + m_2 D_{25}^3 Z_2)$ is divisible by t^2.

Proposition 9.5 (First condition)**.** *Let $m \in \mathbb{R}^5_{>0}$, let γ be a Diagram-6 germ in the numbering above, and let p be its limit (Proposition 9.3). Then:*

(i) $\Sigma(p) = 0$, *and* $\Sigma(p) = -(m_1 m_3 D_{13} + m_1 m_4 D_{14} + m_2 m_3 D_{23} + m_2 m_4 D_{24})(p)$;
(ii) $m_1 = m_2$ *and* $m_3 = m_4$; *this is relation (32) of* [1, §8.6];
(iii) $(D_{13}, D_{14}, D_{23}, D_{24})(p) = (2i\eta_{13},\, 2\eta_{14},\, 2\eta_{23},\, 2i\eta_{24})$ *with signs* $\eta_{kl} \in \{\pm 1\}$ *satisfying* $\eta_{14} = -\eta_{23}$ *and* $\eta_{13} = -\eta_{24}$.

In another numbering, wings $\{a, b, c\}$ and $\{d, e, c\}$ force $m_a = m_b$ and $m_d = m_e$.

Proof. All rows vanish along $\tilde{\gamma}$, so (17) gives $t\Sigma = t^2 Q$, and $t \neq 0$ gives $\Sigma = tQ$. By Proposition 9.3, Q is bounded and $t \to 0$, so $\Sigma(p) = 0$. By Lemma 9.2(iii), $D_{kl}^2 Z_k W_l = -1$ at p for the crossed pairs, so $\Sigma(p) = \sum m_k m_l D_{kl}(D_{kl}^2 Z_k W_l) = -\sum m_k m_l D_{kl}$; this is (i).

By Lemma 9.2(iv), $Z_1 W_3 = m_2 m_4 / s_0^2$, $Z_1 W_4 = -m_2 m_3 / s_0^2$, $Z_2 W_3 = -m_1 m_4 / s_0^2$ and $Z_2 W_4 = m_1 m_3 / s_0^2$ at p, where $s_0 > 0$, $s_0^2 = (m_1 + m_2)(m_3 + m_4)$. Write $m_k = \mu_k^2$ with $\mu_k > 0$. Then $D_{kl}^2 = -1/(Z_k W_l)$ gives

$$D_{13} = \frac{i\eta_{13} s_0}{\mu_2 \mu_4}, \quad D_{14} = \frac{\eta_{14} s_0}{\mu_2 \mu_3}, \quad D_{23} = \frac{\eta_{23} s_0}{\mu_1 \mu_4}, \quad D_{24} = \frac{i\eta_{24} s_0}{\mu_1 \mu_3}$$

with signs $\eta_{kl} \in \{\pm 1\}$, and

$$\Sigma(p) = -\frac{s_0}{\mu_1 \mu_2 \mu_3 \mu_4}(P_{\mathrm{re}} + iP_{\mathrm{im}}), \qquad P_{\mathrm{re}} = \eta_{14}\mu_1^3\mu_4^3 + \eta_{23}\mu_2^3\mu_3^3, \quad P_{\mathrm{im}} = \eta_{13}\mu_1^3\mu_3^3 + \eta_{24}\mu_2^3\mu_4^3.$$

Up to the factor $\pm i$, $P_{\mathrm{re}} + iP_{\mathrm{im}}$ is the left-hand side of (12) of [1, §5.1] written in the μ_k, and each of its eight sign choices arises from exactly two of the sixteen sign classes η. Since the μ_k and s_0 are real, $\Sigma(p) = 0$ forces $P_{\mathrm{re}} = P_{\mathrm{im}} = 0$. If $\eta_{14} = \eta_{23}$, then $P_{\mathrm{re}} \neq 0$, and if $\eta_{13} = \eta_{24}$, then $P_{\mathrm{im}} \neq 0$; this excludes twelve classes. In the other four, $P_{\mathrm{re}} = \eta_{14}(x - y)(x^2 + xy + y^2)$ with $x = \mu_1\mu_4$, $y = \mu_2\mu_3$ and $x^2 + xy + y^2 > 0$, so $\mu_1\mu_4 = \mu_2\mu_3$; likewise $P_{\mathrm{im}} = 0$ gives $\mu_1\mu_3 = \mu_2\mu_4$. Then $\mu_3(\mu_1\mu_4 - \mu_2\mu_3) - \mu_4(\mu_1\mu_3 - \mu_2\mu_4) = \mu_2(\mu_4 - \mu_3)(\mu_4 + \mu_3)$ vanishes, so $\mu_3 = \mu_4$ and then $\mu_1 = \mu_2$: this is (ii). With $m_1 = m_2$ and $m_3 = m_4$ the displayed values of D_{kl} become those of (iii). □

At $m_1 = m_2$, $m_3 = m_4$ the expression $\Sigma(p)$ vanishes for exactly the four classes of (iii). Albouy and Kaloshin [1, §8.6] obtain (32) "as in 5.1" from the estimate $S \preccurlyeq \epsilon^4$ in (31); the proof above makes this exact in the chart.

9.4. **The finite wing atlas.** From now on $m = (a, a, b, b, c)$ with $a, b, c > 0$.

Lemma 9.6 (computer-assisted)**.** *The points of $\mathcal{B}$ are exactly the points with $Z_1 = W_3 = -\frac{1}{2}$, $Z_2 = W_4 = \frac{1}{2}$,*

$$(D_{12}, D_{15}, D_{25}) = (d, \lambda d, \omega\lambda d),$$
$$W_1 = -d(a + 2c\lambda), \quad W_2 = d(a + 2c\omega\lambda), \quad W_5 = ad(4\lambda^{-2} - 1),$$
$$(D_{34}, D_{35}, D_{45}) = (e, \kappa e, \omega'\kappa e),$$
$$Z_3 = -e(b + 2c\kappa), \quad Z_4 = e(b + 2c\omega'\kappa), \quad Z_5 = be(4\kappa^{-2} - 1),$$
$$(D_{13}, D_{14}, D_{23}, D_{24}) = (2i\eta_{13},\, 2\eta_{14},\, 2\eta_{23},\, 2i\eta_{24}), \quad \eta_{kl} \in \{\pm 1\},$$

where

(18)
$$\omega^3 = 1, \quad \lambda^2 = 2(1 + \omega), \quad (2a + c\lambda^3)d^3 = 1, \qquad \omega'^3 = 1, \quad \kappa^2 = 2(1 + \omega'), \quad (2b + c\kappa^3)e^3 = 1.$$

For each ω both roots λ are nonzero. The scale denominator $2a + c\lambda^3$ vanishes only for $(\omega, \lambda) = (1, -2)$ and $a = 4c$: for $\omega = 1$, $\lambda = \pm 2$; for $\omega \neq 1$, $\lambda^6 = -8$ and the two denominators have product $4a^2 + 8c^2$. Hence a wing has 18 classes (ω, λ, d) if $a \neq 4c$ and 15 if $a = 4c$, and $\mathcal{B}$

has $18 \cdot 18 \cdot 16 = 5184$ *points if* $a \neq 4c \neq b$, 4320 *if exactly one of* a, b *equals* $4c$, *and* 3600 *if* $a = b = 4c$.

The proof is an exact computation over the field $\mathbb{Q}(i)(a, b, c)$, with the rows reduced modulo (18) by a Gröbner basis, and follows the argument below (Appendix A). By Lemma 9.2(iv),(v), $Z_1 = -\frac{1}{2}$, $Z_2 = \frac{1}{2}$, and $\mathcal{B}$ is the product of the two wing fibres and the crossed roots. On the z-wing put $\lambda = D_{15}/D_{12}$ and $\omega = D_{25}/D_{15}$. At $t = 0$, $2(R_1^z + R_2^z)/c = D_{15}^3 - D_{25}^3$ gives $\omega^3 = 1$; the wing distance rows give $D_{12}^{-2} = W_2 - W_1 = 2D_{15}^{-2} + 2D_{25}^{-2}$, that is, $\lambda^2 = 2(1 + \omega)$; and $2R_1^z = (2a + c\lambda^3)d^3 - 1$. The row R_1^w and the distance rows of $12, 15$ are linear in (W_1, W_2, W_5) with determinant $d^4\lambda^2/2 \neq 0$; conversely the displayed point satisfies all eight z-wing rows modulo (18). The w-wing follows by the wing exchange below.

Germs at infinity are complex (Remark 3.7), so all eighteen classes are needed, not only the positive real class $(\omega, \lambda) = (1, 2)$, $D_{15} = D_{25} = 2D_{12}$.

Definition 9.7. A *label* is a point of $\mathcal{B}$, that is, a choice of (ω, λ, d), (ω', κ, e) and η in Lemma 9.6. The wing *shapes* (ω, λ) and (ω', κ) are *Gaussian* if $\omega = 1$ (so $\lambda = \pm 2$) and *primitive* otherwise (so $\lambda^4 - 2\lambda^2 + 4 = 0$); each wing has two Gaussian and four primitive shapes. A label is *corank-one* if $\eta_{14} = -\eta_{23}$ and $\eta_{13} = -\eta_{24}$, and *regular* otherwise. For $a \neq 4c \neq b$ there are $12 \cdot 324 = 3888$ regular and $4 \cdot 324 = 1296$ corank-one labels.

Symmetries. At the masses (a, a, b, b, c), $F^{-1}(0)$ is preserved by the exchange of 1 and 2 combined with $t \mapsto it$, $(Z_1, Z_2, W_1, W_2, W_5) \mapsto (-Z_2, -Z_1, -W_2, -W_1, -W_5)$ and $D_{kl} \mapsto -iD_{kl}$ on crossed pairs, by $(t, D_{13}, D_{14}, D_{23}, D_{24}) \mapsto -(t, D_{13}, D_{14}, D_{23}, D_{24})$, and by complex conjugation; the *wing exchange* ($z \leftrightarrow w$, $1 \leftrightarrow 3$, $2 \leftrightarrow 4$) maps the chart at (a, a, b, b, c) to that at (b, b, a, a, c). On the 576 *states* (z-wing shape, w-wing shape, η) these maps have 29 orbits, 21 regular and 8 corank-one (Appendix A). The computations below express the masses through the scales, $a = \frac{1}{2}(d^{-3} - c\lambda^3)$, $b = \frac{1}{2}(e^{-3} - c\kappa^3)$, so one symbolic computation covers the nine choices of (d, e) of a state at every positive (a, b, c).

Remark 9.8 (Regular labels). Regular labels are already excluded by Proposition 9.5(iii). Independently, DF has rank 21 at every regular label, which is therefore an isolated point of $F^{-1}(0)$: at one state of each regular orbit the minor omitting R_1^z is a unit in $\{\pm 1, \pm i, \pm 1 \pm i\}$ times $-\frac{81}{128}a^2b\,d^2e^4H(\lambda)H(\kappa)(2a + 4c\lambda)/(2a + c\lambda^3)$, where $H(x) = x^3(x^2 - 2)^2((x^2 - 2)^2 + 4)$, and this is nonzero on finite labels. The minors are computed in exact arithmetic over the field $\mathbb{Q}(i, \sqrt{2}, \sqrt{3})(c, d, e)$, and their nonvanishing is checked with resultants and norms (Appendix A).

Lemma 9.9 (computer-assisted)**.** *Let* p *be a corank-one label.*

(i) *The* 20 *rows of* F *other than* R_2^w *and* R_4^w, *together with the coordinate* t, *have invertible derivative at* p. *Hence near* p *the common zeros of these* 20 *rows form a smooth curve* C_p, *the graph of an analytic map of* t.

(ii) *Along* C_p, $R_2^w = c_k t^k + O(t^{k+1})$ *with* $c_k \neq 0$, *where* k *and* c_k *are as in Table 4.*

(iii) *Consequently* p *is an isolated point of* $F^{-1}(0)$.

The proof of (i) and (ii) is an exact computation at one representative state of each of the eight corank-one orbits, over the field $\mathbb{Q}(i)(c, d, e)$ if both wings are Gaussian and over its extension by a root κ of $\kappa^4 - 2\kappa^2 + 4$ otherwise (Appendix A). In (i) the derivative is reduced to the Schur complement of the diagonal block of the distance rows. For two Gaussian wings its determinant (rows multiplied by d^3e^3) is $82944\,d^{61}e^{59}(cd^3 - \frac{\lambda}{16})(ce^3 - \frac{\kappa}{16})$, with $cd^3 - \frac{\lambda}{16} = -\frac{\lambda a}{8(2a + c\lambda^3)} \neq 0$. Otherwise the complement, after removal of the unit row of t, is block diagonal with one 5×5 block per wing (Lemma 9.2(v)); we check that an off-diagonal block vanishes and evaluate both diagonal blocks. For one Gaussian and one primitive wing the two determinants are nonzero multiples of $8cd^3 - \frac{\lambda}{2} = -\lambda A/(2A + \lambda^3)$ and of a polynomial in ce^3 and κ whose product over the primitive κ is $(\frac{9}{16})^4 \cdot 64B^4(B^4 - 8B^2 + 64)/(B^2 + 2)^4$, where $B^4 - 8B^2 + 64 = (B^2 - 4)^2 + 48$. This product and those in Table 4 are obtained from resultants with $\kappa^4 - 2\kappa^2 + 4$. For (ii) the

wing shapes $(z,\, w)$	labels	k	c_k is a nonzero multiple of
Gaussian, Gaussian	144	4	$Q_{2,2} = -\frac{2a+2b+c}{60(a+4c)(b+4c)}$, $Q_{2,-2} = \frac{2a+2b+c}{4(a+4c)(b-4c)}$, $Q_{-2,-2} = \frac{2a+2b+c}{68(a-4c)(b-4c)}$ for $(\lambda,\kappa) = (2,2),(2,-2),(-2,-2);\ (-2,2)$ by wing exchange
one Gaussian, one primitive	576	3	$P = A\kappa^3 - 4A\kappa + B\kappa^3 - B\kappa^2 - 4B\kappa + B$ for $\lambda = \pm 2$, κ primitive; product over the primitive κ: $9(8A^2 + 16AB + 9B^2)^2$
primitive, primitive: orbit of (λ_0, λ_0)	288	2	$1 - 2\kappa^3$; product over the primitive κ: 1089
primitive, primitive: orbits of $(\lambda_0, -\lambda_0)$	288	2	$2B/(2B + \kappa^3)$; product over the primitive κ: $B^4/(B^2+2)^2$

TABLE 4. Order k and leading coefficient of R_2^w along C_p at the corank-one labels (Lemma 9.9), at one representative state of each orbit. Counts are for $a \neq 4c \neq b$; $A = a/c$, $B = b/c$; $\lambda_0 = \frac{1}{2}(\sqrt{6} + i\sqrt{2})$ is the primitive ratio for $\omega = \frac{1}{2}(-1 + i\sqrt{3})$. In the first row R_4^w has the same order.

jets of C_p are computed by solving the 20 rows of (i) order by order in t, and are checked by substitution into all 22 rows. Item (iii) follows from (i) and (ii): near p, $F^{-1}(0) \subset C_p$, and on C_p the row R_2^w vanishes only at $t = 0$, that is, at p. With Remark 9.8, every label is thus an isolated point of $F^{-1}(0)$; for $a \neq 4c \neq b$ these are $3888 + 1296 = 5184$ labels.

9.5. **Exclusion.**

Theorem 9.10. *Let $m \in \mathbb{R}^5_{>0}$. No curve in A_m has a Diagram-6 germ, for any numbering of the bodies and either colouring. In particular the case $r = 6$ of Theorem 4.1(a) holds, at the complex level.*

Proof. By the symmetries described at the beginning of this section we may use the numbering above. Let $\gamma(s)$ be a Diagram-6 germ on a curve in A_m. By Proposition 9.3 the chart germ $\tilde{\gamma}(s)$ has $t(s) \to 0$ with $t(s) \neq 0$ for $s \neq 0$, bounded coordinates, and a limit $p \in \mathcal{B}$. By Proposition 9.5, $m = (a, a, b, b, c)$ and the crossed signs of p satisfy $\eta_{14} = -\eta_{23}$, $\eta_{13} = -\eta_{24}$; by Lemma 9.6, p is a label, hence a corank-one label. By Lemma 9.9, p is an isolated point of $F^{-1}(0)$. The coordinates of $\tilde{\gamma}$ are Puiseux series in s without negative exponents, so after a substitution $s = \sigma^N$ they are convergent power series in σ, and $\sigma \mapsto \tilde{\gamma}(\sigma^N)$ is a holomorphic map of a disc into $F^{-1}(0)$ with value p at $\sigma = 0$. Since p is isolated, this map is constant near $\sigma = 0$, contradicting $t \not\equiv 0$. □

The argument covers every ramification, every crossed sign class and the faces $a = 4c$, $b = 4c$, since Lemma 9.6 describes the whole fibre $\mathcal{B}$.

Remark 9.11. On the face $a = 4c$ the wing class $(\omega, \lambda) = (1, -2)$ has no finite scale; it lies at $D_{12} = \infty$. The proof of Theorem 9.10 never meets this class: by Proposition 9.3, D_{12} and D_{34} have finite nonzero limits along every Diagram-6 germ. In a one-wing projective chart with $h = 1/D_{12}$, and on $a = b = 4c$ in a double-projective ratio chart, one can show that no analytic branch through a point of $\{t = h = 0\}$ leaves the divisor $h = 0$, though not that these points are isolated (Appendix A). We do not use this.

10. DIAGRAM 7

Diagram 7 is the butterfly of [1, §7.4] without circles. Albouy and Kaloshin derive for it the mass relations (36), (39) and (40) in [1, §§8.7.1–8.7.3]. We use none of these relations.

Theorem 10.1. *Let $m \in \mathbb{R}^5_{>0}$. No Diagram 7 germ with an unbounded four-product lies on a curve through a real point: if an irreducible curve $\Gamma \subset A_m$ carries such a germ, then Γ contains no real point.*

This is part (b) of Theorem 4.1 for $r = 7$. Germs with $a + b \geq 0$ in Proposition 10.2 have no unbounded four-product (Lemma 10.5); the others lie in the open cone of a nested chart, where an identity valid modulo (1) gives $I \to 0$ (Lemma 10.4), so $I = U \equiv 0$ on Γ, which is impossible at a real point.

10.1. **The diagram and the nested chart.** In the numbering of [1, §8.7], the *upper wing* $\{1, 2, 5\}$ is an isolated triangle of z-edges without z-circles, the *lower wing* $\{3, 4, 5\}$ is an isolated triangle of w-edges without w-circles, and the four *crossed* pairs kj ($k \in \{1, 2\}$, $j \in \{3, 4\}$) are not edges. Since Diagram 7 has no circles, Rule 1e shows that neither wing has a maximal stroke (see the proof of Proposition 10.2(i)). By Definition 3.4 and the convention after Theorem 4.1, we may use this numbering and colouring.

Orders. As in Lemma 3.3 we renormalize a germ so that Z and W have a common order, and reparametrize it by a power t of its Puiseux parameter so that this order is -2; thus t plays the role of ϵ in [1, §3], and ord is the order in t. Renormalizations $(z, w) \mapsto (\lambda z, \lambda^{-1} w)$ preserve the diagram, the ties, I, U and every $r_{kl}^2 = z_{kl} w_{kl}$.

The nested chart with exponents P, Q, R has the coordinates $A_1, A_2, C, u_3, u_4, B_3, B_4, D, v_1, v_2$ and D_{kl} ($k < l$):

$$\begin{aligned} z_1 = A_1, \quad z_2 = A_2, \qquad & w_3 = t^{-Q} B_3, \quad w_4 = t^{-Q} B_4, \\ z_j = C + t^P m_5 u_j \ \ (j = 3, 4), \qquad & z_5 = C - t^P (m_3 u_3 + m_4 u_4), \\ w_k = t^{-Q} D + t^R m_5 v_k \ \ (k = 1, 2), \qquad & w_5 = t^{-Q} D - t^R (m_1 v_1 + m_2 v_2), \end{aligned}$$

and $\delta_{kl} = t^{e_{kl}} D_{kl}$, where $e_{kl} = -R/2$ on the upper wing, $Q/2$ on the crossed pairs and $(Q - P)/2$ on the lower wing (D without indices is not an inverse distance). So C and $t^{-Q} D$ are the centres of mass of the z-cluster $\{3, 4, 5\}$ and of the w-cluster $\{1, 2, 5\}$, P and R are their internal orders, and $-Q$ is the order of the common w-scale. The exponents e_{kl} balance all ten ties $\delta_{kl}^2 z_{kl} w_{kl} = 1$; those of the pairs 12 and 34 read exactly

$$m_5 D_{12}^2 (A_2 - A_1)(v_2 - v_1) = 1, \qquad m_5 D_{34}^2 (u_4 - u_3)(B_4 - B_3) = 1.$$

The *open cone* is $P > Q > 0$, $R > 0$.

Gauges. The chart does not impose the gauge $z_{12} = w_{12}$. By the displayed ties, $A_2 - A_1$ and $B_4 - B_3$ never vanish, so they could serve as gauges for the rotations and for the rescalings of t. We fix none, since I, U, the ties and the zero set of the rows of (1) are invariant under the rotations.

10.2. **Coverage.**

Proposition 10.2 (Coverage)**.** *Let γ be a Diagram 7 germ. There are a, b with $-2 < a < 1$ and $-2 < b < 1$ such that the following hold.*

(i) *On the upper wing* $\operatorname{ord} z_{kl} = a$ *and* $\operatorname{ord} w_{kl} = (4 - a)/3$*; on the lower wing* $\operatorname{ord} w_{kl} = b$ *and* $\operatorname{ord} z_{kl} = (4 - b)/3$*; on the crossed pairs* $\operatorname{ord} z_{kl} = a$ *and* $\operatorname{ord} w_{kl} = b$*. Moreover* $\operatorname{ord} z_k \geq a$ *and* $\operatorname{ord} w_k \geq b$ *for every k.*

(ii) *After the rotation $(z, w) \mapsto (t^{-a} z, t^a w)$, γ is a germ of the nested chart with exponents*

$$P = \frac{4 - b}{3} - a, \qquad Q = -(a + b), \qquad R = \frac{4 + 2a}{3};$$

all chart coordinates have nonnegative order, so they converge, and every D_{kl} has order zero. Moreover $R = \frac{2}{3}(a + 2) > 0$ and $P - Q = \frac{2}{3}(b + 2) > 0$, so (P, Q, R) lies in the open cone if and only if $a + b < 0$.

Proof. (i) We use Fact AK2. By Rule 2e [1, p. 552] the three z-separations of a wing have one order, and so do its three w-separations; let a be the z-order of the upper wing and b the w-order of the lower wing. As $\delta_{kl}^2 z_{kl} w_{kl} = 1$, $\operatorname{ord} Z_{kl} = -(\operatorname{ord} z_{kl} + 3 \operatorname{ord} w_{kl})/2$ and $\operatorname{ord} W_{kl} = -(3 \operatorname{ord} z_{kl} + \operatorname{ord} w_{kl})/2$; since the strokes have order -2, this gives the other two wing orders. The upper wing is an isolated component of the z-diagram without z-circles, so it has no maximal z-stroke (Rule 1e, p. 546), and along a z-stroke Estimate 2 (p. 548) gives $-2 \le \operatorname{ord} z_{kl} \le 1$, with -2 exactly for a maximal z-edge and 1 exactly for a zw-edge. Hence $-2 < a < 1$, and dually $-2 < b < 1$. As $z_{13} = z_{15} + z_{53}$ with $\operatorname{ord} z_{53} = (4-b)/3 > 1 > a$, and $w_{13} = w_{15} + w_{53}$ with $\operatorname{ord} w_{15} = (4-a)/3 > 1 > b$, the crossed orders are a and b; likewise for 14, 23, 24. Finally $\sum_k m_k z_k = \sum_k m_k w_k = 0$ on A_m (proof of Lemma 3.1), so each z_k is a combination of z-differences, all of order at least a; likewise $\operatorname{ord} w_k \ge b$.

(ii) After the rotation the z-differences have order 0 on the upper wing and the crossed pairs and P on the lower wing, the w-differences have order R on the upper wing and $-Q$ elsewhere, and $\operatorname{ord} z_k \ge 0$, $\operatorname{ord} w_k \ge -Q$. Inverting the chart map, $A_k = z_k$, $B_j = t^Q w_j$, C and $t^{-Q}D$ are the centres of the two clusters, $m_5 t^P u_j = z_j - C$ and $m_5 t^R v_k = w_k - t^{-Q}D$ are combinations of differences inside them, and $D_{kl} = t^{-e_{kl}}\delta_{kl}$, where $\operatorname{ord}\delta_{kl} = -(\operatorname{ord} z_{kl} + \operatorname{ord} w_{kl})/2 = e_{kl}$. So every coordinate has nonnegative order and $\operatorname{ord} D_{kl} = 0$. □

Thus a germ lies in the open cone exactly when $a + b < 0$, and the germs with $a + b \ge 0$ have no unbounded four-product (Lemma 10.5). In the terminology of [1, §8.7], a germ with a wing in the general case has $a + b < 0$; we shall not need this.

10.3. **The inertia identity.** Write

$$F_k^z = z_k - \sum_{l\neq k} m_l \delta_{kl}^3 (z_k - z_l), \qquad F_k^w = w_k - \sum_{l\neq k} m_l \delta_{kl}^3 (w_k - w_l)$$

for the coordinate rows of (1) and $T_{kl} = \delta_{kl}^2 z_{kl} w_{kl} - 1$ for the ties. Then $I - U = \sum_k m_k w_k F_k^z + \sum_{k<l} m_k m_l \delta_{kl} T_{kl}$, so $U = I$ on A_m [1, p. 541].

Lemma 10.3 (Inertia identity)**.** *Let*

$$S_5 = \sum_{j=3,4} \sum_{l=1,2} m_j m_l \delta_{jl}^3 ((z_j - z_5)(w_l - w_5) - (w_j - w_5)(z_l - z_5)).$$

Then, as polynomials in the masses and in (z, w, δ),

$$I = \sum_{k=1,2} m_k z_k (w_k - w_5) + \sum_{j=3,4} m_j w_j (z_j - z_5) + S_5 \\ - \sum_{j=3,4} m_j ((z_j - z_5) F_j^w - (w_j - w_5) F_j^z) + w_5 \sum_{k=1}^{5} m_k F_k^z.$$

In particular $I = \sum_{k=1,2} m_k z_k (w_k - w_5) + \sum_{j=3,4} m_j w_j (z_j - z_5) + S_5$ on A_m.

Proof. Put $\hat z_k = z_k - z_5$ and $\hat w_k = w_k - w_5$. By antisymmetry of the force terms, $\sum_k m_k F_k^z = \sum_k m_k z_k$. Since $z_j \hat w_j - w_j \hat z_j = z_5 w_j - w_5 z_j$,

$$\begin{aligned} I &= \sum_k m_k z_k \hat w_k + w_5 \sum_k m_k z_k \\ &= \sum_{k=1,2} m_k z_k \hat w_k + \sum_{j=3,4} m_j (w_j \hat z_j + z_5 w_j - w_5 z_j) + w_5 \sum_k m_k F_k^z. \end{aligned}$$

For $j \in \{3, 4\}$, using $z_j - z_l = \hat z_j - \hat z_l$ and $w_j - w_l = \hat w_j - \hat w_l$,

$$\hat z_j F_j^w - \hat w_j F_j^z = (w_5 z_j - z_5 w_j) + \sum_{l\neq j} m_l \delta_{jl}^3 (\hat z_j \hat w_l - \hat w_j \hat z_l).$$

In $\sum_{j=3,4} m_j(\hat z_j F_j^w - \hat w_j F_j^z)$ the terms with $l=5$ vanish, those with $l\in\{3,4\}$ cancel in pairs, and those with $l\in\{1,2\}$ add up to S_5. Hence $\sum_{j=3,4} m_j(z_5 w_j - w_5 z_j) = S_5 - \sum_{j=3,4} m_j(\hat z_j F_j^w - \hat w_j F_j^z)$, and substituting this into the display gives the identity. □

The term S_5 is the torque about body 5 of the crossed forces on the lower wing; compare [1, (31)].

Lemma 10.4. *Let γ be a Diagram 7 germ on A_m with $a+b<0$ (Proposition 10.2). Then $I(\gamma(t))\to 0$ as $t\to 0$. More precisely, in the nested chart of Proposition 10.2(ii) the three terms $\sum_{k=1,2} m_k z_k(w_k - w_5)$, $\sum_{j=3,4} m_j w_j(z_j - z_5)$ and S_5 are sums of powers t^e times polynomials in the chart coordinates, with exponents*

$$e = R; \qquad e = P - Q; \qquad e \in \{\tfrac{Q}{2},\ P+\tfrac{Q}{2},\ \tfrac{3Q}{2}+R,\ P+\tfrac{3Q}{2}+R\},$$

respectively.

Proof. On A_m the last two sums in Lemma 10.3 vanish, and the three remaining terms are invariant under the rotations, so we may evaluate them on the rotated germ in the chart. There

$$w_k - w_5 = t^R\big((m_k+m_5)v_k + m_{k'}v_{k'}\big), \qquad z_j - z_5 = t^P\big((m_j+m_5)u_j + m_{j'}u_{j'}\big),$$
$$w_j - w_5 = t^{-Q}(B_j - D) + t^R(m_1v_1 + m_2v_2),$$
$$z_l - z_5 = A_l - C + t^P(m_3u_3 + m_4u_4),$$

where $\{k,k'\}=\{1,2\}$, $\{j,j'\}=\{3,4\}$ and $l\in\{1,2\}$, and $\delta^3_{jl} = t^{3Q/2}D^3_{jl}$. Substituting, together with $z_k = A_k$ and $w_j = t^{-Q}B_j$, gives the stated exponents. Since $a+b<0$, Proposition 10.2(ii) puts (P,Q,R) in the open cone, so every exponent is positive, and all chart coordinates are bounded. Hence the three terms, and with them I, tend to zero. □

10.4. **Germs outside the open cone.**

Lemma 10.5. *Let γ be a Diagram 7 germ with $a+b\ge 0$ (Proposition 10.2). Then every four-product tends to zero along γ. If $a+b=0$, then $-1<a<1$, and p_{12}, p_{34} and each of the other eight four-products have the orders $(16-4a)/3$, $(16+4a)/3$ and $8/3$. If $a+b>0$, every r_{kl} tends to zero, and so does I.*

Proof. By Proposition 10.2(i), $r_{kl}^2 = z_{kl}w_{kl}$ has order $(4+2a)/3>0$ on the upper wing, $(4+2b)/3>0$ on the lower wing and $a+b\ge 0$ on the crossed pairs. The four-product p_{ij} contains the three squared sides of the triangle $\{k,l,m\}$ complementary to $\{i,j\}$, and this triangle contains a wing pair: if $5\in\{k,l,m\}$, both pairs through 5 are wing pairs; otherwise $\{k,l,m\}\subset\{1,2,3,4\}$ contains $\{1,2\}$ or $\{3,4\}$ (the crossed pairs form the bipartite graph $K_{2,2}$). Hence $\operatorname{ord} p_{ij} \ge \min\{(4+2a)/3,(4+2b)/3\}>0$. If $a+b=0$, the bounds $-2<a,b<1$ give $-1<a<1$, and adding orders gives the stated values. If $a+b>0$, every r_{kl}^2 has positive order, and the Lagrange identity $MI = (\sum_k m_k z_k)(\sum_k m_k w_k) + \sum_{k<l} m_k m_l z_{kl} w_{kl}$, with $\sum_k m_k z_k = \sum_k m_k w_k = 0$, gives $I\to 0$. □

Remark 10.6. The germs with $a+b=0$ are not excluded, and need not be: they carry no unbounded four-product, so Proposition 3.6 never produces them.

10.5. **End of the proof.**

Proof of Theorem 10.1. Let γ be a Diagram 7 germ with an unbounded four-product on an irreducible curve $\Gamma\subset A_m$, numbered as above, and let a,b be as in Proposition 10.2. Some four-product has negative order along γ (Definition 3.4), so $a+b<0$ by Lemma 10.5, and $I(\gamma(t))\to 0$ by Lemma 10.4. By Lemma 2.1, $I=U$ is constant on Γ, with value $\lim_{t\to 0} I(\gamma(t)) = 0$. At a real point $w_k = \bar z_k$, so $I = \sum_k m_k|z_k|^2$, which is positive because $m_k>0$ and the tie of the pair 12 forces $z_1\ne z_2$. Hence Γ contains no real point. □

Remark 10.7. The proof of Theorem 10.1 uses no relation among the masses: Lemma 10.3 is a polynomial identity, Proposition 10.2 and Lemma 10.5 use only Rules 1e, 2e and Estimate 2 of [1, §3], Lemma 10.4 only the signs of the exponents, and the last step only Lemma 2.1 and $m_k > 0$. So the theorem does not depend on whether the masses satisfy (36), (39) or (40), and no mass condition is derived for Diagram 7.

11. Diagram 8

Diagram 8 is the butterfly of [1, §7.4] with exactly one circled wing (Fact AK4). We use the numbering of [1, §8.8]: the upper wing $\{1,2,5\}$ is an isolated triangle of z-edges without z-circles, the lower wing $\{3,4,5\}$ is a triangle of w-edges with w-circles at 3 and 4 (vertex 5 is never circled [1, §7.4], and Rules 1a, 1d place the circles of a circled wing at its other two vertices), and the crossed pairs 13, 14, 23, 24 are not edges. The other circling becomes this one after exchanging z and w and renumbering $1 \leftrightarrow 3$, $2 \leftrightarrow 4$; both operations are symmetries of (1) (the second permutes the masses), so it suffices to treat this numbering for every $m \in \mathbb{R}^5_{>0}$. The mass condition of the diagram is relation (40) of [1, §8.8],

$$m_5^2(m_1+m_2)^4 = m_1^3 m_2^3, \tag{19}$$

which [1, p. 578] obtain "by the same arguments" as for Diagram 7. For the other circling it reads $m_5^2(m_3+m_4)^4 = m_3^3 m_4^3$, which is their (39). We show that every Diagram 8 germ is a ray germ of a chart with a finite lower wing (Proposition 11.3), derive (19) from the boundary of the chart (Proposition 11.4), and exclude all germs, complex ones included, by an order-six pairing, an order-nine pairing and a second-order elimination on one curve of masses (Theorem 11.9).

Two further symmetries are used tacitly. Scaling the masses by $\lambda > 0$ and (z, w, δ) by $(\lambda^{1/3}, \lambda^{1/3}, \lambda^{-1/3})$ maps A_m onto $A_{\lambda m}$ and multiplies every coordinate and every component of $\mathcal{Z}$, $\mathcal{W}$ by a nonzero constant, so it preserves orders and hence diagrams (Lemma 3.3). The equations of (1) other than $z_{12} = w_{12}$ are invariant under $z \mapsto cz$, $w \mapsto c^{-1}w$; this shifts all z-orders by $\operatorname{ord} c$ and all w-orders by $-\operatorname{ord} c$, so it preserves diagrams [1, p. 547].

11.1. **The ratio chart.** Let γ be a Diagram 8 germ. We rotate it by z_{12}^{-1}, so that $z_{12} = 1$ along γ. The *ratio chart* has two parameters t, h and nineteen coordinates, the positions and shapes $A_1, A_2, C, B_3, B_4, D, u, v, R_{12}, R_{25}, P, Q$ and the inverse distances $D_{13}, D_{14}, D_{23}, D_{24}, D_{34}, D_{35}, D_{45}$; we write x for all of them. It is given by

$$\begin{gathered}
z_1 = A_1, \quad z_2 = A_2, \quad z_3 = C + m_5 t^6 u, \quad z_4 = C + m_5 t^6 v, \\
z_5 = C - t^6(m_3 u + m_4 v), \quad w_3 = t^{-6} B_3, \quad w_4 = t^{-6} B_4, \\
w_1 = t^{-6} D + m_5 t^4 h^{-2} P, \quad w_2 = t^{-6} D + m_5 t^4 h^{-2} Q, \\
w_5 = t^{-6} D - t^4 h^{-2}(m_1 P + m_2 Q), \quad \delta_{15} = t^{-2} h, \\
\delta_{12} = t^{-2} h R_{12}, \quad \delta_{25} = t^{-2} h R_{25}, \\
\delta_{bc} = t^3 D_{bc} \ \ (b \in \{1,2\},\ c \in \{3,4\}), \qquad \delta_{kl} = D_{kl} \ \ (k,l \in \{3,4,5\}).
\end{gathered}$$

For $th \neq 0$ it is invertible: C and $t^{-6}D$ are the centres of mass of z_3, z_4, z_5 and of w_1, w_2, w_5, $h = t^2\delta_{15}$, $R_{12} = \delta_{12}/\delta_{15}$, $R_{25} = \delta_{25}/\delta_{15}$, $u = (z_3 - C)/(m_5 t^6)$, $P = h^2(w_1 - t^{-6}D)/(m_5 t^4)$, and similarly for the other coordinates. The rows of the chart are the gauges $E_{20} := A_2 - A_1 - 1$ and $E_{21} := B_4 - B_3 - 1$ (that is, $z_{12} = 1$ and $w_4 - w_3 = t^{-6}$) and $E_0, \dots, E_{19}$, obtained by substituting the chart into the twenty equations of (1) other than $z_{12} = w_{12}$ and multiplying each by a monomial in t and $h^{1/2}$: E_{2k-2} and E_{2k-1} come from the z- and the w-equation of body k, and $E_{10}, \dots, E_{19}$ from the ties of the pairs $12, 13, 14, 15, 23, 24, 25, 34, 35, 45$. In the corner coordinates

$$a := th^{-1/2}, \qquad \eta := t^2 h^{1/2} \qquad (\text{so } t^3 = a\eta,\ h^{3/2} = \eta\, a^{-2})$$

we use the monomials

$$\mu := a^6 = t^6h^{-3}, \qquad \nu := a^2\eta^2 = t^6, \qquad T := a\eta = t^3, \qquad \eta^2 = t^4h,$$

and we write $X := A_1 - C$, $Y := A_2 - C$, $M_{12} := m_1 + m_2$, $M := m_3 + m_4 + m_5$, and $C_5 := C - \nu(m_3u + m_4v)$, which is z_5.

Lemma 11.1 (computer-assisted)**.** *The following hold identically in the masses.*

(i) *Every E_j is a polynomial in a, η and x, and its* face *$f_j := E_j|_{a=\eta=0}$ is*

$$\begin{aligned}
f_0 &= -\big(m_2(X-Y)R_{12}^3 + m_5X\big), & f_8 &= m_1X + m_2YR_{25}^3,\\
f_2 &= -\big(m_1(Y-X)R_{12}^3 + m_5YR_{25}^3\big), & f_{20} &= A_2 - A_1 - 1,\\
f_{10} &= m_5(X-Y)(P-Q)R_{12}^2 - 1, & f_{13} &= X\big((m_1+m_5)P + m_2Q\big) - 1,\\
f_{16} &= YR_{25}^2\big(m_1P + (m_2+m_5)Q\big) - 1, & f_4 &= f_6 = C, \quad f_1 = f_3 = D,\\
f_{11} &= X(D-B_3)D_{13}^2 - 1, & f_{12} &= X(D-B_4)D_{14}^2 - 1,\\
f_{14} &= Y(D-B_3)D_{23}^2 - 1, & f_{15} &= Y(D-B_4)D_{24}^2 - 1,\\
f_5 &= B_3 + m_4(B_4-B_3)D_{34}^3 - m_5(B_3-D)D_{35}^3, & f_{21} &= B_4 - B_3 - 1,\\
f_7 &= B_4 - m_3(B_4-B_3)D_{34}^3 - m_5(B_4-D)D_{45}^3,\\
f_9 &= D + m_3(B_3-D)D_{35}^3 + m_4(B_4-D)D_{45}^3,\\
f_{17} &= m_5(B_3-B_4)(u-v)D_{34}^2 - 1,\\
f_{18} &= (B_3-D)\big((m_3+m_5)u + m_4v\big)D_{35}^2 - 1,\\
f_{19} &= (B_4-D)\big(m_3u + (m_4+m_5)v\big)D_{45}^2 - 1.
\end{aligned}$$

(ii) $m_1E_0 + m_2E_2 + m_5E_8 + a^6(m_3E_4 + m_4E_6) = a^6(m_1A_1 + m_2A_2 + MC)$.

(iii) $E_1 = D + \eta^2\beta_1 + T^3\tau_1 + \sum_{i\ge6,\,j\ge2} a^i\eta^j(\cdots)$, *and E_3 is the same with β_3, τ_3, where*

$$\begin{aligned}
\beta_1 &= -m_5\big(m_2(P-Q)R_{12}^3 + (m_1+m_5)P + m_2Q\big),\\
\beta_3 &= -m_5\big(m_1(Q-P)R_{12}^3 + (m_1P + (m_2+m_5)Q)R_{25}^3\big),\\
\tau_1 &= m_3(B_3-D)D_{13}^3 + m_4(B_4-D)D_{14}^3,\\
\tau_3 &= m_3(B_3-D)D_{23}^3 + m_4(B_4-D)D_{24}^3;
\end{aligned}$$

the monomials of $E_1 - E_3$ are η^2, T^3, $a^6\eta^2$ and $a^9\eta^5$.

(iv) *For $k \in \{0, 2, 8, 10, 13, 16, 20\}$, $E_k = f_k(\mathcal{I}) + \sum_{i\ge6} a^i\eta^j(\cdots)$, where f_k is read as a function of $(X, Y, R_{12}, R_{25}, P, Q)$ and $\mathcal{I} := (A_1 - C_5,\ A_2 - C_5,\ R_{12},\ R_{25},\ P,\ Q)$.*

(v) *$E_4 = C + \nu L_3 + T^3F_3 + T^5G_3$ exactly, where $L_3 = -m_5\big(m_4(u-v)D_{34}^3 + ((m_3+m_5)u + m_4v)D_{35}^3 - u\big)$, $F_3 = -\big(m_1(C-A_1)D_{13}^3 + m_2(C-A_2)D_{23}^3\big)$ and $G_3 = -m_5u(m_1D_{13}^3 + m_2D_{23}^3)$.*

The lemma is an exact computation over $\mathbb{Q}$, with the masses, a, η and the chart coordinates as indeterminates (Appendix A). For (iv) the computation shows that the ν-coefficient of E_k is $-(\partial_C f_k)(m_3u + m_4v)$ and that every other non-face monomial has $i \ge 6$; since each of these f_k is affine in (X, Y), this is (iv).

Definition 11.2. A germ in the ratio chart is a *ray germ* if along it $t \to 0$, x converges, $\operatorname{ord} a > 0$ and $\operatorname{ord}\eta > 0$. Its *ray* is $r := \operatorname{ord}\eta/\operatorname{ord} a \in (0, \infty)$ and its *exponent* is $p := \operatorname{ord} h/\operatorname{ord} t$.

Since $t^3 = a\eta$ and $h^{3/2} = \eta a^{-2}$, $p = (2r-4)/(r+1)$, an increasing bijection of $(0, \infty)$ onto $(-4, 2)$; the rays $r = \frac{1}{2}, 2, 3$ are $p = -2, 0, \frac{1}{2}$, and the faces $\{a = 0\}$, $\{\eta = 0\}$ of the corner chart are $p = -4$ and $p = 2$.

11.2. **Coverage.**

Proposition 11.3 (Coverage)**.** *Every Diagram 8 germ is a ray germ of the ratio chart with exponent $p \in (-4, 2)$ and a finite lower wing. Precisely: let γ be a Diagram 8 germ, rotated so that $z_{12} = 1$, and put $t := (w_4 - w_3)^{-1/6}$ and $h := t^2\delta_{15}$ along γ. Let θ be the order of z_{12} before the rotation, in the normalization $Z = W = \epsilon^{-2}$ with orders in ϵ. Then $-2 < \theta < 1$; every coordinate in x converges along γ, and R_{12}, R_{25}, $D_{13}, D_{14}, D_{23}, D_{24}$, D_{34}, D_{35}, D_{45} have nonzero limits; and*

$$\operatorname{ord} t = \tfrac{2-\theta}{6}, \qquad \operatorname{ord} h = -\tfrac{2\theta}{3}, \qquad p = -\tfrac{4\theta}{2-\theta},$$
$$\operatorname{ord} a = \tfrac{2+\theta}{6}, \qquad \operatorname{ord} \eta = \tfrac{2-2\theta}{3}, \qquad r = \tfrac{4(1-\theta)}{2+\theta}.$$

Proof. By Fact AK2 and Lemma 3.3 the rules and estimates of [1, §3] hold for the orders along γ. *Lower wing.* $\{3, 4, 5\}$ is an isolated component of the w-diagram with w-circles, so it has a maximal w-stroke [1, Rule 1e, p. 546]. By Rule 2e its three w-separations have one order, so all three w-strokes are maximal, and the right equality case of Estimate 2 [1, p. 548], in the colour w, gives $\operatorname{ord} w_{kl} = -2$, $\operatorname{ord} z_{kl} = 2$, $\operatorname{ord} r_{kl} = 0$ for $kl \in \{34, 35, 45\}$. *Upper wing.* $\{1, 2, 5\}$ is an isolated component of the z-diagram without circles, so it has no maximal z-stroke (Rule 1e); it carries no w-stroke, hence no zw-edge. So neither equality case of Estimate 2 holds, and by Rule 2e the three z-separations have one order θ with $-2 < \theta < 1$. The z-strokes have $\operatorname{ord} Z_{kl} = -2$, so $\operatorname{ord} r_{kl} = (\theta + 2)/3$ and $\operatorname{ord} w_{kl} = (4 - \theta)/3$. *Crossed pairs.* For $b \in \{1, 2\}$, $c \in \{3, 4\}$, $z_{bc} = z_{b5} + z_{5c}$ has order θ (as $\theta < 2$) and $w_{bc} = w_{b5} + w_{5c}$ has order -2 (as $(4 - \theta)/3 > -2$), so $\operatorname{ord} r_{bc} = (\theta - 2)/2$. *Positions.* Every w_k has order ≥ -2, with equality for $k = 3, 4$ (w-circles). Let C be the centre of mass of z_3, z_4, z_5. Then $(m_1 + m_2 + M)C = -m_1(z_1 - C) - m_2(z_2 - C)$, and $z_b - C = z_{5b} + (z_5 - C)$ with $\operatorname{ord} z_{5b} = \theta$ and $\operatorname{ord}(z_5 - C) \geq 2$; so C, z_1, z_2 have order $\geq \theta$, and $z_k - C$ has order ≥ 2 for $k = 3, 4, 5$. *Chart.* The rotation lowers all z-orders by θ and raises all w-orders by θ, so $\operatorname{ord}(w_4 - w_3) = \theta - 2$ and $\operatorname{ord} t = (2 - \theta)/6 > 0$. By the inverse formulas of the chart: A_1, A_2, C have order ≥ 0; u, v have order $\geq (2 - \theta) - 6 \operatorname{ord} t = 0$; B_3, B_4 have order 0 and D order ≥ 0; $\operatorname{ord} h = 2 \operatorname{ord} t - \operatorname{ord} r_{15} = -2\theta/3$; R_{12}, R_{25} have order 0; P, Q have order $\geq 2 \operatorname{ord} h + \frac{4+2\theta}{3} - 4 \operatorname{ord} t = 0$, because $w_1 - t^{-6}D = (m_2 w_{21} + m_5 w_{51})/(m_1 + m_2 + m_5)$ and similarly for w_2; $D_{bc} = t^{-3}\delta_{bc}$ has order $\frac{2-\theta}{2} - 3 \operatorname{ord} t = 0$; and $D_{kl} = \delta_{kl}$ has order 0. Hence every coordinate converges (Lemma 3.3), the inverse-distance coordinates to nonzero limits, and $p = \operatorname{ord} h / \operatorname{ord} t = -4\theta/(2 - \theta) \in (-4, 2)$. Finally $\operatorname{ord} a = \operatorname{ord} t - \frac{1}{2} \operatorname{ord} h = (2 + \theta)/6$ and $\operatorname{ord} \eta = 2 \operatorname{ord} t + \frac{1}{2} \operatorname{ord} h = (2 - 2\theta)/3$, both positive. □

In particular no Diagram 8 germ reaches the lower projective end of the chart, where the lower inverse distances degenerate.

11.3. **The first condition.**

Proposition 11.4 (First condition)**.** *At positive masses every Diagram 8 germ satisfies* (19)*: $m_5^2(m_1 + m_2)^4 = m_1^3 m_2^3$. Moreover, in the chart of Proposition 11.3 its limit x_0 has $A_1 = -m_2/M_{12}$, $A_2 = m_1/M_{12}$, $C = 0$, $R_{25} = 1$, $R_{12} = -m_5 M_{12}/(m_1 m_2)$, $(P, Q) = (P_0, Q_0)$ the unique solution of $f_{10} = f_{13} = 0$, $D = 0$, $B_3 = -m_4/(m_3 + m_4)$, $B_4 = m_3/(m_3 + m_4)$, and, with $L := (M_{12}(m_3 + m_4))^{1/2} > 0$ and signs $s_{bc} \in \{\pm 1\}$,*

$$D_{13} = \frac{i s_{13} L}{\sqrt{m_2 m_4}}, \qquad D_{14} = \frac{s_{14} L}{\sqrt{m_2 m_3}},$$
$$D_{23} = \frac{s_{23} L}{\sqrt{m_1 m_4}}, \qquad D_{24} = \frac{i s_{24} L}{\sqrt{m_1 m_3}}.$$

We write $U := (A_1, A_2, C, R_{12}, R_{25}, P, Q)$, U_0 for its limit, and $\mathcal{I}_0 := (X_0, Y_0, R_{12}^0, 1, P_0, Q_0)$, with $X_0 = -m_2/M_{12}$, $Y_0 = m_1/M_{12}$ and $R_{12}^0 = -m_5 M_{12}/(m_1 m_2)$, for the limit of $\mathcal{I}$ (note $C_5 \to C \to 0$).

Proof. Let $m \in \mathbb{R}^5_{>0}$ and let γ be a Diagram 8 germ, in the chart of Proposition 11.3, with limit x_0. By Lemma 11.1 (i) and Proposition 11.3, $E_j = f_j(x) + O(|a| + |\eta|)$ along γ, so $f_j(x_0) = 0$ for all j. *Positions.* By Lemma 11.1 (ii) and $a \neq 0$, $m_1A_1 + m_2A_2 + MC = 0$ along γ. With $f_4 = C$ and f_{20} this linear system, of determinant M_{12}, gives $A_1 = -m_2/M_{12}$, $A_2 = m_1/M_{12}$, $C = 0$ at x_0; so $X = A_1$, $Y = A_2$ and $X - Y = -1$ there. *Force rows.* Then $f_8 = \frac{m_1m_2}{M_{12}}(R_{25}^3 - 1)$ and $f_0 = \frac{m_2}{M_{12}}F$ with $F := M_{12}R_{12}^3 + m_5$, so $R_{25}^3 = 1$ and $F = 0$. *Closure of the w-triangle* 125. Identically,

$$\begin{aligned}(X-Y)R_{12}^2Xf_{16} - (X-Y)R_{12}^2YR_{25}^2f_{13} + XYR_{25}^2f_{10}\\ = (X-Y)R_{12}^2(YR_{25}^2 - X) - XYR_{25}^2,\end{aligned}$$

which is free of P, Q. At x_0 the right side is $-G/M_{12}^2$ with $G := M_{12}R_{12}^2(m_1R_{25}^2+m_2)-m_1m_2R_{25}^2$; so $G = 0$. *Elimination.* Let $H := M_{12}(m_1 + m_2R_{25})R_{12}^2 - m_1m_2$ and $\Phi(s) := m_5^2M_{12}(m_1 + m_2s)^3 - (m_1m_2)^3$. Identically

$$\begin{aligned}H &= R_{25}G + (1 - R_{25}^3)\, m_1(M_{12}R_{12}^2 - m_2),\\ \Phi(R_{25}) &= H\big(H^2 + 3m_1m_2H + 3m_1^2m_2^2\big) - M_{12}(m_1 + m_2R_{25})^3F(F - 2m_5),\end{aligned}$$

so $H = 0$ and $\Phi(R_{25}) = 0$ at x_0. *Phase.* If $R_{25} \neq 1$, then $R_{25}^2 + R_{25} + 1 = 0$ and $\Phi(R_{25}) = \Phi_0 + \Phi_1R_{25}$ with real Φ_0, Φ_1; as $R_{25} \notin \mathbb{R}$, $\Phi_0 = \Phi_1 = 0$. But $\Phi_1 = 3m_1m_2m_5^2M_{12}(m_1 - m_2)$ forces $m_1 = m_2$, and then $\Phi_0 = -m_1^4(m_1^2 + 2m_5^2) < 0$. Hence $R_{25} = 1$, and $\Phi(1) = 0$ is (19). *The limit point.* At $R_{25} = 1$, $F = 0$ and $H = M_{12}^2R_{12}^2 - m_1m_2 = 0$ give $R_{12} = R_{12}^3/R_{12}^2 = -m_5M_{12}/(m_1m_2)$. The faces f_{10}, f_{13} are linear in (P, Q) with determinant $m_2m_5(m_1 + m_2 + m_5)R_{12}^2/M_{12} \neq 0$. Next $f_1 = D$ gives $D = 0$; the identity $m_1f_1 + m_2f_3 + m_3f_5 + m_4f_7 + m_5f_9 = (m_1 + m_2 + m_5)D + m_3B_3 + m_4B_4$ and f_{21} give B_3, B_4; and $f_{11}, f_{12}, f_{14}, f_{15}$ give $D_{13}^2 = -M_{12}(m_3+m_4)/(m_2m_4)$, $D_{14}^2 = M_{12}(m_3+m_4)/(m_2m_3)$, $D_{23}^2 = M_{12}(m_3+m_4)/(m_1m_4)$, $D_{24}^2 = -M_{12}(m_3 + m_4)/(m_1m_3)$. □

Conversely, a direct computation shows that on (19) the whole face system has solutions, with the lower wing at the phase $D_{45} = D_{35}$ and any crossed signs; so (19) is exactly the solvability condition of the faces together with the centre of mass. We do not use this.

Remark 11.5. [1, §8.8] place the cluster z_3, z_4, z_5 at the centre of mass of z_1, z_2, invoke their relation (34), $(m_1 + m_2)W_{12} \sim m_5W_{51} + m_5W_{25}$, and refer to their §8.7.3. In the chart $(m_1 + m_2)W_{12} - m_5(W_{51} + W_{25}) = t^{-2}h(\beta_1 - \beta_3) = t^{-6}\eta^2(\beta_1 - \beta_3)$, and $\eta^2(\beta_1 - \beta_3)$ is the η^2-part of $E_1 - E_3$; so (34) holds at the limit of every germ (Lemma 11.1 (iii)), and given $R_{25}^3 = 1$, the ties and the centre of mass it is equivalent to $F = 0$. The centre of mass cannot be dropped: at $(m_1, m_2, m_5) = (4, 1, \frac{4}{9})$, which violates (19), the point $A_1 = R_{12} = -\frac{2}{3}$, $A_2 = \frac{1}{3}$, $C = 0$, $R_{25} = 2$, with P, Q from $f_{10} = f_{13} = 0$, satisfies every upper face and (34), but $m_1A_1+m_2A_2+MC = -\frac{7}{3}$. Relation (39) is not used: at $D = 0$, $B_3 = -m_4/(m_3 + m_4)$, $B_4 = m_3/(m_3 + m_4)$, with $D_{35} = g$, $D_{45} = \omega g$, $D_{34} = \lambda g$, the face f_5 reads $g^3((m_3 + m_4)\lambda^3 + m_5) = 1$, and the escape $(m_3+m_4)\lambda^3+m_5 = 0$ of the lower scale forces $\omega = 1$ and (39); by Proposition 11.3 no Diagram 8 germ escapes.

11.4. **Exclusion.** The germ is excluded in three steps: an order-six pairing for $0 < p < 2$; a lemma on the coefficients of $t^9 = T^3$ in E_1 and E_3, which for $-4 < p < \frac{1}{2}$ forces the order-nine pairing Ξ below to vanish; and a second-order elimination on the locus $\Xi = 0$.

Lemma 11.6 (computer-assisted)**.** *Let $m \in \mathbb{R}^5_{>0}$ satisfy* (19)*, and let S be the vector of the rows $E_0, E_2, E_4, E_{10}, E_{13}, E_{16}, E_{20}$ and $E_Z := m_1A_1 + m_2A_2 + MC$.*

- (i) *The rows E_0 and E_2 consist of their faces and multiples of μ, ν, $a^9\eta^3$, $a^{11}\eta^5$; E_4 of its face and multiples of ν, T^3, T^5; E_{13} and E_{16} of their faces and multiples of ν; and E_{10}, E_{20}, E_Z equal their faces. The faces of S depend on U only.*
- (ii) *The faces of S vanish at U_0, and their Jacobian J with respect to U there (an 8×7 matrix) has rank 7.*

(iii) *Let S_6 be the vector of μ-coefficients of S at x_0 and let $\ell \neq 0$ satisfy $\ell J = 0$. If $m_1 \neq m_2$, scale the masses to $m_2 = 1$, $m_1 = \alpha^2$ (so $m_5 = \alpha^3/(\alpha^2+1)^2$) and normalize ℓ by $\ell_{E_4} = \alpha^2 - 1$; then*

$$\ell \cdot S_6 = -\frac{2(\alpha^2+1)^4}{\alpha\,\Delta},$$
$$\Delta := (m_3+m_4)(\alpha^2+1)^2 + \alpha^6 + 3\alpha^4 + \alpha^3 + 3\alpha^2 + 1 > 0.$$

If $m_1 = m_2$ $(= 4m_5)$, then ℓ is a multiple of $(1, -1, 0, \frac{3}{2}m_5, -\frac{3}{4}m_5, -\frac{3}{4}m_5, 0, 0)$, and $\ell \cdot S_6 = -1$ for this vector.

Consequently no Diagram 8 germ at the masses m has $0 < p < 2$.

Parts (i)–(iii) are exact computations, with α, m_3, m_4 as indeterminates if $m_1 \neq m_2$, and with m_3, m_4, m_5 as indeterminates if $m_1 = m_2 = 4m_5$ (Appendix A). For $m_1 \neq m_2$ the rank in (ii) and the null vector in (iii) are computed over the field $\mathbb{Q}(\alpha, m_3, m_4)$, and they specialize to every $\alpha > 0$, $\alpha \neq 1$. The maximal minors of J that involve α only have the greatest common divisor $\alpha(\alpha^2+\alpha+1)(\alpha^4-\alpha^3+3\alpha^2-\alpha+1)$, which has no positive zero, so J has rank 7 at every $\alpha > 0$. The denominators of the entries of J, of the normalized ℓ and of S_6 are products of α, α^2+1 and Δ, which do not vanish for $\alpha > 0$ and $m_3, m_4 > 0$. Hence the identities $\ell J = 0$ and the formula for $\ell \cdot S_6$ hold at every such α, where $\ell \neq 0$ because $\ell_{E_4} = \alpha^2 - 1 \neq 0$.

Proof of the last assertion. Let γ be a Diagram 8 germ with $0 < p < 2$, that is $r > 2$. Then $\nu/\mu = (\eta/a^2)^2$, $T^3/\mu = (\eta/a)^3$, $T^5/\mu = \eta^5/a$, $a^9\eta^3/\mu$ and $a^{11}\eta^5/\mu$ tend to 0, because $\operatorname{ord}\eta > 2\operatorname{ord} a$. By (i) and the convergence of x, $S = S^{(0)}(U) + \mu S_6(x) + o(\mu)$ along γ, where $S^{(0)}$ is the vector of faces. As $S = 0$, $S^{(0)}(U) = O(\mu)$, and (ii) gives $U - U_0 = O(\mu)$. Pairing with ℓ and using $\ell \cdot S^{(0)}(U) = O(|U-U_0|^2)$, we get $0 = \mu\big(\ell \cdot S_6(x_0) + o(1)\big)$, which contradicts (iii) since $\mu = a^6 \neq 0$. □

Lemma 11.7 (computer-assisted)**.** *Let $m \in \mathbb{R}^5_{>0}$ satisfy (19); let x_0, $\mathcal{I}_0$, U_0 and the signs s_{bc} be as in Proposition 11.4, $\beta := \beta_1 - \beta_3$, $b := (f_0, f_2, f_{10}, f_{13}, f_{16}, f_{20})$ as a function of $(X, Y, R_{12}, R_{25}, P, Q)$, and $B := \partial b/\partial\mathcal{I}$ at $\mathcal{I}_0$.*

(a) $\beta(\mathcal{I}_0) = \dfrac{m_1^3 m_2^3 - m_5^2(m_1+m_2)^4}{m_1 m_2 m_5 (m_1+m_2)^2}$, *which is 0 by (19).*

(b) *If $m_1 \neq m_2$, then B is invertible. If $m_1 = m_2$ $(= 4m_5)$, then $\ker B$ is spanned by a vector k_0, equal to $(\frac{27}{272}, \frac{27}{272}, \frac{9}{272}, -\frac{9}{68}, -\frac{19}{17}, 1)$ at $(m_1, m_2, m_5) = (1, 1, \frac{1}{4})$; $d\beta(\mathcal{I}_0)\,k_0 = 0$; and the Jacobian of the eight functions $f_0, f_2, f_{10}, f_{13}, f_{16}, f_{20}$ (evaluated at $X = A_1 - C$, $Y = A_2 - C$), C and $m_1A_1 + m_2A_2 + MC$ with respect to U at U_0 has rank 7. (These three facts are computed at $(m_1, m_2, m_5) = (1, 1, \frac{1}{4})$ with m_3, m_4 free, which suffices by the mass scaling.)*

(c) *With $\Xi := s_{14}(m_1m_4)^{3/2} + s_{23}(m_2m_3)^{3/2} + i\big(s_{13}(m_1m_3)^{3/2} + s_{24}(m_2m_4)^{3/2}\big)$,*

$$(\tau_1 - \tau_3)(x_0) = \frac{m_3 m_4}{m_3+m_4}\, L^3\, (m_1m_2m_3m_4)^{-3/2}\, \Xi.$$

Consequently, if a Diagram 8 germ at the masses m has $-4 < p < \frac{1}{2}$, then $\Xi = 0$, that is, $m_1 = m_2 = 4m_5$, $m_3 = m_4$, $s_{13} = -s_{24}$ and $s_{14} = -s_{23}$.

The lemma is an exact computation (Appendix A); (a) is an identity in the masses, and (c), for each choice of the signs, an identity in their square roots. For $m_1 \neq m_2$, in the scaling $m_2 = 1$, $m_1 = \alpha^2$ and after clearing denominators row by row, $\det B$ factors as $(\alpha-1)(\alpha+1)\alpha^{13}(\alpha^2+\alpha+1)(\alpha^2+1)^6(\alpha^4-\alpha^3+3\alpha^2-\alpha+1)$, and its only positive root is $\alpha = 1$.

Proof of the last assertion. Let γ be such a germ. Along γ, $\mathcal{I} \to \mathcal{I}_0$ and $b(\mathcal{I}) = O(a^6)$ by Lemma 11.1 (iv). We claim

$$\beta(\mathcal{I}) = O(a^6 + a^4\eta^4). \tag{20}$$

If $m_1 \neq m_2$, (b) and the inverse function theorem give $\mathcal{I} - \mathcal{I}_0 = O(a^6)$, and (20) follows from (a). If $m_1 = m_2$, the eight functions in (b) are $O(a^6 + \nu + T^3)$ along γ: the first six because $f_k(A_1 - C, A_2 - C, \dots) - f_k(\mathcal{I}) = O(\nu)$, the seventh by Lemma 11.1 (v), and the last vanishes by Lemma 11.1 (ii) as $a \neq 0$. So $U - U_0$ and $\mathcal{I} - \mathcal{I}_0$ are $O(a^6 + \nu + T^3)$. Since $B(\mathcal{I} - \mathcal{I}_0) = O(a^6 + |\mathcal{I} - \mathcal{I}_0|^2)$, the component of $\mathcal{I} - \mathcal{I}_0$ transverse to k_0 is $O(a^6 + |\mathcal{I} - \mathcal{I}_0|^2)$; as $\beta(\mathcal{I}_0) = 0$ and $d\beta(\mathcal{I}_0)k_0 = 0$, we get $\beta(\mathcal{I}) = O(a^6 + |\mathcal{I} - \mathcal{I}_0|^2) = O(a^6 + \nu^2 + T^6)$, which is (20) because $\nu^2 = a^4\eta^4$ and $T^6 = a^6\eta^6$.

Now $E_1 = E_3 = 0$ along γ, so Lemma 11.1 (iii) and (20) give $T^3(\tau_1 - \tau_3)(x) = O(a^6\eta^2 + a^4\eta^6)$, that is, $(\tau_1 - \tau_3)(x) = O(a^3\eta^{-1} + a\eta^3)$. As $-4 < p < \frac{1}{2}$ means $r < 3$, $\operatorname{ord}(a^3\eta^{-1}) = (3 - r)\operatorname{ord} a > 0$; so $(\tau_1 - \tau_3)(x_0) = 0$, and $\Xi = 0$ by (c). Since all $m_k > 0$, the real part $s_{14}(m_1m_4)^{3/2} + s_{23}(m_2m_3)^{3/2}$ of Ξ vanishes only if $s_{14} = -s_{23}$ and $m_1m_4 = m_2m_3$, and its imaginary part only if $s_{13} = -s_{24}$ and $m_1m_3 = m_2m_4$. These give $m_3 = m_4$ and $m_1 = m_2$, and then (19) gives $m_1 = 4m_5$. □

On the locus $\Xi = 0$, which is the *cancelling curve* $m = (4, 4, y, y, 1)$, $y > 0$, up to scaling, together with the four sign classes $s_{13} = -s_{24}$, $s_{14} = -s_{23}$, the order-nine pairing gives nothing, and we expand to second order.

Lemma 11.8 (computer-assisted)**.** *Let $m_1 = m_2 = 4m_5$ and $m_3 = m_4$, scaled to $(1, 1, m, m, \frac{1}{4})$ with $m > 0$. Let ∇ and ∇^2 denote gradient and Hessian with respect to $\mathcal{I}$ at $\mathcal{I}_0$, and let k_0 be as in Lemma 11.7 (b).*

(a) Lower block. *The faces $f_1, f_5, f_7, f_9, f_{17}, f_{18}, f_{19}, f_{21}$ involve only*

$$x' := (B_3, B_4, D, u, v, D_{34}, D_{35}, D_{45}).$$

At each common zero with $g := D_{35} \neq 0$, writing $D_{45} = \omega g$ and $D_{34} = \lambda g$: $D = 0$, $B_3 = -\frac{1}{2}$, $B_4 = \frac{1}{2}$, $\omega^3 = 1$, $2\lambda^2(\omega^2 + 1) = \omega^2$, $g^3(8m\lambda^3 + 1) = 4$, f_{18}, f_{19} determine u, v, and the Jacobian determinant of these eight faces with respect to x' is

$$-9g^7\lambda m(8m + 1)(\omega^2 + 1)(8m\lambda^3 + 1)/(128\,\omega) \neq 0.$$

(b) Lower forcing. *In $E_1, E_3, E_5, E_7, E_9, E_{17}, E_{18}, E_{19}, E_{21}$ and in the crossed rows $E_{11}, E_{12}, E_{14}, E_{15}$, every monomial $a^i\eta^j$ other than the face has $i \geq 2$ and $i + 2j \geq 6$, except η^2 in E_1, E_3, E_9, whose coefficients vanish at $\mathcal{I}_0$.*

(c) Upper rows. *Below a^8 the rows $E_0, E_2, E_8, E_{10}, E_{13}, E_{16}, E_{20}$ consist of $f_k(\mathcal{I})$ and the terms μA_1, μA_2, μC of E_0, E_2, E_8. The row E_1 (resp. E_3) consists of D, $\eta^2\beta_1$, $T^3\tau_1$, $m_5\mu\eta^2 P$ (resp. D, $\eta^2\beta_3$, $T^3\tau_3$, $m_5\mu\eta^2 Q$) and a multiple of $a^9\eta^5$. The Jacobian $\nabla(f_0, f_2, f_8, f_{10}, f_{13}, f_{16}, f_{20})$ has rank 5 and kernel $\mathbb{C}k_0$. The Jacobian B_5 of $(f_2, f_{10}, f_{13}, f_{16}, f_{20})$ with respect to $(X, Y, R_{12}, R_{25}, P)$ has $\det B_5 = -\frac{17}{16}$, and $\varphi := m_1X + m_2Y$ satisfies $\varphi(k_0) = \frac{27}{136}$.*

(d) Second-order constants. *Let $\delta_\mu, \delta_2 \in \mathbb{C}^6$ have last entry 0 and first five entries $-B_5^{-1}(Y_0, 0, 0, 0, 0)^{\mathsf{T}}$ and $-B_5^{-1}(\frac{1}{2}k_0^{\mathsf{T}}\nabla^2 f_k\, k_0)_{k=2,10,13,16,20}$, and put*

$$c_0 := X_0 + \nabla f_0 \cdot \delta_\mu, \qquad q_2 := \tfrac{1}{2}k_0^{\mathsf{T}}\nabla^2 f_0\, k_0 + \nabla f_0 \cdot \delta_2,$$
$$e_2 := \nabla\beta \cdot \delta_2 + \tfrac{1}{2}k_0^{\mathsf{T}}\nabla^2\beta\, k_0, \qquad c_{62} := \nabla\beta \cdot \delta_\mu + m_5(P_0 - Q_0).$$

Then $c_0 = -1$, $q_2 = \frac{243}{73984}$, $e_2 = \frac{1215}{18496}$, $c_{62} = -4$ and $c_0e_2 - q_2c_{62} = -\frac{243}{4624}$.

Consequently no Diagram 8 germ at these masses has a limit with $s_{13} = -s_{24}$ and $s_{14} = -s_{23}$.

The lemma is an exact computation over the field $\mathbb{Q}(m)$; in (a) the Jacobian determinant is reduced modulo $\omega^3 - 1$, and in (c) and (d) all constants are rational numbers (Appendix A).

Proof of the last assertion. Let γ be such a germ, with limit x_0. Then $\Xi = 0$, so $(\tau_1 - \tau_3)(x_0) = 0$ by Lemma 11.7 (c), and $D_{35}(x_0) \neq 0$ by Proposition 11.3. Write $\mathcal{I} = \mathcal{I}_0 + \rho k_0 + \delta$ with $\rho := Q - Q_0$ and δ of last entry 0. All estimates are along γ.

Upper rows. By (c), $E_2, E_{10}, E_{13}, E_{16}, E_{20}$ give, for the first five entries of δ, $B_5\delta = -\mu(Y_0, 0, 0, 0, 0)^{\mathsf{T}} - \frac{1}{2}\rho^2(k_0^{\mathsf{T}}\nabla^2 f_k k_0)_k + o(\mu + \rho^2)$, so $\delta = \mu\delta_\mu + \rho^2\delta_2 + o(\mu + \rho^2)$, and $E_0 = 0$ becomes

$$c_0\mu + q_2\rho^2 = o(\mu + \rho^2). \tag{21}$$

Lower rows. By (a), (b) and the implicit function theorem (each crossed face $f_{11}, f_{12}, f_{14}, f_{15}$ contains one crossed unknown D_{bc}, with derivative $2/D_{bc} \neq 0$ at a zero), the lower and crossed coordinates differ from their limits by $O(\nu + \mu + |\mathcal{I} - \mathcal{I}_0|) = O(\nu + \mu + |\rho|)$.

Amplitude. The centre of mass (Lemma 11.1 (ii)), $E_4 = 0$ (Lemma 11.1 (v)) and $C_5 = C - \nu(m_3u + m_4v)$ give the identity

$$\varphi(\mathcal{I}) = (M_{12} + M)(\nu\Psi + T^3F_3 + T^5G_3),$$

where $\Psi := L_3 + M_{12}(m_3u + m_4v)/(M_{12} + M)$. As $\varphi(\mathcal{I}_0) = 0$ and $\varphi(\delta) = O(\mu + \rho^2)$, first $\rho = O(\nu + T^3 + \mu)$, and then, by the lower estimate, $\rho = \rho_\nu\nu + O(T^3 + \nu^2 + \mu)$ with $\rho_\nu := (M_{12} + M)\Psi(x_0)/\varphi(k_0)$.

Case $\Psi(x_0) = 0$ (it contains the lower phase $\omega = 1$). Then $\rho^2 = O(T^6 + \nu^4 + \mu^2) = o(\mu)$, and (21) reads $\mu(c_0 + o(1)) = 0$, which is impossible.

Case $\Psi(x_0) \neq 0$. Then $\rho^2 = \rho_\nu^2 a^4\eta^4 + o(a^6 + a^4\eta^4)$, because $\nu T^3 = a^5\eta^5 = o(a^6 + a^4\eta^4)$; so (21) reads $c_0a^6 + q_2\rho_\nu^2 a^4\eta^4 = o(a^6 + a^4\eta^4)$. Both coefficients are nonzero, so the two monomials have the same order, that is $r = \frac{1}{2}$ ($p = -2$), and $c_0a_0^2 + q_2\rho_\nu^2\eta_0^4 = 0$ for the leading coefficients $a_0, \eta_0 \neq 0$ of a, η. On this ray, (c) gives $E_1 - E_3 = \eta^2\beta(\mathcal{I}) + T^3(\tau_1 - \tau_3)(x) + m_5\mu\eta^2(P - Q) + O(a^9\eta^5)$, where $\beta(\mathcal{I}) = \mu\,\nabla\beta \cdot \delta_\mu + e_2\rho^2 + o(\mu + \rho^2)$ (because $\beta(\mathcal{I}_0) = 0$ and $\nabla\beta \cdot k_0 = 0$, Lemma 11.7) and $T^3(\tau_1 - \tau_3)(x) = O(T^3(\nu + \mu + |\rho|)) = O(a^5\eta^5 + a^6\eta^6 + a^9\eta^3)$. With weight 1 for a and $\frac{1}{2}$ for η, the monomials $a^6\eta^2$ and $a^4\eta^6$ have weight 7, while $a^5\eta^5$, $a^6\eta^6$, $a^9\eta^3$, $a^9\eta^5$ have weights $\frac{15}{2}, 9, \frac{21}{2}, \frac{23}{2}$. Hence $E_1 - E_3 = 0$ gives $c_{62}a_0^2 + e_2\rho_\nu^2\eta_0^4 = 0$. Since $c_0e_2 - q_2c_{62} \neq 0$, the two linear equations force $a_0 = 0$, a contradiction. □

Theorem 11.9. *No Diagram 8 germ exists at positive masses: for $m \in \mathbb{R}^5_{>0}$ no curve in A_m carries a Diagram 8 germ, complex germs included. In particular Theorem 4.1 (a) holds for $r = 8$.*

Proof. By the symmetries at the beginning of this section we may use the numbering of [1, §8.8]. Let γ be a Diagram 8 germ at $m \in \mathbb{R}^5_{>0}$. By Proposition 11.3 it is a ray germ with ray $r \in (0, \infty)$, and by Proposition 11.4 the masses satisfy (19) and the limit has signs s_{bc}. If $r > 2$, that is $0 < p < 2$, Lemma 11.6 is contradicted. If $r \leq 2$, that is $-4 < p \leq 0$, Lemma 11.7 gives $m_1 = m_2 = 4m_5$, $m_3 = m_4$, $s_{13} = -s_{24}$ and $s_{14} = -s_{23}$, and Lemma 11.8 is contradicted. So γ does not exist. A Diagram 8 germ with an unbounded four-product is in particular a Diagram 8 germ. □

Remark 11.10. (a) The window is sharp. Along rays of the ratio chart with a finite lower wing, the strokes and circles are exactly those of Diagram 8 for $-4 < p < 2$; at $p = 2$ the z-circles at 1 and 2 join the upper z-strokes, which is the pattern of Diagram 6; for $p > 2$ there would be z-circles at $1, 2$ without z-strokes, against Rule 1a; and for $p \leq -4$ the upper w-strokes reach the order of the lower w-stroke 34. So the face $p = 2$ of the chart is never reached by a Diagram 8 germ (Proposition 11.3); nothing is excluded or delegated there, and germs on that face are not Diagram 8 germs.

(b) Along a ray germ the squared distances have orders $4 - 2p$ (upper wing), -6 (crossed pairs) and 0 (lower wing) in t, so the least four-product has order $-8 - 2p < 0$: every Diagram 8 germ has an unbounded four-product. We do not use this.

Both items follow from the orders in t of the components of $\mathcal{Z}$ and $\mathcal{W}$ along the rays (Appendix A).

Appendix A. The computations

The computer-assisted steps are carried out by programs written in Python 3.14 with the libraries SymPy 1.14 and mpmath 1.3. Each program has tests that compare its output with the statement in the text. Statements proved by hand in the text were also checked by computer where this was easy; these checks are not listed here.

Exact algebra. Identities, resultants, greatest common divisors, Gröbner bases, ranks and determinants are computed over $\mathbb{Q}$, over number fields such as $\mathbb{Q}(i)$, $\mathbb{Q}(\omega)$ and $\mathbb{Q}(\sqrt{3}, i, 3^{-1/3})$, and over fields of rational functions. An identity is proved by expanding the difference of its two sides to the zero polynomial. Real roots are isolated by Sturm sequences, and signs of polynomials on boxes are obtained from Bernstein coefficients. A rank or a nonvanishing obtained over a field of rational functions is used at a point only after the relevant denominators and minors have been shown not to vanish there, as in Lemma 11.6. A computation modulo a prime is used to show that a quantity is nonzero only through a minor that is a unit modulo that prime, as in Lemma 7.11.

Interval and ball arithmetic. Real and complex numbers are enclosed by interval arithmetic with outward rounding. Roots are enclosed in discs by the Krawczyk and Pellet tests and by Rouché's theorem. The covers of Section 6.5 also use complex balls built on IEEE binary64 arithmetic, with relative error bounds $2u + 2u^2$ for the modulus, $2.5u$ for a product, $3u + 7u^2$ for an inverse and $3.5u + 8u^2$ for a square root, where $u = 2^{-53}$; these bounds assume round-to-nearest and no overflow. The ball operations use a common $8u$ allowance for these primitives. Sums of n terms are enlarged by the factors $(1 + u)^n$ and $\gamma_n = nu/(1 - nu)$, every radius is increased by 10^{-300} to account for underflow, and nonzero arguments of modulus below 2^{-1000} are rejected.

Exact linear programming. The systems of rational linear equations and inequalities of Section 8.2 are decided by the simplex method over $\mathbb{Q}$, with Bland's rule and free variables; strict inequalities are encoded with a common slack $0 < s \leq 1$.

Stored results. The tables of Lemmas 8.20 and 8.21, the zero classes of Lemma 8.10, the representatives of Lemma 6.16 and the jets of the curves C_p of Lemma 9.9 are stored with the programs. Designated verification checks compare pinned witnesses with their recorded SHA-256 digests; an individual helper may read a witness before that separate check runs. A matching digest establishes the identity of the stored data, not the correctness of its calculation. The pinned checks test selected identities and witness entries; a separate full replay recomputes the tables of Section 8.2 and the representatives of Lemma 6.16 from scratch. The zero classes are checked by root isolation, and the jets by substitution into the equations.

Programs and running times. Table 5 lists the programs used for the computer-assisted statements, including supplementary remarks not needed for the main theorem; their names begin with `ak_`, which is omitted. The selected proof verification checks do not replay every supplementary remark or the cross-check of Appendix B. On a computer with eight logical processors, the tests of these programs take about 15 hours of processor time, and the recomputation of the tables of Section 8.2 about 4 more hours.

Appendix B. Cross-check through the exceptional set

The exceptional set of [1] is not used in the proof. As an independent check of the case analysis we rebuilt it. The pairs of first and second limits in the proof of [1, Thm. 6], with all labellings of the bodies, give 171 components of codimension two in the mass space and 661 orbits of pairs of diagrams under the symmetries. Of these orbits, 255 contain no positive mass vector, and each of the other 406 is excluded by Sections 5–11 applied to its first limit; no orbit requires an argument about its second limit.

Statement	Programs
Lemmas 5.1, 5.11	`diagram1_separated_center_cone`, `diagram1_virial_exclusion`, `diagram5_valuation_reduction`
Lemma 5.4	`central_body_slice`
Lemma 5.8	`isolated_triangle_shape_derivation`
Lemma 5.9	`diagram5_valuation_reduction`, `diagram1_edge_triangle`
Lemma 6.2, Definition 6.3	`blowup`, `resultants`, `p_source_completeness`, `p_general`, `diagram2_symmetry`
Lemma 6.10	`l_block_symmetric_germs`, `p_sheet_second_order`
Lemma 6.12	`l4_same_sign`, `resultants`, `exceptional`, `l5_obstruction`
Lemmas 6.13, 6.8	`l5_obstruction`
Lemmas 6.14, 6.15	`exceptional_branch_positive_feasibility`, `l1_complex_fibre`, `l_block_symmetric_germs`
Lemma 6.16	`k1_curve`, `k1_complex_fibre`, `p4_curve`, `p5_completion`, `resultants`, `diagram2_symmetry`
Lemmas 6.17, 6.18	`p_sheet_second_order`, `p_sheet_special_ends`, `p_sheet_e5_corner`, `p_sheet_euler_closed_form`, `p_sheet_p5_exact`
Remark 6.20	`p_general`, `diagram2_symmetry`
Lemmas 7.3–7.9	`diagram3_blowup`
Lemma 7.10	`diagram3_fifth_order`
Lemmas 7.11, 7.12	`diagram3_equal_shape`
Lemma 8.4	`diagram4_valuation_coverage`
Lemma 8.10	`diagram4_first_order_formula`, `diagram4_sign_census`, `diagram4_sign_orbits`, `diagram4_zero_uniqueness`, `diagram4_third_order_formula`
Lemmas 8.16, 8.19	`diagram5_morse_bott_lemma`, `diagram5_morse_bott_uniformity`
Lemmas 8.20, 8.21	`diagram5_outer_census`, `diagram5_census_model`, `diagram5_first_limit_relevance`, `tropical_census`
Lemmas 9.2, 9.4	`diagram6_blowup`, `diagram6_valuation_coverage`, `diagram6_relation32_derivation`
Lemma 9.6, Definition 9.7, Remarks 9.8, 9.11	`diagram6_blowup`
Lemma 9.9	`diagram6_blowup`, `diagram6_primitive_cubic`, `diagram6_primitive_quadratic`, `diagram6_primitive_opposite`
Lemma 11.1	`diagram8_relation40_derivation`, `diagram8_kappa_row`
Lemma 11.6	`diagram8_order_six_pairing`, `diagram8_equal_pair_germs`
Lemma 11.7	`diagram8_kappa_row`
Lemma 11.8	`diagram8_cancelling_curve`
Remark 11.10	`diagram8_ray_diagrams`, `diagram8_pattern_boundaries`

TABLE 5. Programs used for the computer-assisted statements.

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